

GRADE 6 COMMON CORE MATH

Answer Key

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[82 question sets](#) • [2465 questions](#)

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Write a ratio that describes the relationship between two quantities

MAIN	Grade 6 main question	6.RP.A.1
BACKWARD 1	Reading counts from a table or graph	3.MD.B.3
BACKWARD 2	Parts and totals	2.OA.A.1
BACKWARD 3	Fraction of a whole	3.NF.A.1
FORWARD 1	Ratios and unit rates with fractional quantities	7.RP.A.1
FORWARD 2	The unit rate as the slope of a graph	8.EE.B.5

SET 1 • 6.RP.A.1

MAIN

Grade 6 main question • Grade 6 • Ratios & Proportional Relationships •
6.RP.A.1

A fruit bowl has 5 pears and 8 plums.

Write the ratio of pears to plums.

Ratio: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 1 • MAIN • Q1/5

9

QUESTION ID

S1-M-Q1

TYPE

Short answer

CORRECT ANSWER

5 : 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 1 • MAIN • Q1/5

A choir has 14 sopranos and 9 altos.

Write the ratio of altos to sopranos.

Ratio: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 1 • MAIN • Q2/5

10

QUESTION ID

S1-M-Q2

TYPE

Short answer

CORRECT ANSWER

9 : 14

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 1 • MAIN • Q2/5

The table shows the pets owned by students in a class.

Write the ratio of dogs to fish.

Pet	Dogs	Cats	Fish
Students	11	6	4

Ratio: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 1 • MAIN • Q3/5

11

QUESTION ID

S1-M-Q3

TYPE

Short answer

CORRECT ANSWER

11 : 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 1 • MAIN • Q3/5

A parking lot has 13 cars and 7 trucks. There are no other vehicles.

Write the ratio of trucks to all of the vehicles.

Ratio: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 1 • MAIN • Q4/5

12

QUESTION ID

S1-M-Q4

TYPE

Short answer

CORRECT ANSWER

7 : 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 1 • MAIN • Q4/5

A bag has 20 marbles. 12 are green and the rest are yellow.

Write the ratio of yellow marbles to green marbles.

Ratio: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 1 • MAIN • Q5/5

13

QUESTION ID

S1-M-Q5

TYPE

Short answer

CORRECT ANSWER

8 : 12

GRADING NOTE Equivalent ratio 2 : 3 is also correct

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 1 • MAIN • Q5/5

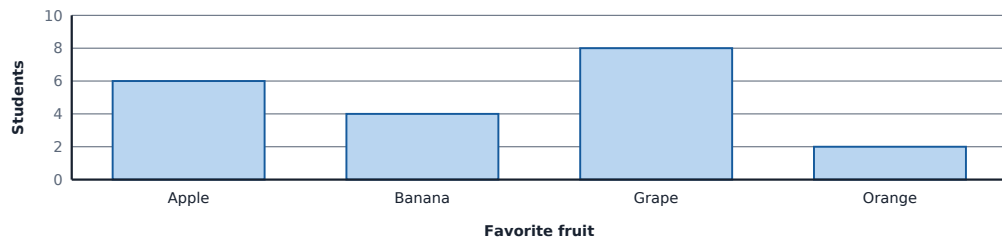
SET 1 • 6.RP.A.1

BACKWARD 1

Reading counts from a table or graph • Grade 3 • Measurement & Data •
3.MD.B.3

The graph shows the favorite fruits of a class.

How many students chose grape?



Students: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q1

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 1 • BACKWARD 1 • Q1/5

15

QUESTION ID

S1-B1-Q1

TYPE

Short answer

CORRECT ANSWER

8

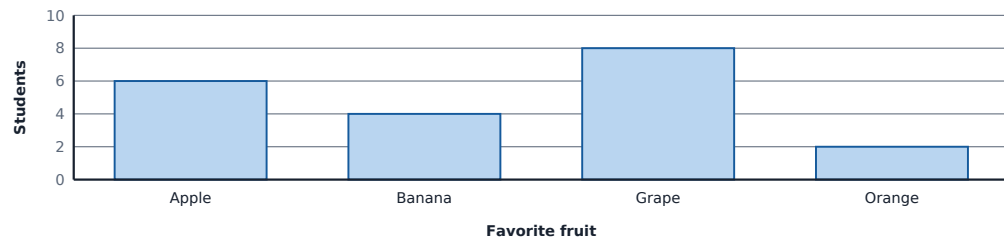
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q1

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 1 • BACKWARD 1 • Q1/5

The graph shows the favorite fruits of a class.

How many students chose banana?



Students: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q2

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 1 • BACKWARD 1 • Q2/5

16

QUESTION ID

S1-B1-Q2

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q2

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 1 • BACKWARD 1 • Q2/5

The table shows the pets owned by students.

How many students own cats?

Pet	Dogs	Cats	Fish
Students	11	6	4

Students: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q3

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 1 • BACKWARD 1 • Q3/5

17

QUESTION ID

S1-B1-Q3

TYPE

Short answer

CORRECT ANSWER

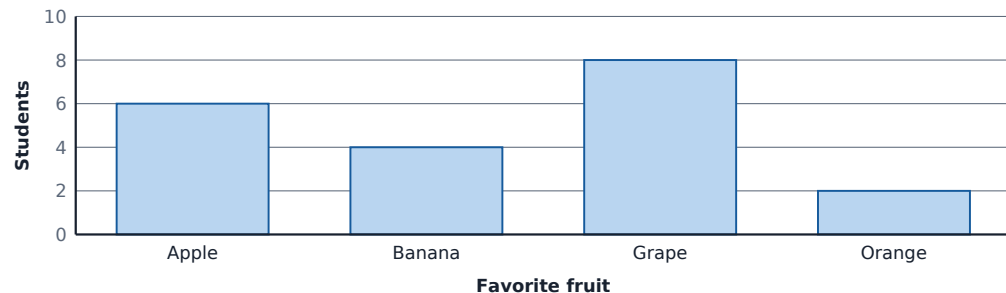
6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q3

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 1 • BACKWARD 1 • Q3/5

Which fruit was chosen by 2 students?



A. Grape B. Orange C. Banana D. Apple

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q4

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 1 • BACKWARD 1 • Q4/5

18

QUESTION ID

S1-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. Orange

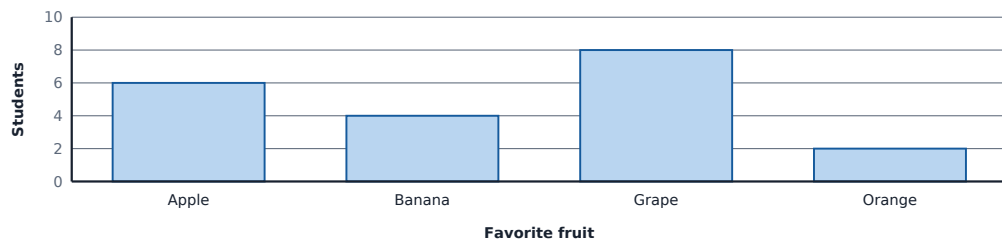
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q4

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 1 • BACKWARD 1 • Q4/5

True or false?

6 students chose apple.



A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q5

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 1 • BACKWARD 1 • Q5/5

19

QUESTION ID

S1-B1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q5

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 1 • BACKWARD 1 • Q5/5

SET 1 • 6.RP.A.1

BACKWARD 2

Parts and totals • Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1

A team has 9 boys and 6 girls.

How many players are on the team?

Players: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q1

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 1 • BACKWARD 2 • Q1/5

21

QUESTION ID

S1-B2-Q1

TYPE

Short answer

CORRECT ANSWER

15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q1

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 1 • BACKWARD 2 • Q1/5

A box holds 24 crayons. 15 are broken.

How many crayons are NOT broken?

Crayons: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q2

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 1 • BACKWARD 2 • Q2/5

22

QUESTION ID

S1-B2-Q2

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q2

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 1 • BACKWARD 2 • Q2/5

True or false?

A shelf has 7 red books and 12 blue books. There are 19 books in all.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q3

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 1 • BACKWARD 2 • Q3/5

23

QUESTION ID

S1-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q3

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 1 • BACKWARD 2 • Q3/5

A garden has 30 plants. 18 are flowers and the rest are herbs.

How many herbs are there?

A. 12

B. 18

C. 22

D. 48

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q4

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 1 • BACKWARD 2 • Q4/5

24

QUESTION ID

S1-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q4

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 1 • BACKWARD 2 • Q4/5

A bus has 17 adults and 26 children.

How many people are on the bus?

People: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q5

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 1 • BACKWARD 2 • Q5/5

25

QUESTION ID

S1-B2-Q5

TYPE

Short answer

CORRECT ANSWER

43

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q5

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 1 • BACKWARD 2 • Q5/5

SET 1 • 6.RP.A.1

BACKWARD 3

Fraction of a whole • Grade 3 • Number & Operations–Fractions • 3.NF.A.1

The bar is divided into equal parts.

What fraction of the bar is shaded?



Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q1

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 1 • BACKWARD 3 • Q1/5

27

QUESTION ID

S1-B3-Q1

TYPE

Short answer

CORRECT ANSWER

3/7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q1

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 1 • BACKWARD 3 • Q1/5

A pizza is cut into 6 equal slices. Ava eats 5 slices.

What fraction of the pizza does Ava eat?

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q2

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 1 • BACKWARD 3 • Q2/5

28

QUESTION ID

S1-B3-Q2

TYPE

Short answer

CORRECT ANSWER

5/6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q2

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 1 • BACKWARD 3 • Q2/5

True or false?

A bar is cut into 4 equal parts and 1 part is shaded. The shaded part is $\frac{1}{4}$ of the bar.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q3

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 1 • BACKWARD 3 • Q3/5

29

QUESTION ID

S1-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q3

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 1 • BACKWARD 3 • Q3/5

What fraction of the bar is shaded?



A. $\frac{2}{5}$

B. $\frac{3}{5}$

C. $\frac{2}{3}$

D. $\frac{5}{2}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q4

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 1 • BACKWARD 3 • Q4/5

30

QUESTION ID

S1-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 2/5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q4

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 1 • BACKWARD 3 • Q4/5

A field has 8 equal sections. Corn grows in 3 of the sections.

What fraction of the field has corn?

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q5

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 1 • BACKWARD 3 • Q5/5

31

QUESTION ID

S1-B3-Q5

TYPE

Short answer

CORRECT ANSWER

3/8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q5

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 1 • BACKWARD 3 • Q5/5

SET 1 • 6.RP.A.1

FORWARD 1

Ratios and unit rates with fractional quantities • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1

A dressing uses $\frac{1}{2}$ cup of oil for every $\frac{3}{4}$ cup of vinegar.

How many cups of oil are used per cup of vinegar?

Cups of oil: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 1 • FORWARD 1 • Q1/5

33

QUESTION ID

S1-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$\frac{2}{3}$ cup

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 1 • FORWARD 1 • Q1/5

Mia paints $\frac{2}{3}$ of a fence in $\frac{1}{2}$ hour.

How much of the fence does she paint per hour?

Fence per hour: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 1 • FORWARD 1 • Q2/5

34

QUESTION ID

S1-F1-Q2

TYPE

Short answer

CORRECT ANSWER

**$\frac{4}{3} = 1 \frac{1}{3}$ fences
per hour**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 1 • FORWARD 1 • Q2/5

A garden uses $\frac{3}{4}$ bag of soil for every $\frac{3}{8}$ bag of mulch.

How many bags of soil are used per bag of mulch?

A. $\frac{9}{32}$

B. $\frac{1}{2}$

C. $1\frac{1}{8}$

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 1 • FORWARD 1 • Q3/5

35

QUESTION ID

S1-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 1 • FORWARD 1 • Q3/5

A cyclist rides $2\frac{1}{2}$ miles in $\frac{1}{6}$ hour.

What is the speed in miles per hour?

Speed: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 1 • FORWARD 1 • Q4/5

36

QUESTION ID

S1-F1-Q4

TYPE

Short answer

CORRECT ANSWER

15 miles per hour

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 1 • FORWARD 1 • Q4/5

A pump moves $\frac{5}{8}$ gallon of water every $\frac{1}{4}$ minute.

How many gallons does it move per minute?

Gallons per minute: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 1 • FORWARD 1 • Q5/5

37

QUESTION ID

S1-F1-Q5

TYPE

Short answer

CORRECT ANSWER

**$5/2 = 2 \frac{1}{2}$ gallons
per minute**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 1 • FORWARD 1 • Q5/5

SET 1 • 6.RP.A.1

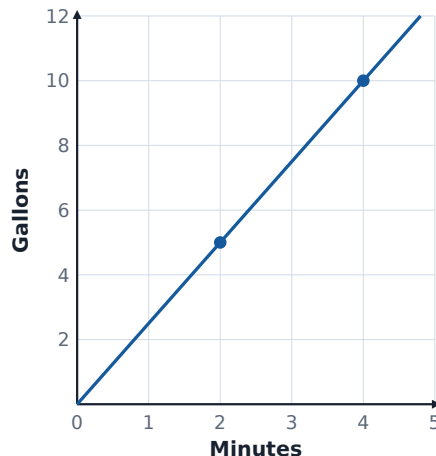
FORWARD 2

The unit rate as the slope of a graph • Grade 8 • Expressions & Equations •
8.EE.B.5

The graph shows a proportional relationship between minutes and gallons.

What is the slope of the line?

Slope: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 1 • FORWARD 2 • Q1/5

39

QUESTION ID

S1-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$5/2 = 2.5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 1 • FORWARD 2 • Q1/5

The graph of a proportional relationship passes through (0, 0) and (6, 9).

What is the slope?

Slope: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 1 • FORWARD 2 • Q2/5

40

QUESTION ID

S1-F2-Q2

TYPE

Short answer

CORRECT ANSWER

$3/2 = 1.5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

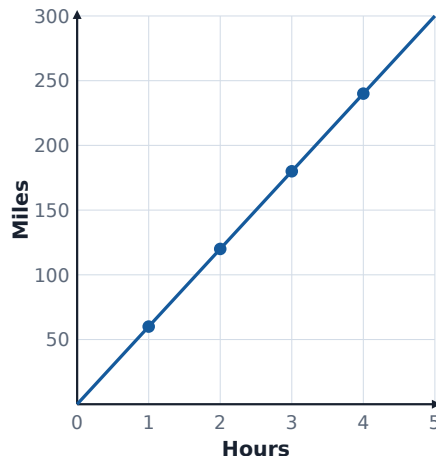
ID S1-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 1 • FORWARD 2 • Q2/5

The graph shows the distance a car travels.

What is the slope of the line in miles per hour?

Slope: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 1 • FORWARD 2 • Q3/5

41

QUESTION ID

S1-F2-Q3

TYPE

Short answer

CORRECT ANSWER

60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 1 • FORWARD 2 • Q3/5

A proportional relationship has a unit rate of 4 dollars per pound.

Which point is on its graph?

- A.** (3, 12) **B.** (4, 1)
C. (3, 7) **D.** (12, 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 1 • FORWARD 2 • Q4/5

42

QUESTION ID

S1-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. (3, 12)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 1 • FORWARD 2 • Q4/5

True or false?

For $y = 2.5x$, the unit rate is 2.5, and the slope of its graph is also 2.5.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 1 • FORWARD 2 • Q5/5

43

QUESTION ID

S1-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 1 • FORWARD 2 • Q5/5

Interpret ratio language ("for every," "to")

MAIN	Grade 6 main question	6.RP.A.1
BACKWARD 1	Equal groups	3.OA.A.1
BACKWARD 2	Multiplicative comparison statements	4.OA.A.1
FORWARD 1	Constant of proportionality from a description	7.RP.A.2.b
FORWARD 2	Compare proportional relationships represented in different ways	8.EE.B.5

SET 2 • 6.RP.A.1

MAIN

Grade 6 main question • Grade 6 • Ratios & Proportional Relationships •
6.RP.A.1

A recipe uses 5 cups of flour for every 2 cups of sugar.

Which statement describes the same ratio?

- A. For every 7 cups of sugar, there are 5 cups of flour.
- B. The ratio of sugar to flour is 2 : 5.
- C. For every 2 cups of flour, there are 5 cups of sugar.
- D. The ratio of flour to sugar is 2 : 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 2 • MAIN • Q1/5

46

QUESTION ID

S2-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. The ratio of sugar to flour is 2 : 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 2 • MAIN • Q1/5

At an aquarium, the ratio of sharks to rays is 4 : 9.

Which statement is true?

- A. For every 4 rays, there are 9 sharks.
- B. For every 9 sharks, there are 4 rays.
- C. For every 13 sharks, there are 4 rays.
- D. For every 9 rays, there are 4 sharks.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 2 • MAIN • Q2/5

47

QUESTION ID

S2-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

**D. For every 9 rays,
there are 4 sharks.**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 2 • MAIN • Q2/5

The ratio of red beads to blue beads is 3 : 7.

Which statement is true?

- A. For every 10 blue beads, there are 3 red beads.
- B. For every 7 red beads, there are 3 blue beads.
- C. For every 3 red beads, there are 7 blue beads.
- D. For every 3 blue beads, there are 7 red beads.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 2 • MAIN • Q3/5

48

QUESTION ID

S2-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. For every 3 red beads, there are 7 blue beads.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 2 • MAIN • Q3/5

At a camp, there are 2 counselors for every 9 campers.

What is the ratio of campers to counselors?

- A. 2 : 9
- B. 11 : 9
- C. 9 : 2
- D. 2 : 11

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 2 • MAIN • Q4/5

49

QUESTION ID

S2-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 9 : 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 2 • MAIN • Q4/5

In a class vote, there were 4 votes for pizza for every 3 votes for tacos.

Which statement is true?

- A.** The ratio of pizza votes to taco votes is 3 : 4. **B.** The ratio of pizza votes to all votes is 4 : 3.
- C.** There were 4 taco votes for every 3 pizza votes. **D.** The ratio of taco votes to pizza votes is 3 : 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 2 • MAIN • Q5/5

50

QUESTION ID

S2-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

D. The ratio of taco votes to pizza votes is 3 : 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 2 • MAIN • Q5/5

SET 2 • 6.RP.A.1

BACKWARD 1

Equal groups • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1

True or false?

4×6 can mean 4 groups with 6 in each group.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 2 • BACKWARD 1 • Q1/5

52

QUESTION ID

S2-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 2 • BACKWARD 1 • Q1/5

There are 3 bags with 5 apples in each bag.

Which expression shows the total?

A. 3×5

B. $5 - 3$

C. $3 + 5$

D. $5 \div 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 2 • BACKWARD 1 • Q2/5

53

QUESTION ID

S2-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. 3×5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 2 • BACKWARD 1 • Q2/5

There are 7 tables with 4 chairs at each table.

How many chairs are there?

Chairs: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 2 • BACKWARD 1 • Q3/5

54

QUESTION ID

S2-B1-Q3

TYPE

Short answer

CORRECT ANSWER

28

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 2 • BACKWARD 1 • Q3/5

True or false?

"2 wheels for every bike" means each bike has 2 wheels.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 2 • BACKWARD 1 • Q4/5

55

QUESTION ID

S2-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 2 • BACKWARD 1 • Q4/5

What does 6×2 mean?

- A.** 6 groups of 2
- B.** 62
- C.** 6 more than 2
- D.** 6 minus 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 2 • BACKWARD 1 • Q5/5

56

QUESTION ID

S2-B1-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. 6 groups of 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 2 • BACKWARD 1 • Q5/5

SET 2 • 6.RP.A.1

BACKWARD 2

Multiplicative comparison statements • Grade 4 • Operations & Algebraic Thinking • 4.OA.A.1

True or false?

24 is 3 times as many as 8.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.1 • Set 2 • BACKWARD 2 • Q1/5

58

QUESTION ID

S2-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.1 • Set 2 • BACKWARD 2 • Q1/5

Which equation shows "42 is 6 times as many as 7"?

A. $42 = 6 + 7$

B. $42 = 6 \times 7$

C. $42 - 7 = 6$

D. $7 = 6 \times 42$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.1 • Set 2 • BACKWARD 2 • Q2/5

59

QUESTION ID

S2-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. $42 = 6 \times 7$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.1 • Set 2 • BACKWARD 2 • Q2/5

True or false?

18 is 6 times as many as 3.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.1 • Set 2 • BACKWARD 2 • Q3/5

60

QUESTION ID

S2-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.1 • Set 2 • BACKWARD 2 • Q3/5

Fill in the blank.

45 is ____ times as many as 9.

Blank: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.1 • Set 2 • BACKWARD 2 • Q4/5

61

QUESTION ID

S2-B2-Q4

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.1 • Set 2 • BACKWARD 2 • Q4/5

True or false?

20 is 4 times as many as 16.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.1 • Set 2 • BACKWARD 2 • Q5/5

62

QUESTION ID

S2-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.1 • Set 2 • BACKWARD 2 • Q5/5

SET 2 • 6.RP.A.1

FORWARD 1

Constant of proportionality from a description • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b

A candle burns 2 inches every 3 hours at a constant rate.

What is the constant of proportionality in inches per hour?

$k =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 2 • FORWARD 1 • Q1/5

64

QUESTION ID

S2-F1-Q1

TYPE

Short answer

CORRECT ANSWER

2/3 inch per hour

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 2 • FORWARD 1 • Q1/5

For every 5 laps, Jo swims 250 meters.

What is the constant of proportionality in meters per lap?

$k =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 2 • FORWARD 1 • Q2/5

65

QUESTION ID

S2-F1-Q2

TYPE

Short answer

CORRECT ANSWER

50 meters per lap

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 2 • FORWARD 1 • Q2/5

A machine fills 18 bottles for every 4 minutes.

What is the constant of proportionality in bottles per minute?

A. $\frac{2}{9}$

B. 4.5

C. 14

D. 72

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 2 • FORWARD 1 • Q3/5

66

QUESTION ID

S2-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 4.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 2 • FORWARD 1 • Q3/5

For every 8 cups of water, a recipe uses 3 cups of rice.

What is the constant of proportionality in cups of rice per cup of water?

$k =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 2 • FORWARD 1 • Q4/5

67

QUESTION ID

S2-F1-Q4

TYPE

Short answer

CORRECT ANSWER

3/8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 2 • FORWARD 1 • Q4/5

True or false?

For every 6 pounds of apples, the cost is \$9. The constant of proportionality is 1.5 dollars per pound.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 2 • FORWARD 1 • Q5/5

68

QUESTION ID

S2-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 2 • FORWARD 1 • Q5/5

SET 2 • 6.RP.A.1

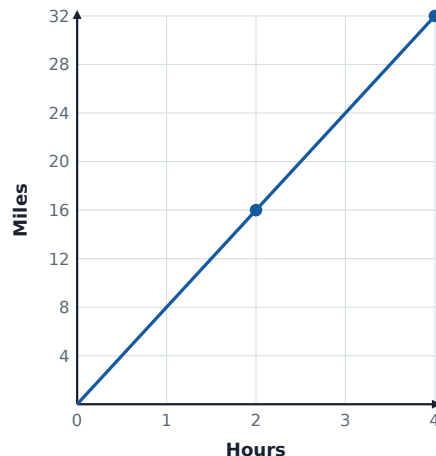
FORWARD 2

Compare proportional relationships represented in different ways • Grade 8 •
Expressions & Equations • 8.EE.B.5

Cyclist A's distance is shown in the graph. Cyclist B rides according to $d = 7.5t$, where d is miles and t is hours.

Which cyclist is faster?

- A. Cyclist A
- B. Cyclist B
- C. They ride at the same speed.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 2 • FORWARD 2 • Q1/5

70

QUESTION ID

S2-F2-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. Cyclist A

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 2 • FORWARD 2 • Q1/5

Tank A fills according to $y = 12x$, where x is minutes and y is gallons. The graph of Tank B is a line through $(0, 0)$ and $(3, 45)$.

Which tank fills faster, and by how many gallons per minute?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 2 • FORWARD 2 • Q2/5

71

QUESTION ID

S2-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Tank B is faster, by 3 gallons per minute

GRADING NOTE Tank B rate = $45/3 = 15$ gal/min; Tank A = 12 gal/min

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 2 • FORWARD 2 • Q2/5

Store A: 2 pounds of apples cost \$3.50.

Store B: the cost is $c = 1.6p$ for p pounds.

Which store charges less per pound?

- A. Store A
- B. Store B
- C. They charge the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 2 • FORWARD 2 • Q3/5

72

QUESTION ID

S2-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. Store B

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 2 • FORWARD 2 • Q3/5

Printer A prints 3 pages for every 2 seconds. Printer B prints according to $p = 1.4s$.

Which printer is faster?

- A. Printer A
- B. Printer B
- C. They are the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 2 • FORWARD 2 • Q4/5

73

QUESTION ID

S2-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. Printer A

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 2 • FORWARD 2 • Q4/5

Car A travels 150 miles in 3 hours at a constant speed.

Car B travels according to $d = 48t$.

How much faster is Car A, in miles per hour?

Miles per hour: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 2 • FORWARD 2 • Q5/5

74

QUESTION ID

S2-F2-Q5

TYPE

Short answer

CORRECT ANSWER

2 miles per hour

GRADING NOTE Car A = $150/3 = 50$ mph; Car B = 48 mph

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 2 • FORWARD 2 • Q5/5

Find and describe a unit rate

MAIN

Grade 6 main question

6.RP.A.2

BACKWARD 1

Whole-number division

4.NBT.B.6

BACKWARD 2

Fraction as division

5.NF.B.3

BACKWARD 3

Meaning of "per" (the amount for each one)

3.OA.A.2

FORWARD 1

Unit rates with fractions

7.RP.A.1

FORWARD 2

Rate of change of a linear function

8.F.B.4

SET 3 • 6.RP.A.2

MAIN

Grade 6 main question • Grade 6 • Ratios & Proportional Relationships •
6.RP.A.2

A car travels 150 miles in 3 hours at a constant speed.

What is the unit rate in miles per hour?

Unit rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 3 • MAIN • Q1/5

77

QUESTION ID

S3-M-Q1

TYPE

Short answer

CORRECT ANSWER

50 miles per hour

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 3 • MAIN • Q1/5

A 4-pound bag of rice costs \$12.

What is the price per pound?

Price per pound: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 3 • MAIN • Q2/5

78

QUESTION ID

S3-M-Q2

TYPE

Short answer

CORRECT ANSWER

\$3 per pound

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 3 • MAIN • Q2/5

A punch recipe uses 5 cups of juice for every 8 cups of water.

How many cups of juice are used for each cup of water?

Cups of juice: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 3 • MAIN • Q3/5

79

QUESTION ID

S3-M-Q3

TYPE

Short answer

CORRECT ANSWER

5/8 cup

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 3 • MAIN • Q3/5

There are 2 teachers for every 30 students on a field trip.

How many students are there per teacher?

A. 15

B. 28

C. 30

D. 60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 3 • MAIN • Q4/5

80

QUESTION ID

S3-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 3 • MAIN • Q4/5

A printer prints 90 pages in 6 minutes.

How many pages does it print per minute?

Pages per minute: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 3 • MAIN • Q5/5

81

QUESTION ID

S3-M-Q5

TYPE

Short answer

CORRECT ANSWER

15 pages per minute

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 3 • MAIN • Q5/5

SET 3 • 6.RP.A.2

BACKWARD 1

Whole-number division • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.6

Find the quotient.

$$150 \div 3$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 3 • BACKWARD 1 • Q1/5

83

QUESTION ID

S3-B1-Q1

TYPE

Short answer

CORRECT ANSWER

50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 3 • BACKWARD 1 • Q1/5

Which is equal to $96 \div 4$?

- A.** 22
- B.** 23
- C.** 24
- D.** 26

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 3 • BACKWARD 1 • Q2/5

84

QUESTION ID

S3-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. 24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 3 • BACKWARD 1 • Q2/5

True or false?

$$84 \div 6 = 14$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 3 • BACKWARD 1 • Q3/5

85

QUESTION ID

S3-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 3 • BACKWARD 1 • Q3/5

Find the quotient.

$$212 \div 4$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 3 • BACKWARD 1 • Q4/5

86

QUESTION ID

S3-B1-Q4

TYPE

Short answer

CORRECT ANSWER

53

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 3 • BACKWARD 1 • Q4/5

Find the quotient.

$$135 \div 5$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 3 • BACKWARD 1 • Q5/5

87

QUESTION ID

S3-B1-Q5

TYPE

Short answer

CORRECT ANSWER

27

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 3 • BACKWARD 1 • Q5/5

SET 3 • 6.RP.A.2

BACKWARD 2

Fraction as division • Grade 5 • Number & Operations–Fractions • 5.NF.B.3

Write $5 \div 8$ as a fraction.

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.3 • Set 3 • BACKWARD 2 • Q1/5

89

QUESTION ID

S3-B2-Q1

TYPE

Short answer

CORRECT ANSWER

5/8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.3 • Set 3 • BACKWARD 2 • Q1/5

True or false?

$\frac{2}{5}$ means the same as $2 \div 5$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.3 • Set 3 • BACKWARD 2 • Q2/5

90

QUESTION ID

S3-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.3 • Set 3 • BACKWARD 2 • Q2/5

Four friends share 3 sandwiches equally.

How much sandwich does each friend get?

Each friend gets: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.3 • Set 3 • BACKWARD 2 • Q3/5

91

QUESTION ID

S3-B2-Q3

TYPE

Short answer

CORRECT ANSWER

3/4 of a sandwich

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.3 • Set 3 • BACKWARD 2 • Q3/5

Which division expression is equal to $\frac{7}{8}$?

- A. $8 \div 7$
- B. $8 - 7$
- C. $7 \div 8$
- D. 7×8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.3 • Set 3 • BACKWARD 2 • Q4/5

92

QUESTION ID

S3-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. $7 \div 8$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.3 • Set 3 • BACKWARD 2 • Q4/5

A 5-yard ribbon is cut into 6 equal pieces.

How long is each piece?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.3 • Set 3 • BACKWARD 2 • Q5/5

93

QUESTION ID

S3-B2-Q5

TYPE

Short answer

CORRECT ANSWER

5/6 yard

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.3 • Set 3 • BACKWARD 2 • Q5/5

SET 3 • 6.RP.A.2

BACKWARD 3

Meaning of "per" (the amount for each one) • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2

12 apples are packed equally into 4 bags.

How many apples are in each bag?

Apples: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B3-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 3 • BACKWARD 3 • Q1/5

95

QUESTION ID

S3-B3-Q1

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B3-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 3 • BACKWARD 3 • Q1/5

True or false?

"5 miles per hour" means 5 miles for each 1 hour.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B3-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 3 • BACKWARD 3 • Q2/5

96

QUESTION ID

S3-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B3-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 3 • BACKWARD 3 • Q2/5

Which phrase means the same as "\$2 per pound"?

- A.** \$2 for each pound
- B.** \$2 for all of the pounds
- C.** 2 pounds for each dollar
- D.** \$2 more than a pound

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B3-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 3 • BACKWARD 3 • Q3/5

97

QUESTION ID

S3-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. \$2 for each pound

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B3-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 3 • BACKWARD 3 • Q3/5

A farmer puts 24 eggs into cartons, 6 eggs per carton.

How many cartons does the farmer fill?

Cartons: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B3-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 3 • BACKWARD 3 • Q4/5

98

QUESTION ID

S3-B3-Q4

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B3-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 3 • BACKWARD 3 • Q4/5

Jen reads 40 pages in 5 days. She reads the same number of pages each day.

How many pages does she read per day?

Pages: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B3-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 3 • BACKWARD 3 • Q5/5

99

QUESTION ID

S3-B3-Q5

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B3-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 3 • BACKWARD 3 • Q5/5

SET 3 • 6.RP.A.2

FORWARD 1

Unit rates with fractions • Grade 7 • Ratios & Proportional Relationships •
7.RP.A.1

Kim walks $\frac{3}{4}$ mile in $\frac{1}{4}$ hour.

What is her speed in miles per hour?

Speed: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 3 • FORWARD 1 • Q1/5

101

QUESTION ID

S3-F1-Q1

TYPE

Short answer

CORRECT ANSWER

3 miles per hour

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 3 • FORWARD 1 • Q1/5

A painter paints $\frac{2}{3}$ of a wall in $\frac{1}{3}$ hour.

At this rate, how many walls can the painter paint per hour?

Walls per hour: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 3 • FORWARD 1 • Q2/5

102

QUESTION ID

S3-F1-Q2

TYPE

Short answer

CORRECT ANSWER

2 walls per hour

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 3 • FORWARD 1 • Q2/5

A snail crawls $\frac{5}{6}$ foot in $\frac{1}{3}$ minute.

How many feet does it crawl per minute?

Feet per minute: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 3 • FORWARD 1 • Q3/5

103

QUESTION ID

S3-F1-Q3

TYPE

Short answer

CORRECT ANSWER

$5/2 = 2 \frac{1}{2}$ feet per minute

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 3 • FORWARD 1 • Q3/5

A machine fills $\frac{3}{4}$ of a tank in $\frac{1}{2}$ hour.

How many tanks does it fill per hour?

A. $\frac{1}{4}$

B. $\frac{3}{8}$

C. $1\frac{1}{4}$

D. $1\frac{1}{2}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 3 • FORWARD 1 • Q4/5

104

QUESTION ID

S3-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 1 1/2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 3 • FORWARD 1 • Q4/5

Leo mows $\frac{7}{8}$ of an acre in $\frac{1}{2}$ hour.

How many acres does he mow per hour?

Acres per hour: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 3 • FORWARD 1 • Q5/5

105

QUESTION ID

S3-F1-Q5

TYPE

Short answer

CORRECT ANSWER

$7/4 = 1 \frac{3}{4}$ acres per hour

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 3 • FORWARD 1 • Q5/5

SET 3 • 6.RP.A.2

FORWARD 2

Rate of change of a linear function • Grade 8 • Functions • 8.F.B.4

The table shows a linear function.

What is the rate of change?

x	0	1	2	3
y	5	8	11	14

Rate of change: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q1

Grade 8 • Functions • 8.F.B.4 • Set 3 • FORWARD 2 • Q1/5

107

QUESTION ID

S3-F2-Q1

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q1

Grade 8 • Functions • 8.F.B.4 • Set 3 • FORWARD 2 • Q1/5

A plumber charges a \$40 fee plus \$65 per hour.

Write a function for the cost C after h hours.

$C =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q2

Grade 8 • Functions • 8.F.B.4 • Set 3 • FORWARD 2 • Q2/5

108

QUESTION ID

S3-F2-Q2

TYPE

Short answer

CORRECT ANSWER

$C = 65h + 40$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q2

Grade 8 • Functions • 8.F.B.4 • Set 3 • FORWARD 2 • Q2/5

A line passes through (2, 7) and (6, 19).

What is the rate of change?

Rate of change: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q3

Grade 8 • Functions • 8.F.B.4 • Set 3 • FORWARD 2 • Q3/5

109

QUESTION ID

S3-F2-Q3

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q3

Grade 8 • Functions • 8.F.B.4 • Set 3 • FORWARD 2 • Q3/5

A tank drains at a constant rate.

What is the rate of change in gallons per minute?

Minutes	0	5	10
Gallons	200	170	140

Rate of change: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q4

Grade 8 • Functions • 8.F.B.4 • Set 3 • FORWARD 2 • Q4/5

110

QUESTION ID

S3-F2-Q4

TYPE

Short answer

CORRECT ANSWER

**–6 gallons per
minute**

GRADING NOTE The tank loses 6 gallons each minute; "6 gallons per minute, decreasing" is also correct

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q4

Grade 8 • Functions • 8.F.B.4 • Set 3 • FORWARD 2 • Q4/5

A taxi ride costs \$3 plus \$2.50 per mile.

Which function gives the cost y for a ride of x miles?

A. $y = 3x + 2.5$

B. $y = 2.5x - 3$

C. $y = 2.5x + 3$

D. $y = 5.5x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q5

Grade 8 • Functions • 8.F.B.4 • Set 3 • FORWARD 2 • Q5/5

111

QUESTION ID

S3-F2-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. $y = 2.5x + 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q5

Grade 8 • Functions • 8.F.B.4 • Set 3 • FORWARD 2 • Q5/5

Find missing values in tables of equivalent ratios

MAIN Grade 6 main question

BACKWARD 1 Equivalent fractions

BACKWARD 2 Two related number patterns

BACKWARD 3 Multiples of a number

FORWARD 1 Constant of proportionality from a table

FORWARD 2 Rate of change and initial value from a table

6.RP.A.3.a

4.NF.A.1

5.OA.B.3

4.OA.B.4

7.RP.A.2.b

8.F.B.4

SET 4 • 6.RP.A.3.a

MAIN

Grade 6 main question • Grade 6 • Ratios & Proportional Relationships •
6.RP.A.3.a

The table shows equivalent ratios.

Find the missing value.

Cups of juice	2	4	6	10
Cups of water	3	6	9	?

Missing value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 4 • MAIN • Q1/5

114

QUESTION ID

S4-M-Q1

TYPE

Short answer

CORRECT ANSWER

15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 4 • MAIN • Q1/5

Paint is mixed in the same ratio in every column.

Find the missing value.

Blue (cups)	3	6	?	15
Yellow (cups)	2	4	8	10

Missing value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 4 • MAIN • Q2/5

115

QUESTION ID

S4-M-Q2

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 4 • MAIN • Q2/5

A swimmer keeps the same pace.

Find the missing value.

Laps	3	6	?	15
Minutes	8	16	24	40

Missing value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 4 • MAIN • Q3/5

116

QUESTION ID

S4-M-Q3

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 4 • MAIN • Q3/5

The table shows equivalent ratios.

What is the missing value?

Tickets	4	8	12	?
Cost (\$)	10	20	30	45

A. 16

B. 18

C. 20

D. 35

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 4 • MAIN • Q4/5

117

QUESTION ID

S4-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 4 • MAIN • Q4/5

A recipe uses 5 cups of oats for every 2 cups of raisins.

Find the missing values A and B.

Oats (cups)	5	10	A	25
Raisins (cups)	2	B	6	10

A = _____

B = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 4 • MAIN • Q5/5

118

QUESTION ID

S4-M-Q5

TYPE

Short answer

CORRECT ANSWER

A = 15, B = 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 4 • MAIN • Q5/5

SET 4 • 6.RP.A.3.a

BACKWARD 1

Equivalent fractions • Grade 4 • Number & Operations–Fractions • 4.NF.A.1

True or false?

$\frac{2}{3}$ is equivalent to $\frac{8}{12}$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 4 • BACKWARD 1 • Q1/5

120

QUESTION ID

S4-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 4 • BACKWARD 1 • Q1/5

Find the missing number.

$$\frac{3}{4} = \frac{?}{20}$$

Missing number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 4 • BACKWARD 1 • Q2/5

121

QUESTION ID

S4-B1-Q2

TYPE

Short answer

CORRECT ANSWER

15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 4 • BACKWARD 1 • Q2/5

Which fraction is equivalent to $\frac{5}{6}$?

A. $\frac{5}{12}$

B. $\frac{10}{12}$

C. $\frac{6}{7}$

D. $\frac{10}{11}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 4 • BACKWARD 1 • Q3/5

122

QUESTION ID

S4-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 10/12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 4 • BACKWARD 1 • Q3/5

True or false?

$\frac{4}{5}$ is equivalent to $\frac{6}{7}$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 4 • BACKWARD 1 • Q4/5

123

QUESTION ID

S4-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 4 • BACKWARD 1 • Q4/5

Find the missing number.

$$\frac{2}{5} = \frac{6}{?}$$

Missing number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 4 • BACKWARD 1 • Q5/5

124

QUESTION ID

S4-B1-Q5

TYPE

Short answer

CORRECT ANSWER

15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 4 • BACKWARD 1 • Q5/5

SET 4 • 6.RP.A.3.a

BACKWARD 2

Two related number patterns • Grade 5 • Operations & Algebraic Thinking •
5.OA.B.3

Pattern A starts at 0 and adds 2.

Pattern B starts at 0 and adds 5.

What is the 5th number in Pattern B?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 4 • BACKWARD 2 • Q1/5

126

QUESTION ID

S4-B2-Q1

TYPE

Short answer

CORRECT ANSWER

20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 4 • BACKWARD 2 • Q1/5

True or false?

Each number in Pattern B is 2 times the matching number in Pattern A.

Pattern A (add 4)	0	4	8	12
Pattern B (add 8)	0	8	16	24

A. True**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 4 • BACKWARD 2 • Q2/5

127

QUESTION ID

S4-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 4 • BACKWARD 2 • Q2/5

Pattern A starts at 0 and adds 7.

Pattern B starts at 0 and adds 14.

What is the 4th number in Pattern A?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 4 • BACKWARD 2 • Q3/5

128

QUESTION ID

S4-B2-Q3

TYPE

Short answer

CORRECT ANSWER

21

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 4 • BACKWARD 2 • Q3/5

Pattern A: 0, 2, 4, 6, 8

Pattern B: 0, 6, 12, 18, 24

How does each number in Pattern B compare to the matching number in Pattern A?

- A.** 2 times as large **B.** 3 times as large
C. 4 more **D.** 6 times as large

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 4 • BACKWARD 2 • Q4/5

129

QUESTION ID

S4-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 3 times as large

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 4 • BACKWARD 2 • Q4/5

Pattern A: 0, 1, 2, 3, ...

Pattern B: 0, 4, 8, 12, ...

What number in Pattern B matches 5 in Pattern A?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 4 • BACKWARD 2 • Q5/5

130

QUESTION ID

S4-B2-Q5

TYPE

Short answer

CORRECT ANSWER

20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 4 • BACKWARD 2 • Q5/5

SET 4 • 6.RP.A.3.a

BACKWARD 3

Multiples of a number • Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4

List the first four multiples of 3.

Multiples: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B3-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 4 • BACKWARD 3 • Q1/5

132

QUESTION ID

S4-B3-Q1

TYPE

Short answer

CORRECT ANSWER

3, 6, 9, 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B3-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 4 • BACKWARD 3 • Q1/5

True or false?

40 is a multiple of 8.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B3-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 4 • BACKWARD 3 • Q2/5

133

QUESTION ID

S4-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B3-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 4 • BACKWARD 3 • Q2/5

Which number is a multiple of 5?

- A.** 32
- B.** 35
- C.** 48
- D.** 52

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B3-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 4 • BACKWARD 3 • Q3/5

134

QUESTION ID

S4-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 35

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B3-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 4 • BACKWARD 3 • Q3/5

What is the 6th multiple of 4?

Multiple: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B3-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 4 • BACKWARD 3 • Q4/5

135

QUESTION ID

S4-B3-Q4

TYPE

Short answer

CORRECT ANSWER

24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B3-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 4 • BACKWARD 3 • Q4/5

True or false?

27 is a multiple of 6.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B3-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 4 • BACKWARD 3 • Q5/5

136

QUESTION ID

S4-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B3-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 4 • BACKWARD 3 • Q5/5

SET 4 • 6.RP.A.3.a

FORWARD 1

Constant of proportionality from a table • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b

The table shows a proportional relationship.

What is the constant of proportionality ($\frac{y}{x}$)?

x	2	5	8
y	7	17.5	28

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 4 • FORWARD 1 • Q1/5

138

QUESTION ID

S4-F1-Q1

TYPE

Short answer

CORRECT ANSWER

3.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 4 • FORWARD 1 • Q1/5

The table shows the cost of cheese.

What is the constant of proportionality in dollars per pound?

Pounds	3	6	9
Cost (\$)	7.50	15.00	22.50

$k =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 4 • FORWARD 1 • Q2/5

139

QUESTION ID

S4-F1-Q2

TYPE

Short answer

CORRECT ANSWER

\$2.50 per pound

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 4 • FORWARD 1 • Q2/5

The table shows a proportional relationship.

What is the constant of proportionality?

x	4	8	12
y	3	6	9

A. $\frac{3}{4}$

B. $\frac{4}{3}$

C. 3

D. 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 4 • FORWARD 1 • Q3/5

140

QUESTION ID

S4-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 3/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 4 • FORWARD 1 • Q3/5

The table shows how far a toy car rolls.

What is the constant of proportionality in meters per second?

Seconds	2	3	5
Meters	1.2	1.8	3.0

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 4 • FORWARD 1 • Q4/5

141

QUESTION ID

S4-F1-Q4

TYPE

Short answer

CORRECT ANSWER

0.6 meter per second

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 4 • FORWARD 1 • Q4/5

True or false?

The constant of proportionality for the table is 6.

x	1	2	4
y	6	12	24

- A. True
- B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 4 • FORWARD 1 • Q5/5

142

QUESTION ID

S4-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 4 • FORWARD 1 • Q5/5

SET 4 • 6.RP.A.3.a

FORWARD 2

Rate of change and initial value from a table • Grade 8 • Functions • 8.F.B.4

The table shows a linear function.

What are the rate of change and the initial value?

x	0	1	2	3
y	4	7	10	13

Rate of change: _____

Initial value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q1

Grade 8 • Functions • 8.F.B.4 • Set 4 • FORWARD 2 • Q1/5

144

QUESTION ID

S4-F2-Q1

TYPE

Short answer

CORRECT ANSWER

**Rate of change: 3;
Initial value: 4**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q1

Grade 8 • Functions • 8.F.B.4 • Set 4 • FORWARD 2 • Q1/5

The table shows the height of a plant.

What is the rate of change in cm per week?

Weeks	0	2	4
Height (cm)	6	9	12

Rate of change: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q2

Grade 8 • Functions • 8.F.B.4 • Set 4 • FORWARD 2 • Q2/5

145

QUESTION ID

S4-F2-Q2

TYPE

Short answer

CORRECT ANSWER

1.5 cm per week

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q2

Grade 8 • Functions • 8.F.B.4 • Set 4 • FORWARD 2 • Q2/5

The table shows a linear function.

Which equation matches the table?

x	0	1	2	3
y	5	7	9	11

A. $y = x + 5$

B. $y = 5x + 2$

C. $y = 2x + 5$

D. $y = 2x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q3

Grade 8 • Functions • 8.F.B.4 • Set 4 • FORWARD 2 • Q3/5

146

QUESTION ID

S4-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $y = 2x + 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q3

Grade 8 • Functions • 8.F.B.4 • Set 4 • FORWARD 2 • Q3/5

The table shows a linear function.

Write an equation for the function.

x	1	2	3	4
y	9	13	17	21

$y =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q4

Grade 8 • Functions • 8.F.B.4 • Set 4 • FORWARD 2 • Q4/5

147

QUESTION ID

S4-F2-Q4

TYPE

Short answer

CORRECT ANSWER

$$y = 4x + 5$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q4

Grade 8 • Functions • 8.F.B.4 • Set 4 • FORWARD 2 • Q4/5

True or false?

The initial value of the function in the table is 20.

x	0	1	2
y	20	17	14

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q5

Grade 8 • Functions • 8.F.B.4 • Set 4 • FORWARD 2 • Q5/5

148

QUESTION ID

S4-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q5

Grade 8 • Functions • 8.F.B.4 • Set 4 • FORWARD 2 • Q5/5

Plot pairs of values from a ratio table on the coordinate plane

MAIN	Grade 6 main question	6.RP.A.3.a
BACKWARD 1	Ordered pairs from two related patterns	5.OA.B.3
BACKWARD 2	Axes and ordered pairs	5.G.A.1
BACKWARD 3	Plotting points in the first quadrant	5.G.A.2
BACKWARD 4	Reading an axis marked by 2s, 3s, or 5s	3.MD.B.3
FORWARD 1	Use a graph to decide whether a relationship is proportional	7.RP.A.2.a
FORWARD 2	Graph proportional relationships; interpret the slope	8.EE.B.5

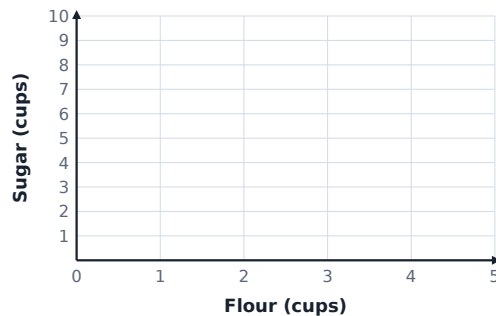
SET 5 • 6.RP.A.3.a

MAIN

Grade 6 main question • Grade 6 • Ratios & Proportional Relationships •
6.RP.A.3.a

Plot the pairs of values from the table on the coordinate plane.

Flour (cups)	1	2	3	4
Sugar (cups)	2	4	6	8



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 5 • MAIN • Q1/5

151

QUESTION ID

S5-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Points plotted at (1, 2), (2, 4), (3, 6), (4, 8)

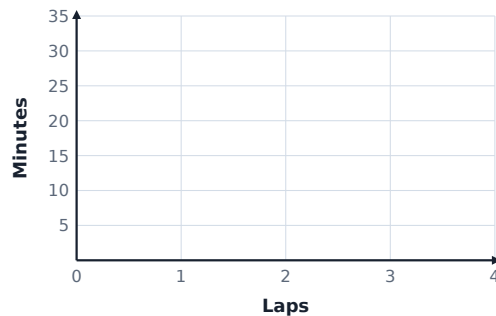
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 5 • MAIN • Q1/5

Plot the pairs of values from the table on the coordinate plane.

Laps	1	2	3
Minutes	10	20	30



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 5 • MAIN • Q2/5

152

QUESTION ID

S5-M-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

Points plotted at (1, 10), (2, 20), (3, 30)

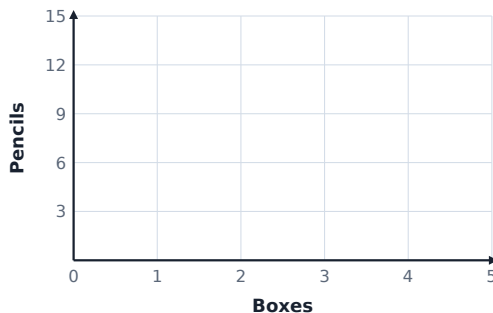
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 5 • MAIN • Q2/5

Plot the pairs of values from the table on the coordinate plane.

Boxes	1	2	3	4
Pencils	3	6	9	12



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 5 • MAIN • Q3/5

153

QUESTION ID

S5-M-Q3

TYPE

Drawing / plotting

CORRECT ANSWER

Points plotted at (1, 3), (2, 6), (3, 9), (4, 12)

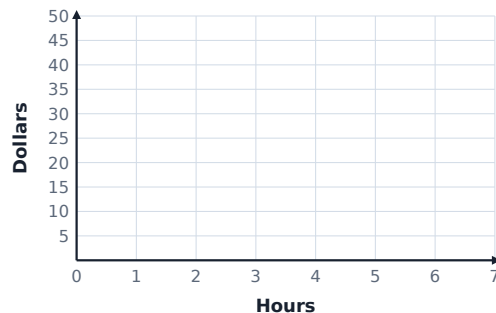
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 5 • MAIN • Q3/5

Plot the pairs of values from the table on the coordinate plane.

Hours	2	4	6
Dollars	15	30	45



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 5 • MAIN • Q4/5

154

QUESTION ID

S5-M-Q4

TYPE

Drawing / plotting

CORRECT ANSWER

Points plotted at (2, 15), (4, 30), (6, 45)

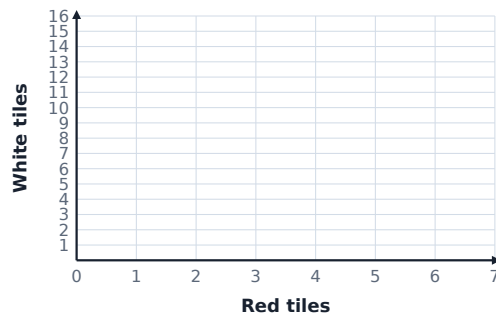
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 5 • MAIN • Q4/5

Plot the pairs of values from the table on the coordinate plane.

Red tiles	2	4	6
White tiles	5	10	15



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 5 • MAIN • Q5/5

155

QUESTION ID

S5-M-Q5

TYPE

Drawing / plotting

CORRECT ANSWER

Points plotted at (2, 5), (4, 10), (6, 15)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 5 • MAIN • Q5/5

SET 5 • 6.RP.A.3.a

BACKWARD 1

Ordered pairs from two related patterns • Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3

Pattern A: 0, 1, 2, 3

Pattern B: 0, 3, 6, 9

Write the ordered pair formed by the 3rd numbers of the patterns.

Ordered pair: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 5 • BACKWARD 1 • Q1/5

157

QUESTION ID

S5-B1-Q1

TYPE

Short answer

CORRECT ANSWER

(2, 6)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 5 • BACKWARD 1 • Q1/5

Which ordered pairs come from the table?

x	1	2	3
y	5	10	15

- A.** (1, 5), (2, 10), (3, 15) **B.** (5, 1), (10, 2), (15, 3)
C. (1, 2), (5, 10), (15, 3) **D.** (1, 10), (2, 5), (3, 15)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 5 • BACKWARD 1 • Q2/5

158

QUESTION ID

S5-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. (1, 5), (2, 10), (3, 15)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 5 • BACKWARD 1 • Q2/5

True or false?

The table gives the ordered pair (4, 12).

x	2	4	6
y	6	12	18

- A. True
- B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 5 • BACKWARD 1 • Q3/5

159

QUESTION ID

S5-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 5 • BACKWARD 1 • Q3/5

Write the ordered pair from the last column of the table.

x	1	2	3	4
y	7	14	21	28

Ordered pair: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 5 • BACKWARD 1 • Q4/5

160

QUESTION ID

S5-B1-Q4

TYPE

Short answer

CORRECT ANSWER

(4, 28)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 5 • BACKWARD 1 • Q4/5

True or false?

In an ordered pair, the first number comes from the top row of the table.

x	1	2
y	4	8

A. True**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 5 • BACKWARD 1 • Q5/5

161

QUESTION ID

S5-B1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 5 • BACKWARD 1 • Q5/5

SET 5 • 6.RP.A.3.a

BACKWARD 2

Axes and ordered pairs • Grade 5 • Geometry • 5.G.A.1

True or false?

To plot (3, 7), start at (0, 0), move 3 units right, then 7 units up.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q1

Grade 5 • Geometry • 5.G.A.1 • Set 5 • BACKWARD 2 • Q1/5

163

QUESTION ID

S5-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q1

Grade 5 • Geometry • 5.G.A.1 • Set 5 • BACKWARD 2 • Q1/5

In the ordered pair (6, 2), which number tells how far to move up?

- A.** 2
- B.** 4
- C.** 6
- D.** 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q2

Grade 5 • Geometry • 5.G.A.1 • Set 5 • BACKWARD 2 • Q2/5

164

QUESTION ID

S5-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q2

Grade 5 • Geometry • 5.G.A.1 • Set 5 • BACKWARD 2 • Q2/5

What are the coordinates of the origin?

Origin: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q3

Grade 5 • Geometry • 5.G.A.1 • Set 5 • BACKWARD 2 • Q3/5

165

QUESTION ID

S5-B2-Q3

TYPE

Short answer

CORRECT ANSWER

(0, 0)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q3

Grade 5 • Geometry • 5.G.A.1 • Set 5 • BACKWARD 2 • Q3/5

True or false?

(4, 9) and (9, 4) name the same point.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q4

Grade 5 • Geometry • 5.G.A.1 • Set 5 • BACKWARD 2 • Q4/5

166

QUESTION ID

S5-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q4

Grade 5 • Geometry • 5.G.A.1 • Set 5 • BACKWARD 2 • Q4/5

Which axis is vertical?

- A.** The x-axis
- B.** The y-axis

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q5

Grade 5 • Geometry • 5.G.A.1 • Set 5 • BACKWARD 2 • Q5/5

167

QUESTION ID

S5-B2-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. The y-axis

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q5

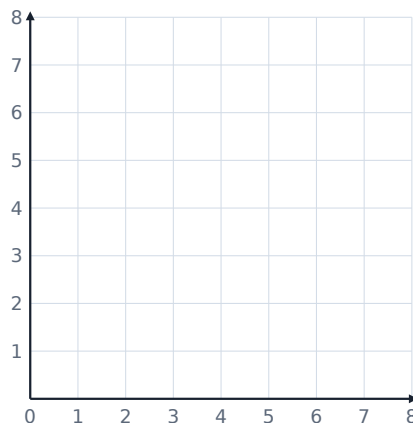
Grade 5 • Geometry • 5.G.A.1 • Set 5 • BACKWARD 2 • Q5/5

SET 5 • 6.RP.A.3.a

BACKWARD 3

Plotting points in the first quadrant • Grade 5 • Geometry • 5.G.A.2

Plot and label point A at (3, 5).



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q1

Grade 5 • Geometry • 5.G.A.2 • Set 5 • BACKWARD 3 • Q1/5

169

QUESTION ID

S5-B3-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

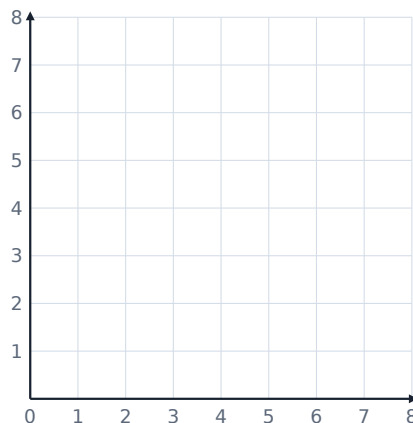
Point A plotted at (3, 5) and labeled

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q1

Grade 5 • Geometry • 5.G.A.2 • Set 5 • BACKWARD 3 • Q1/5

Plot and label point B at (6, 2).



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q2

Grade 5 • Geometry • 5.G.A.2 • Set 5 • BACKWARD 3 • Q2/5

170

QUESTION ID

S5-B3-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

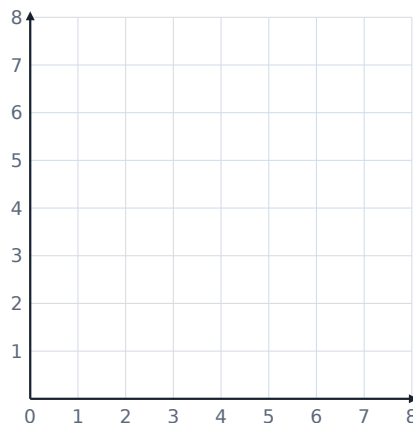
Point B plotted at (6, 2) and labeled

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q2

Grade 5 • Geometry • 5.G.A.2 • Set 5 • BACKWARD 3 • Q2/5

Plot and label point C at (0, 4).



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q3

Grade 5 • Geometry • 5.G.A.2 • Set 5 • BACKWARD 3 • Q3/5

171

QUESTION ID

S5-B3-Q3

TYPE

Drawing / plotting

CORRECT ANSWER

Point C plotted at (0, 4) on the y-axis and labeled

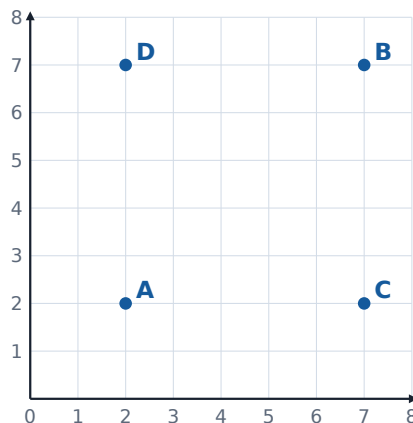
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q3

Grade 5 • Geometry • 5.G.A.2 • Set 5 • BACKWARD 3 • Q3/5

Which point is at (2, 7)?

- A. Point A
- B. Point B
- C. Point C
- D. Point D



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q4

Grade 5 • Geometry • 5.G.A.2 • Set 5 • BACKWARD 3 • Q4/5

172

QUESTION ID

S5-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

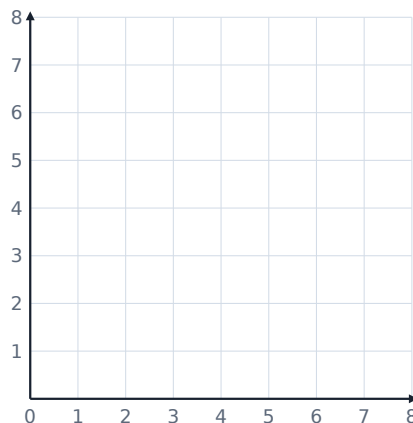
D. Point D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q4

Grade 5 • Geometry • 5.G.A.2 • Set 5 • BACKWARD 3 • Q4/5

Plot and label point D at (5, 5).



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q5

Grade 5 • Geometry • 5.G.A.2 • Set 5 • BACKWARD 3 • Q5/5

173

QUESTION ID

S5-B3-Q5

TYPE

Drawing / plotting

CORRECT ANSWER

Point D plotted at (5, 5) and labeled

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q5

Grade 5 • Geometry • 5.G.A.2 • Set 5 • BACKWARD 3 • Q5/5

SET 5 • 6.RP.A.3.a

BACKWARD 4

Reading an axis marked by 2s, 3s, or 5s • Grade 3 • Measurement & Data •
3.MD.B.3

On the y-axis, the grid lines are numbered 0, 5, 10, 15.

Where would 20 go?

- A. At the 2nd grid line
- B. At the 20th grid line
- C. One grid line above 15
- D. Halfway between 10 and 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B4-Q1

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 5 • BACKWARD 4 • Q1/5

175

QUESTION ID

S5-B4-Q1

TYPE

Multiple choice

CORRECT ANSWER

**C. One grid line
above 15**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B4-Q1

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 5 • BACKWARD 4 • Q1/5

True or false?

On an axis numbered 0, 3, 6, 9, the value 12 is one grid line above 9.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B4-Q2

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 5 • BACKWARD 4 • Q2/5

176

QUESTION ID

S5-B4-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B4-Q2

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 5 • BACKWARD 4 • Q2/5

On an axis numbered 0, 2, 4, 6, 8, where does the value 5 go?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B4-Q3

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 5 • BACKWARD 4 • Q3/5

177

QUESTION ID

S5-B4-Q3

TYPE

Short answer

CORRECT ANSWER

**Halfway between 4
and 6**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B4-Q3

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 5 • BACKWARD 4 • Q3/5

An axis is numbered 0, 5, 10, 15, 20.

How many units does each grid line stand for?

- A.** 1
- B.** 5
- C.** 10
- D.** 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B4-Q4

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 5 • BACKWARD 4 • Q4/5

178

QUESTION ID

S5-B4-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B4-Q4

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 5 • BACKWARD 4 • Q4/5

True or false?

On an axis numbered by 10s, the value 35 is halfway between 30 and 40.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B4-Q5

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 5 • BACKWARD 4 • Q5/5

179

QUESTION ID

S5-B4-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B4-Q5

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 5 • BACKWARD 4 • Q5/5

SET 5 • 6.RP.A.3.a

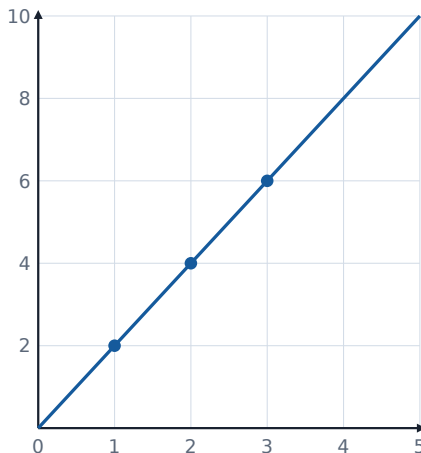
FORWARD 1

Use a graph to decide whether a relationship is proportional • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a

True or false?

The graph shows a proportional relationship.

- A.** True
- B.** False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 5 • FORWARD 1 • Q1/5

181

QUESTION ID

S5-F1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

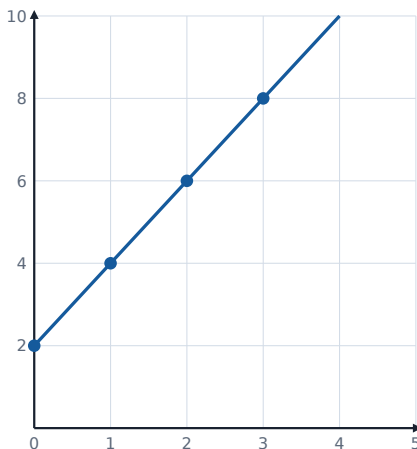
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 5 • FORWARD 1 • Q1/5

Which statement about the graph is true?

- A. It is proportional because y increases by 2.
- B. It is not proportional because the line does not pass through $(0, 0)$.
- C. It is not proportional because the points are not on a line.
- D. It is proportional because it is a straight line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 5 • FORWARD 1 • Q2/5

182

QUESTION ID

S5-F1-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. It is not proportional because the line does not pass through $(0, 0)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

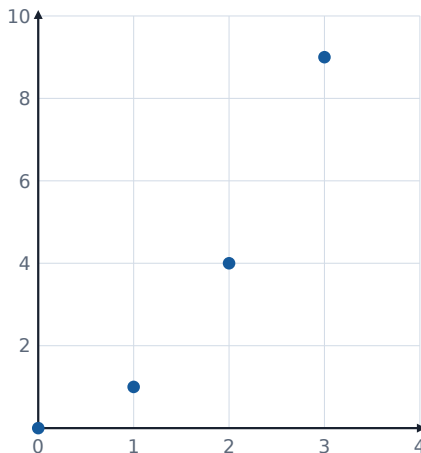
ID S5-F1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 5 • FORWARD 1 • Q2/5

True or false?

The graph shows a proportional relationship.

- A.** True
- B.** False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 5 • FORWARD 1 • Q3/5

183

QUESTION ID

S5-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 5 • FORWARD 1 • Q3/5

The points (2, 5), (4, 10), and (6, 15) are graphed.
Do they lie on a straight line through the origin? Write
yes or no.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 5 • FORWARD 1 • Q4/5

184

QUESTION ID

S5-F1-Q4

TYPE

Short answer

CORRECT ANSWER

Yes

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 5 • FORWARD 1 • Q4/5

Which set of points lies on a straight line through the origin?

- A. (0, 2), (1, 4), (2, 6)
- B. (1, 3), (2, 4), (3, 5)
- C. (2, 3), (4, 6), (6, 9)
- D. (1, 1), (2, 4), (3, 9)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 5 • FORWARD 1 • Q5/5

185

QUESTION ID

S5-F1-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. (2, 3), (4, 6), (6, 9)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 5 • FORWARD 1 • Q5/5

SET 5 • 6.RP.A.3.a

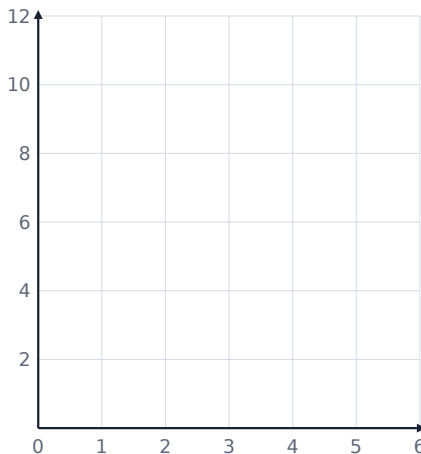
FORWARD 2

Graph proportional relationships; interpret the slope • Grade 8 • Expressions & Equations • 8.EE.B.5

Graph $y = 2x$ on the coordinate plane.

What is the slope of the line?

Slope: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 5 • FORWARD 2 • Q1/5

187

QUESTION ID

S5-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Slope: 2

GRADING NOTE Graph: straight line through (0, 0), (1, 2), (3, 6), (6, 12)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

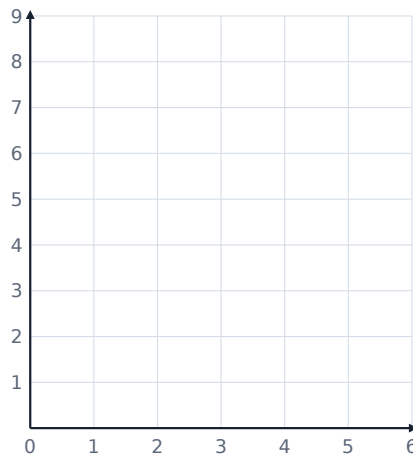
ID S5-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 5 • FORWARD 2 • Q1/5

Graph $y = 1.5x$ on the coordinate plane.

What is the slope of the line?

Slope: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 5 • FORWARD 2 • Q2/5

188

QUESTION ID

S5-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Slope: 1.5

GRADING NOTE Graph: straight line through (0, 0), (2, 3), (4, 6), (6, 9)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

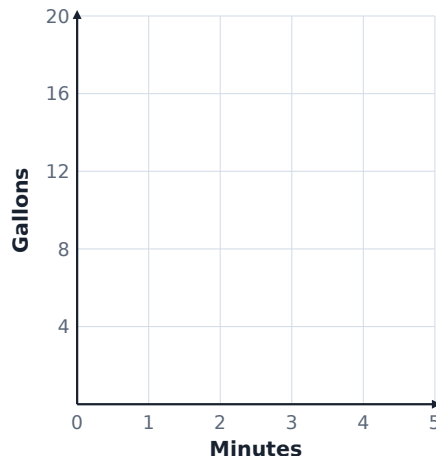
ID S5-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 5 • FORWARD 2 • Q2/5

Water flows at 4 gallons per minute. Graph the gallons y after x minutes.

What is the slope of the line?

Slope: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 5 • FORWARD 2 • Q3/5

189

QUESTION ID

S5-F2-Q3

TYPE

Short answer

CORRECT ANSWER

Slope: 4

GRADING NOTE Graph: straight line through $(0, 0)$, $(1, 4)$, $(5, 20)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 5 • FORWARD 2 • Q3/5

The graph of a proportional relationship passes through (5, 2).

What is the slope?

A. $\frac{2}{5}$

B. 2

C. $\frac{5}{2}$

D. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 5 • FORWARD 2 • Q4/5

190

QUESTION ID

S5-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 2/5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 5 • FORWARD 2 • Q4/5

True or false?

The graph of $y = 3x$ passes through (0, 0) and (2, 6), and its slope is 3.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 5 • FORWARD 2 • Q5/5

191

QUESTION ID

S5-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 5 • FORWARD 2 • Q5/5

Use tables of equivalent ratios to compare ratios

MAIN	Grade 6 main question	6.RP.A.3.a
BACKWARD 1	Comparing fractions	4.NF.A.2
BACKWARD 2	Common multiples	4.OA.B.4
BACKWARD 3	Equivalent fractions	4.NF.A.1
FORWARD 1	Compare unit rates involving fractions	7.RP.A.1
FORWARD 2	Compare proportional relationships represented in different ways	8.EE.B.5

SET 6 • 6.RP.A.3.a

MAIN

Grade 6 main question • Grade 6 • Ratios & Proportional Relationships •
6.RP.A.3.a

Lemonade A uses 2 cups of mix for every 5 cups of water. Lemonade B uses 3 cups of mix for every 7 cups of water.

Use tables of equivalent ratios. Which lemonade has a stronger mix flavor?

A: Mix	2	4	6
A: Water	5	10	15

B: Mix	3	6	9
B: Water	7	14	21

- A.** Lemonade A **B.** Lemonade B **C.** They taste the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 6 • MAIN • Q1/5

194

QUESTION ID

S6-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. Lemonade B

GRADING NOTE $A = 2/5 = 0.4$; $B = 3/7 \approx 0.43$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 6 • MAIN • Q1/5

Class A has 4 boys for every 6 girls. Class B has 6 boys for every 9 girls.

Use tables of equivalent ratios. Which statement is true?

- A. Class A has a greater ratio of boys to girls.
- B. Class B has a greater ratio of boys to girls.
- C. The ratios are equivalent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 6 • MAIN • Q2/5

195

QUESTION ID

S6-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. The ratios are equivalent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 6 • MAIN • Q2/5

Runner A runs 3 laps in 10 minutes. Runner B runs 4 laps in 12 minutes.

Use tables of equivalent ratios. Who runs faster?

- A. Runner A
- B. Runner B
- C. They run at the same speed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 6 • MAIN • Q3/5

196

QUESTION ID

S6-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. Runner B

GRADING NOTE A: 0.3 lap/min; B: 1/3 lap/min

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 6 • MAIN • Q3/5

Paint A uses 1 part blue for every 3 parts white. Paint B uses 2 parts blue for every 5 parts white.

Use tables of equivalent ratios. Which paint is a darker blue?

- A. Paint A
- B. Paint B
- C. They are the same color.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 6 • MAIN • Q4/5

197

QUESTION ID

S6-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. Paint B

GRADING NOTE A: $\frac{1}{3}$; B: $\frac{2}{5}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 6 • MAIN • Q4/5

Store A sells 5 pens for \$3. Store B sells 8 pens for \$5.

Use tables of equivalent ratios. Which store gives more pens for the same amount of money?

- A. Store A
- B. Store B
- C. They are the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 6 • MAIN • Q5/5

198

QUESTION ID

S6-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. Store A

GRADING NOTE A: $\frac{5}{3}$ pens per \$; B: $\frac{8}{5}$ pens per \$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 6 • MAIN • Q5/5

SET 6 • 6.RP.A.3.a

BACKWARD 1

Comparing fractions • Grade 4 • Number & Operations–Fractions • 4.NF.A.2

Which fraction is greater?

A. $\frac{3}{7}$

B. $\frac{3}{5}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.A.2 • Set 6 • BACKWARD 1 • Q1/5

200

QUESTION ID

S6-B1-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. 3/5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.A.2 • Set 6 • BACKWARD 1 • Q1/5

True or false?

$$\frac{5}{8} > \frac{4}{8}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.A.2 • Set 6 • BACKWARD 1 • Q2/5

201

QUESTION ID

S6-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.A.2 • Set 6 • BACKWARD 1 • Q2/5

Which symbol makes the statement true?

$$\frac{2}{3} \text{ — } \frac{3}{4}$$

A. <

B. >

C. =

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.A.2 • Set 6 • BACKWARD 1 • Q3/5

202

QUESTION ID

S6-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. <

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.A.2 • Set 6 • BACKWARD 1 • Q3/5

True or false?

$$\frac{1}{3} = \frac{2}{6}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.A.2 • Set 6 • BACKWARD 1 • Q4/5

203

QUESTION ID

S6-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.A.2 • Set 6 • BACKWARD 1 • Q4/5

Which fraction is greatest?

A. $\frac{1}{2}$

B. $\frac{7}{12}$

C. $\frac{2}{3}$

D. $\frac{5}{6}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.A.2 • Set 6 • BACKWARD 1 • Q5/5

204

QUESTION ID

S6-B1-Q5

TYPE

Multiple choice

CORRECT ANSWER

D. 5/6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.A.2 • Set 6 • BACKWARD 1 • Q5/5

SET 6 • 6.RP.A.3.a

BACKWARD 2

Common multiples • Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4

List the multiples of 5 and the multiples of 7 up to 40.

Which number is on both lists?

Common multiple: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 6 • BACKWARD 2 • Q1/5

206

QUESTION ID

S6-B2-Q1

TYPE

Short answer

CORRECT ANSWER

35

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 6 • BACKWARD 2 • Q1/5

True or false?

30 is a multiple of both 5 and 6.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 6 • BACKWARD 2 • Q2/5

207

QUESTION ID

S6-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 6 • BACKWARD 2 • Q2/5

Which number is a multiple of both 3 and 4?

- A.** 9
- B.** 12
- C.** 14
- D.** 16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 6 • BACKWARD 2 • Q3/5

208

QUESTION ID

S6-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 6 • BACKWARD 2 • Q3/5

Name a number that is a multiple of both 6 and 9.

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 6 • BACKWARD 2 • Q4/5

209

QUESTION ID

S6-B2-Q4

TYPE

Short answer

CORRECT ANSWER

18

GRADING NOTE Any common multiple of 18 (18, 36, 54, ...) is correct

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 6 • BACKWARD 2 • Q4/5

True or false?

20 is a multiple of both 4 and 8.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 6 • BACKWARD 2 • Q5/5

210

QUESTION ID

S6-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 6 • BACKWARD 2 • Q5/5

SET 6 • 6.RP.A.3.a

BACKWARD 3

Equivalent fractions • Grade 4 • Number & Operations–Fractions • 4.NF.A.1

Find the missing number.

$$\frac{2}{5} = \frac{?}{35}$$

Missing number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B3-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 6 • BACKWARD 3 • Q1/5

212

QUESTION ID

S6-B3-Q1

TYPE

Short answer

CORRECT ANSWER

14

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B3-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 6 • BACKWARD 3 • Q1/5

True or false?

$$\frac{3}{7} = \frac{6}{14}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B3-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 6 • BACKWARD 3 • Q2/5

213

QUESTION ID

S6-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B3-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 6 • BACKWARD 3 • Q2/5

Which fraction is equivalent to $\frac{4}{6}$?

A. $\frac{4}{9}$

B. $\frac{6}{9}$

C. $\frac{6}{8}$

D. $\frac{8}{10}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B3-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 6 • BACKWARD 3 • Q3/5

214

QUESTION ID

S6-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 6/9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B3-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 6 • BACKWARD 3 • Q3/5

Find the missing number.

$$\frac{1}{3} = \frac{5}{?}$$

Missing number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B3-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 6 • BACKWARD 3 • Q4/5

215

QUESTION ID

S6-B3-Q4

TYPE

Short answer

CORRECT ANSWER

15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B3-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 6 • BACKWARD 3 • Q4/5

True or false?

$$\frac{3}{10} = \frac{9}{20}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B3-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 6 • BACKWARD 3 • Q5/5

216

QUESTION ID

S6-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B3-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 6 • BACKWARD 3 • Q5/5

SET 6 • 6.RP.A.3.a

FORWARD 1

Compare unit rates involving fractions • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1

Ana walks $\frac{1}{2}$ mile in $\frac{1}{5}$ hour. Ben walks $\frac{3}{4}$ mile in $\frac{1}{3}$ hour.

Who walks faster?

- A. Ana
- B. Ben
- C. They walk at the same speed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 6 • FORWARD 1 • Q1/5

218

QUESTION ID

S6-F1-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. Ana

GRADING NOTE Ana 2 1/2 mph; Ben 2 1/4 mph

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 6 • FORWARD 1 • Q1/5

Mix A uses $\frac{2}{3}$ cup of sugar per $\frac{1}{2}$ cup of water. Mix B uses $\frac{3}{4}$ cup of sugar per $\frac{2}{3}$ cup of water.

Which mix is sweeter?

- A. Mix A
- B. Mix B
- C. They are the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 6 • FORWARD 1 • Q2/5

219

QUESTION ID

S6-F1-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. Mix A

GRADING NOTE A: $\frac{4}{3}$; B: $\frac{9}{8}$ cup sugar per cup water

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 6 • FORWARD 1 • Q2/5

Pump A moves $\frac{3}{8}$ gallon in $\frac{1}{4}$ minute. Pump B moves $\frac{1}{2}$ gallon in $\frac{1}{2}$ minute.

Which pump is faster, and what is its rate in gallons per minute?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 6 • FORWARD 1 • Q3/5

220

QUESTION ID

S6-F1-Q3

TYPE

Short answer

CORRECT ANSWER

**Pump A, 1 1/2
gallons per minute**

GRADING NOTE Pump B = 1 gallon per minute

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 6 • FORWARD 1 • Q3/5

Snail A crawls $\frac{1}{6}$ foot in $\frac{1}{2}$ minute. Snail B crawls $\frac{1}{4}$ foot in $\frac{2}{3}$ minute.

Which snail is faster?

- A. Snail A
- B. Snail B
- C. They are the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 6 • FORWARD 1 • Q4/5

221

GRADING NOTE A: $\frac{1}{3}$ ft/min; B: $\frac{3}{8}$ ft/min

QUESTION ID

S6-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. Snail B

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 6 • FORWARD 1 • Q4/5

True or false?

$\frac{1}{2}$ mile in $\frac{1}{4}$ hour is a faster speed than $\frac{3}{4}$ mile in $\frac{1}{2}$ hour.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 6 • FORWARD 1 • Q5/5

222

QUESTION ID

S6-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 6 • FORWARD 1 • Q5/5

SET 6 • 6.RP.A.3.a

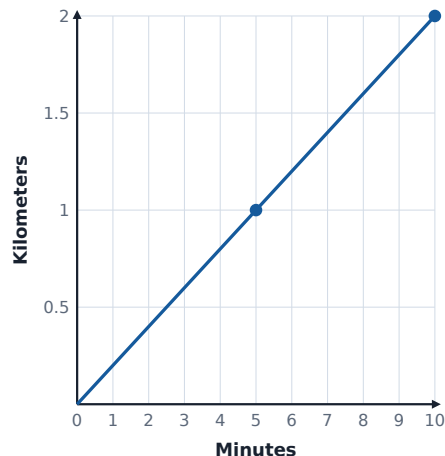
FORWARD 2

Compare proportional relationships represented in different ways • Grade 8 •
Expressions & Equations • 8.EE.B.5

Runner A's distance is shown in the graph. Runner B runs according to $d = 0.15t$, where d is kilometers and t is minutes.

Which runner is faster?

- A. Runner A
- B. Runner B
- C. They are the same.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 6 • FORWARD 2 • Q1/5

224

QUESTION ID

S6-F2-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. Runner A

GRADING NOTE Runner A = 0.2 km per minute

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 6 • FORWARD 2 • Q1/5

Machine A: $y = 45x$ bottles after x minutes. Machine B: the table shows 3 minutes $\rightarrow$ 120 bottles, 6 minutes $\rightarrow$ 240 bottles.

Which machine is faster?

- A. Machine A
- B. Machine B
- C. They are the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 6 • FORWARD 2 • Q2/5

225

QUESTION ID

S6-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. Machine A

GRADING NOTE Machine B = 40 bottles per minute

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 6 • FORWARD 2 • Q2/5

Recipe A uses flour and sugar in the ratio $y = 1.5x$.
Recipe B's graph passes through (0, 0) and (4, 7).
Which recipe uses more flour y per cup of sugar x ?

Recipe: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 6 • FORWARD 2 • Q3/5

226

QUESTION ID

S6-F2-Q3

TYPE

Short answer

CORRECT ANSWER

Recipe B

GRADING NOTE B: $7/4 = 1.75$ per cup vs A: 1.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 6 • FORWARD 2 • Q3/5

Two proportional relationships: P is $y = 6x$. Q is a line through (0, 0) and (2, 13).

Which has the greater slope?

- A. P
- B. Q
- C. They are the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 6 • FORWARD 2 • Q4/5

227

QUESTION ID

S6-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. Q

GRADING NOTE Q slope = 6.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 6 • FORWARD 2 • Q4/5

Plan A: 20 minutes of calls cost \$3.00. Plan B:
 $c = 0.12m$, where m is minutes.

Which plan costs less per minute?

Plan: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 6 • FORWARD 2 • Q5/5

228

QUESTION ID

S6-F2-Q5

TYPE

Short answer

CORRECT ANSWER

Plan B

GRADING NOTE Plan A = \$0.15 per minute; Plan B = \$0.12 per minute

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 6 • FORWARD 2 • Q5/5

Solve unit pricing problems

- MAIN** Grade 6 main question
- BACKWARD 1** Dividing money amounts
- BACKWARD 2** Multiplying money amounts
- BACKWARD 3** Finding the cost of one when each costs the same
- FORWARD 1** Equations for proportional relationships
- FORWARD 2** Build a linear function for a pricing situation

6.RP.A.3.b

5.NBT.B.7

5.NBT.B.7

3.OA.A.2

7.RP.A.2.c

8.F.B.4

SET 7 • 6.RP.A.3.b

MAIN

Grade 6 main question • Grade 6 • Ratios & Proportional Relationships •
6.RP.A.3.b

4 notebooks cost \$10.

At this rate, how much do 7 notebooks cost?

Cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 7 • MAIN • Q1/5

231

QUESTION ID

S7-M-Q1

TYPE

Short answer

CORRECT ANSWER

\$17.50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 7 • MAIN • Q1/5

3 pounds of apples cost \$5.40.

At this rate, how much do 5 pounds cost?

Cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 7 • MAIN • Q2/5

232

QUESTION ID

S7-M-Q2

TYPE

Short answer

CORRECT ANSWER

\$9.00

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 7 • MAIN • Q2/5

6 cans of soup cost \$7.50.

At this rate, how much do 10 cans cost?

Cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 7 • MAIN • Q3/5

233

QUESTION ID

S7-M-Q3

TYPE

Short answer

CORRECT ANSWER

\$12.50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 7 • MAIN • Q3/5

5 movie tickets cost \$42.50.

At this rate, how much do 3 tickets cost?

- A. \$8.50
- B. \$25.50
- C. \$28.33
- D. \$127.50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 7 • MAIN • Q4/5

234

QUESTION ID

S7-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. \$25.50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 7 • MAIN • Q4/5

8 yards of fabric cost \$36.

At this rate, how much do 3 yards cost?

Cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 7 • MAIN • Q5/5

235

QUESTION ID

S7-M-Q5

TYPE

Short answer

CORRECT ANSWER

\$13.50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 7 • MAIN • Q5/5

SET 7 • 6.RP.A.3.b

BACKWARD 1

Dividing money amounts • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.7

ANSWER KEY

Reduced copy of the question page, with its answer

Find the quotient.

$$\$10.00 \div 4$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 7 • BACKWARD 1 • Q1/5

237

QUESTION ID

S7-B1-Q1

TYPE

Short answer

CORRECT ANSWER

\$2.50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 7 • BACKWARD 1 • Q1/5

ANSWER KEY

Reduced copy of the question page, with its answer

Find the quotient.

$$\$5.40 \div 3$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 7 • BACKWARD 1 • Q2/5

238

QUESTION ID

S7-B1-Q2

TYPE

Short answer

CORRECT ANSWER

\$1.80

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 7 • BACKWARD 1 • Q2/5

238

$$\$7.50 \div 6 = ?$$

- A.** \$0.80
- B.** \$1.20
- C.** \$1.25
- D.** \$12.50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 7 • BACKWARD 1 • Q3/5

239

QUESTION ID

S7-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. \$1.25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 7 • BACKWARD 1 • Q3/5

ANSWER KEY

Reduced copy of the question page, with its answer

Find the quotient.

$$\$36 \div 8$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 7 • BACKWARD 1 • Q4/5

240

QUESTION ID

S7-B1-Q4

TYPE

Short answer

CORRECT ANSWER

\$4.50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 7 • BACKWARD 1 • Q4/5

240

True or false?

$$\$42.50 \div 5 = \$8.50$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 7 • BACKWARD 1 • Q5/5

241

QUESTION ID

S7-B1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 7 • BACKWARD 1 • Q5/5

SET 7 • 6.RP.A.3.b

BACKWARD 2

Multiplying money amounts • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.7

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$\$2.50 \times 7$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 7 • BACKWARD 2 • Q1/5

243

QUESTION ID

S7-B2-Q1

TYPE

Short answer

CORRECT ANSWER

\$17.50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 7 • BACKWARD 2 • Q1/5

243

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$\$1.80 \times 5$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 7 • BACKWARD 2 • Q2/5

244

QUESTION ID

S7-B2-Q2

TYPE

Short answer

CORRECT ANSWER

\$9.00

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 7 • BACKWARD 2 • Q2/5

True or false?

$$\$1.25 \times 10 = \$12.50$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 7 • BACKWARD 2 • Q3/5

245

QUESTION ID

S7-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 7 • BACKWARD 2 • Q3/5

$$\$4.50 \times 3 = ?$$

- A. \$1.50
- B. \$7.50
- C. \$12.50
- D. \$13.50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 7 • BACKWARD 2 • Q4/5

246

QUESTION ID

S7-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. \$13.50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 7 • BACKWARD 2 • Q4/5

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$\$8.50 \times 3$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 7 • BACKWARD 2 • Q5/5

247

QUESTION ID

S7-B2-Q5

TYPE

Short answer

CORRECT ANSWER

\$25.50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 7 • BACKWARD 2 • Q5/5

SET 7 • 6.RP.A.3.b

BACKWARD 3

Finding the cost of one when each costs the same • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2

4 toys cost \$12 in all. Each toy costs the same.

How much does 1 toy cost?

Cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B3-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 7 • BACKWARD 3 • Q1/5

249

QUESTION ID

S7-B3-Q1

TYPE

Short answer

CORRECT ANSWER

\$3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B3-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 7 • BACKWARD 3 • Q1/5

5 pencils cost 30 cents in all. Each pencil costs the same.

How much does 1 pencil cost?

Cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B3-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 7 • BACKWARD 3 • Q2/5

250

QUESTION ID

S7-B3-Q2

TYPE

Short answer

CORRECT ANSWER

6 cents

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B3-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 7 • BACKWARD 3 • Q2/5

True or false?

6 stickers cost 18 cents in all. Each costs the same, so 1 sticker costs 3 cents.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B3-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 7 • BACKWARD 3 • Q3/5

251

QUESTION ID

S7-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B3-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 7 • BACKWARD 3 • Q3/5

8 cards cost \$16. Each card costs the same.

How much is 1 card?

- A.** \$2
- B.** \$8
- C.** \$24
- D.** \$128

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B3-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 7 • BACKWARD 3 • Q4/5

252

QUESTION ID

S7-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. \$2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B3-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 7 • BACKWARD 3 • Q4/5

3 books cost \$27 in all. Each book costs the same.

How much does 1 book cost?

Cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B3-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 7 • BACKWARD 3 • Q5/5

253

QUESTION ID

S7-B3-Q5

TYPE

Short answer

CORRECT ANSWER

\$9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B3-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 7 • BACKWARD 3 • Q5/5

SET 7 • 6.RP.A.3.b

FORWARD 1

Equations for proportional relationships • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c

Apples cost \$2.50 per pound.

Write an equation for the cost c of p pounds of apples.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 7 • FORWARD 1 • Q1/5

255

QUESTION ID

S7-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$c = 2.5p$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 7 • FORWARD 1 • Q1/5

The table shows a proportional relationship.

Write an equation for the earnings e after h hours.

Hours (h)	2	3	5
Earnings (e)	36	54	90

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 7 • FORWARD 1 • Q2/5

256

QUESTION ID

S7-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$e = 18h$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 7 • FORWARD 1 • Q2/5

Pens cost \$0.75 each.

Which equation gives the cost c of n pens?

A. $c = \frac{n}{0.75}$

B. $c = n + 0.75$

C. $c = 0.75n$

D. $n = 0.75c$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 7 • FORWARD 1 • Q3/5

257

QUESTION ID

S7-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $c = 0.75n$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 7 • FORWARD 1 • Q3/5

5 pounds of rice cost \$8.

Write an equation for the cost c of p pounds of rice.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 7 • FORWARD 1 • Q4/5

258

QUESTION ID

S7-F1-Q4

TYPE

Short answer

CORRECT ANSWER

$c = 1.6p$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 7 • FORWARD 1 • Q4/5

True or false?

Grapes cost \$3.20 per pound. The equation $c = 3.2p$ gives the cost c of p pounds.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 7 • FORWARD 1 • Q5/5

259

QUESTION ID

S7-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 7 • FORWARD 1 • Q5/5

SET 7 • 6.RP.A.3.b

FORWARD 2

Build a linear function for a pricing situation • Grade 8 • Functions • 8.F.B.4

A store charges \$2.50 per pound of cherries plus a \$4 box fee.

Write a function for the total cost C of p pounds.

$C =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q1

Grade 8 • Functions • 8.F.B.4 • Set 7 • FORWARD 2 • Q1/5

261

QUESTION ID

S7-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$C = 2.5p + 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q1

Grade 8 • Functions • 8.F.B.4 • Set 7 • FORWARD 2 • Q1/5

Tickets cost \$12 each plus a \$5 service fee per order.

Which function gives the cost C of t tickets?

A. $C = 5t + 12$

B. $C = 12t - 5$

C. $C = 12t + 5$

D. $C = 17t$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q2

Grade 8 • Functions • 8.F.B.4 • Set 7 • FORWARD 2 • Q2/5

262

QUESTION ID

S7-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. $C = 12t + 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q2

Grade 8 • Functions • 8.F.B.4 • Set 7 • FORWARD 2 • Q2/5

A delivery costs \$6 plus \$1.75 per mile.

What are the rate of change and the initial value?

Rate of change: _____

Initial value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q3

Grade 8 • Functions • 8.F.B.4 • Set 7 • FORWARD 2 • Q3/5

263

QUESTION ID

S7-F2-Q3

TYPE

Short answer

CORRECT ANSWER

**Rate of change:
\$1.75 per mile;
Initial value: \$6**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q3

Grade 8 • Functions • 8.F.B.4 • Set 7 • FORWARD 2 • Q3/5

The table shows the cost of a gym membership.

Write a function for the cost C after m months.

Months (m)	0	1	2	3
Cost (C)	25	55	85	115

$C =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q4

Grade 8 • Functions • 8.F.B.4 • Set 7 • FORWARD 2 • Q4/5

264

QUESTION ID

S7-F2-Q4

TYPE

Short answer

CORRECT ANSWER

$C = 30m + 25$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q4

Grade 8 • Functions • 8.F.B.4 • Set 7 • FORWARD 2 • Q4/5

True or false?

A function for "\$3 per pound plus a \$2 bag fee" is
 $C = 3p + 2$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q5

Grade 8 • Functions • 8.F.B.4 • Set 7 • FORWARD 2 • Q5/5

265

QUESTION ID

S7-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q5

Grade 8 • Functions • 8.F.B.4 • Set 7 • FORWARD 2 • Q5/5

Solve constant speed problems

MAIN	Grade 6 main question
BACKWARD 1	Whole-number division
BACKWARD 2	Multiplying by a one-digit number
BACKWARD 3	Hours and minutes
FORWARD 1	Speed as a unit rate with fractions
FORWARD 2	Compare speeds from graphs and equations

6.RP.A.3.b

4.NBT.B.6

4.NBT.B.5

4.MD.A.1

7.RP.A.1

8.EE.B.5

SET 8 • 6.RP.A.3.b

MAIN

Grade 6 main question • Grade 6 • Ratios & Proportional Relationships •
6.RP.A.3.b

A runner runs 12 kilometers in 60 minutes at a constant speed.

How far does the runner go in 25 minutes?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 8 • MAIN • Q1/5

268

QUESTION ID

S8-M-Q1

TYPE

Short answer

CORRECT ANSWER

5 km

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 8 • MAIN • Q1/5

A train travels 180 miles in 3 hours at a constant speed.
How far does it travel in 5 hours?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 8 • MAIN • Q2/5

269

QUESTION ID

S8-M-Q2

TYPE

Short answer

CORRECT ANSWER

300 miles

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 8 • MAIN • Q2/5

A cyclist rides 45 miles in 3 hours at a constant speed.
How long does it take to ride 75 miles?

Time: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 8 • MAIN • Q3/5

270

QUESTION ID

S8-M-Q3

TYPE

Short answer

CORRECT ANSWER

5 hours

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 8 • MAIN • Q3/5

A snail moves 18 cm in 6 minutes at a constant speed.

How far does it move in 20 minutes?

- A. 3 cm
- B. 54 cm
- C. 60 cm
- D. 120 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 8 • MAIN • Q4/5

271

QUESTION ID

S8-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 60 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 8 • MAIN • Q4/5

A car travels 220 miles in 4 hours at a constant speed.

How long does it take to travel 385 miles?

Time: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 8 • MAIN • Q5/5

272

QUESTION ID

S8-M-Q5

TYPE

Short answer

CORRECT ANSWER

7 hours

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 8 • MAIN • Q5/5

SET 8 • 6.RP.A.3.b

BACKWARD 1

Whole-number division • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.6

Divide.

$$180 \div 3$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 8 • BACKWARD 1 • Q1/5

274

QUESTION ID

S8-B1-Q1

TYPE

Short answer

CORRECT ANSWER

60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 8 • BACKWARD 1 • Q1/5

ANSWER KEY

Reduced copy of the question page, with its answer

Divide.

$$220 \div 4$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 8 • BACKWARD 1 • Q2/5

275

QUESTION ID

S8-B1-Q2

TYPE

Short answer

CORRECT ANSWER

55

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 8 • BACKWARD 1 • Q2/5

275

True or false?

$$385 \div 55 = 7$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 8 • BACKWARD 1 • Q3/5

276

QUESTION ID

S8-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 8 • BACKWARD 1 • Q3/5

ANSWER KEY

Reduced copy of the question page, with its answer

$$45 \div 3 = ?$$

- A.** 12
- B.** 14
- C.** 15
- D.** 135

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 8 • BACKWARD 1 • Q4/5

277

QUESTION ID

S8-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 8 • BACKWARD 1 • Q4/5

277

ANSWER KEY

Reduced copy of the question page, with its answer

Divide.

$$375 \div 5$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 8 • BACKWARD 1 • Q5/5

278

QUESTION ID

S8-B1-Q5

TYPE

Short answer

CORRECT ANSWER

75

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 8 • BACKWARD 1 • Q5/5

278

SET 8 • 6.RP.A.3.b

BACKWARD 2

Multiplying by a one-digit number • Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$60 \times 5$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 8 • BACKWARD 2 • Q1/5

280

QUESTION ID

S8-B2-Q1

TYPE

Short answer

CORRECT ANSWER

300

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 8 • BACKWARD 2 • Q1/5

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$15 \times 7$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 8 • BACKWARD 2 • Q2/5

281

QUESTION ID

S8-B2-Q2

TYPE

Short answer

CORRECT ANSWER

105

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 8 • BACKWARD 2 • Q2/5

281

True or false?

$$55 \times 4 = 220$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 8 • BACKWARD 2 • Q3/5

282

QUESTION ID

S8-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 8 • BACKWARD 2 • Q3/5

$$48 \times 6 = ?$$

- A.** 54
- B.** 248
- C.** 288
- D.** 2,448

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 8 • BACKWARD 2 • Q4/5

283

QUESTION ID

S8-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 288

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 8 • BACKWARD 2 • Q4/5

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$125 \times 3$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 8 • BACKWARD 2 • Q5/5

284

QUESTION ID

S8-B2-Q5

TYPE

Short answer

CORRECT ANSWER

375

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 8 • BACKWARD 2 • Q5/5

SET 8 • 6.RP.A.3.b

BACKWARD 3

Hours and minutes • Grade 4 • Measurement & Data • 4.MD.A.1

How many minutes are in 3 hours?

Minutes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B3-Q1

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 8 • BACKWARD 3 • Q1/5

286

QUESTION ID

S8-B3-Q1

TYPE

Short answer

CORRECT ANSWER

180

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B3-Q1

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 8 • BACKWARD 3 • Q1/5

True or false?

There are 60 minutes in 1 hour.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B3-Q2

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 8 • BACKWARD 3 • Q2/5

287

QUESTION ID

S8-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B3-Q2

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 8 • BACKWARD 3 • Q2/5

ANSWER KEY

Reduced copy of the question page, with its answer

Complete.

5 hours = ____ minutes

Minutes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B3-Q3

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 8 • BACKWARD 3 • Q3/5

288

QUESTION ID

S8-B3-Q3

TYPE

Short answer

CORRECT ANSWER

300

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B3-Q3

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 8 • BACKWARD 3 • Q3/5

288

Which is longer?

- A.** 1 hour
- B.** 90 minutes
- C.** They are the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B3-Q4

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 8 • BACKWARD 3 • Q4/5

289

QUESTION ID

S8-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 90 minutes

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B3-Q4

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 8 • BACKWARD 3 • Q4/5

How many minutes are in $\frac{1}{2}$ hour?

Minutes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B3-Q5

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 8 • BACKWARD 3 • Q5/5

290

QUESTION ID

S8-B3-Q5

TYPE

Short answer

CORRECT ANSWER

30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B3-Q5

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 8 • BACKWARD 3 • Q5/5

SET 8 • 6.RP.A.3.b

FORWARD 1

Speed as a unit rate with fractions • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1

A hiker walks $\frac{3}{4}$ mile in $\frac{3}{8}$ hour.

What is the hiker's speed in miles per hour?

Speed: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 8 • FORWARD 1 • Q1/5

292

QUESTION ID

S8-F1-Q1

TYPE

Short answer

CORRECT ANSWER

2 miles per hour

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 8 • FORWARD 1 • Q1/5

A turtle crawls $\frac{1}{5}$ mile in $\frac{1}{2}$ hour.

What is its speed in miles per hour?

Speed: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 8 • FORWARD 1 • Q2/5

293

QUESTION ID

S8-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$\frac{2}{5}$ mile per hour

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 8 • FORWARD 1 • Q2/5

A boat travels $2\frac{1}{2}$ miles in $\frac{1}{4}$ hour.

What is its speed in miles per hour?

A. $\frac{5}{8}$

B. $2\frac{3}{4}$

C. 5

D. 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 8 • FORWARD 1 • Q3/5

294

QUESTION ID

S8-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 8 • FORWARD 1 • Q3/5

A skater glides $\frac{2}{3}$ kilometer in $\frac{1}{6}$ hour.

What is the skater's speed in kilometers per hour?

Speed: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 8 • FORWARD 1 • Q4/5

295

QUESTION ID

S8-F1-Q4

TYPE

Short answer

CORRECT ANSWER

**4 kilometers per
hour**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 8 • FORWARD 1 • Q4/5

True or false?

Going $\frac{1}{2}$ mile in $\frac{1}{10}$ hour is a speed of 5 miles per hour.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 8 • FORWARD 1 • Q5/5

296

QUESTION ID

S8-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 8 • FORWARD 1 • Q5/5

SET 8 • 6.RP.A.3.b

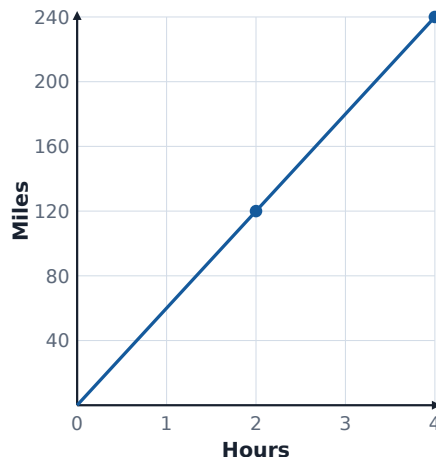
FORWARD 2

Compare speeds from graphs and equations • Grade 8 • Expressions & Equations • 8.EE.B.5

Car A's distance is shown in the graph. Car B travels according to $d = 55t$.

Which car is faster?

- A. Car A
- B. Car B
- C. They are the same.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 8 • FORWARD 2 • Q1/5

ID S8-F2-Q1

298

QUESTION ID

S8-F2-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. Car A

GRADING NOTE Car A = 60 mph

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 8 • FORWARD 2 • Q1/5

ID S8-F2-Q1

Train A travels according to $d = 80t$. Train B's graph is a line through $(0, 0)$ and $(3, 210)$.

Which train is faster, and by how many miles per hour?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 8 • FORWARD 2 • Q2/5

299

QUESTION ID

S8-F2-Q2

TYPE

Short answer

CORRECT ANSWER

**Train A is faster, by
10 miles per hour**

GRADING NOTE Train B = $210/3 = 70$ mph

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 8 • FORWARD 2 • Q2/5

Walker A: $d = 3.5t$. Walker B: the graph passes through (0, 0) and (2, 8).

Who is faster?

- A. Walker A
- B. Walker B
- C. They are the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 8 • FORWARD 2 • Q3/5

300

QUESTION ID

S8-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. Walker B

GRADING NOTE Walker B = 4 per hour

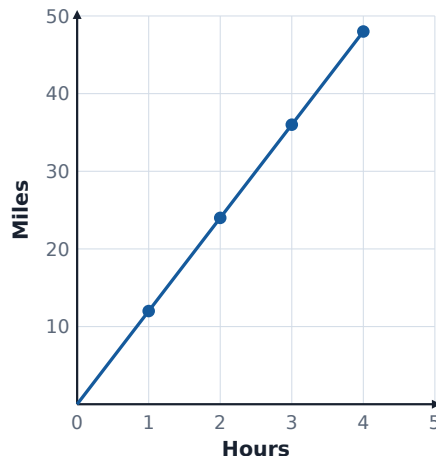
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 8 • FORWARD 2 • Q3/5

What is the speed shown by the graph in miles per hour?

Speed: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 8 • FORWARD 2 • Q4/5

301

QUESTION ID

S8-F2-Q4

TYPE

Short answer

CORRECT ANSWER

12 miles per hour

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 8 • FORWARD 2 • Q4/5

True or false?

A line through (0, 0) and (5, 300) shows a greater speed than $d = 65t$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 8 • FORWARD 2 • Q5/5

302

QUESTION ID

S8-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 8 • FORWARD 2 • Q5/5

Compare unit rates to find the better buy or the faster rate

MAIN	Grade 6 main question	6.RP.A.3.b
BACKWARD 1	Dividing money amounts	5.NBT.B.7
BACKWARD 2	Comparing decimals	5.NBT.A.3.b
BACKWARD 3	Finding the amount for one	3.OA.A.2
FORWARD 1	Compare deals using percents	7.RP.A.3
FORWARD 2	Compare two functions represented in different ways	8.F.A.2

SET 9 • 6.RP.A.3.b

MAIN

Grade 6 main question • Grade 6 • Ratios & Proportional Relationships •
6.RP.A.3.b

Which is the better buy?

A: 6 bottles of water for \$4.50

B: 10 bottles of water for \$7.00

A. A

B. B

C. Both are the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 9 • MAIN • Q1/5

305

QUESTION ID

S9-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. B

GRADING NOTE A: \$0.75 per bottle; B: \$0.70 per bottle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 9 • MAIN • Q1/5

Which is the better buy?

A: a 12-ounce box of cereal for \$3.60

B: an 18-ounce box of cereal for \$4.95

A. A

B. B

C. Both are the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 9 • MAIN • Q2/5

306

QUESTION ID

S9-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. B

GRADING NOTE A: \$0.30 per oz; B: \$0.275 per oz

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 9 • MAIN • Q2/5

Which store has the lower price per pound?

A: 4 pounds of grapes for \$7.00

B: 3 pounds of grapes for \$5.40

A. A

B. B

C. Both are the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 9 • MAIN • Q3/5

307

QUESTION ID

S9-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. A

GRADING NOTE A: \$1.75 per lb; B: \$1.80 per lb

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 9 • MAIN • Q3/5

Which is the better buy?

A: 8 batteries for \$10.00

B: 12 batteries for \$14.40

A. A

B. B

C. Both are the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 9 • MAIN • Q4/5

308

QUESTION ID

S9-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. B

GRADING NOTE A: \$1.25 each; B: \$1.20 each

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 9 • MAIN • Q4/5

Which printer is faster?

A: 90 pages in 6 minutes

B: 140 pages in 10 minutes

A. A

B. B

C. Both are the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 9 • MAIN • Q5/5

309

QUESTION ID

S9-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. A

GRADING NOTE A: 15 pages/min; B: 14 pages/min

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 9 • MAIN • Q5/5

SET 9 • 6.RP.A.3.b

BACKWARD 1

Dividing money amounts • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.7

Find the quotient.

$$\$4.50 \div 6$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 9 • BACKWARD 1 • Q1/5

311

QUESTION ID

S9-B1-Q1

TYPE

Short answer

CORRECT ANSWER

\$0.75

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 9 • BACKWARD 1 • Q1/5

ANSWER KEY

Reduced copy of the question page, with its answer

Find the quotient.

$$\$3.60 \div 12$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 9 • BACKWARD 1 • Q2/5

312

QUESTION ID

S9-B1-Q2

TYPE

Short answer

CORRECT ANSWER

\$0.30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 9 • BACKWARD 1 • Q2/5

312

$$\$7.00 \div 10 = ?$$

- A.** \$0.07
- B.** \$0.70
- C.** \$7.00
- D.** \$70.00

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 9 • BACKWARD 1 • Q3/5

313

QUESTION ID

S9-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. \$0.70

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 9 • BACKWARD 1 • Q3/5

ANSWER KEY

Reduced copy of the question page, with its answer

Find the quotient.

$$\$14.40 \div 12$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 9 • BACKWARD 1 • Q4/5

314

QUESTION ID

S9-B1-Q4

TYPE

Short answer

CORRECT ANSWER

\$1.20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 9 • BACKWARD 1 • Q4/5

314

True or false?

$$\$5.40 \div 3 = \$1.80$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 9 • BACKWARD 1 • Q5/5

315

QUESTION ID

S9-B1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 9 • BACKWARD 1 • Q5/5

SET 9 • 6.RP.A.3.b

BACKWARD 2

Comparing decimals • Grade 5 • Number & Operations in Base Ten •
5.NBT.A.3.b

Which amount is less?

A. \$0.70

B. \$0.75

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 9 • BACKWARD 2 • Q1/5

317

QUESTION ID

S9-B2-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. \$0.70

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 9 • BACKWARD 2 • Q1/5

True or false?

$0.8 > 0.75$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 9 • BACKWARD 2 • Q2/5

318

QUESTION ID

S9-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 9 • BACKWARD 2 • Q2/5

ANSWER KEY

Reduced copy of the question page, with its answer

Write $>$, $<$, or $=$ to compare.

0.275 ____ 0.3

Symbol: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 9 • BACKWARD 2 • Q3/5

319

QUESTION ID

S9-B2-Q3

TYPE

Short answer

CORRECT ANSWER

 $<$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 9 • BACKWARD 2 • Q3/5

319

Which number is greatest?

- A.** 1.075
- B.** 1.705
- C.** 1.75
- D.** 1.8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 9 • BACKWARD 2 • Q4/5

320

QUESTION ID

S9-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 1.8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 9 • BACKWARD 2 • Q4/5

True or false?

$$1.25 < 1.2$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 9 • BACKWARD 2 • Q5/5

321

QUESTION ID

S9-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 9 • BACKWARD 2 • Q5/5

SET 9 • 6.RP.A.3.b

BACKWARD 3

Finding the amount for one • Grade 3 • Operations & Algebraic Thinking •
3.OA.A.2

3 bags hold 21 oranges in all, the same number in each.

How many oranges are in 1 bag?

Oranges: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B3-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 9 • BACKWARD 3 • Q1/5

323

QUESTION ID

S9-B3-Q1

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B3-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 9 • BACKWARD 3 • Q1/5

A worker packs 32 boxes in 4 hours, the same number each hour.

How many boxes per hour?

Boxes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B3-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 9 • BACKWARD 3 • Q2/5

324

QUESTION ID

S9-B3-Q2

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B3-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 9 • BACKWARD 3 • Q2/5

True or false?

9 muffins cost \$18 in all, the same price each, so 1 muffin costs \$2.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B3-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 9 • BACKWARD 3 • Q3/5

325

QUESTION ID

S9-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B3-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 9 • BACKWARD 3 • Q3/5

40 pages are read in 8 days, the same number each day.

How many pages per day?

A. 5

B. 32

C. 48

D. 320

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B3-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 9 • BACKWARD 3 • Q4/5

326

QUESTION ID

S9-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B3-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 9 • BACKWARD 3 • Q4/5

6 packs hold 54 cards in all, the same number in each.

How many cards are in 1 pack?

Cards: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B3-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 9 • BACKWARD 3 • Q5/5

327

QUESTION ID

S9-B3-Q5

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B3-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 9 • BACKWARD 3 • Q5/5

SET 9 • 6.RP.A.3.b

FORWARD 1

Compare deals using percents • Grade 7 • Ratios & Proportional Relationships
• 7.RP.A.3

A \$60 jacket is on sale. Which deal saves more money?

- A. \$10 off
- B. 20% off
- C. They save the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 9 • FORWARD 1 • Q1/5

329

QUESTION ID

S9-F1-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. 20% off

GRADING NOTE 20% of \$60 = \$12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 9 • FORWARD 1 • Q1/5

Which costs less after the discount?

A: a \$45 game at 30% off

B: a \$40 game at 20% off

A. A

B. B

C. They cost the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 9 • FORWARD 1 • Q2/5

330

QUESTION ID

S9-F1-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. A

GRADING NOTE A: \$31.50; B: \$32.00

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 9 • FORWARD 1 • Q2/5

Store A sells a \$80 bike at 15% off. Store B sells the same bike for \$70.

Which store has the lower price, and by how much?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 9 • FORWARD 1 • Q3/5

331

QUESTION ID

S9-F1-Q3

TYPE

Short answer

CORRECT ANSWER

Store A, by \$2

GRADING NOTE Store A price = \$68

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 9 • FORWARD 1 • Q3/5

Which is the better deal on a \$120 phone?

- A.** 25% off
- B.** \$25 off
- C.** They are the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 9 • FORWARD 1 • Q4/5

332

QUESTION ID

S9-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 25% off

GRADING NOTE 25% of \$120 = \$30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 9 • FORWARD 1 • Q4/5

True or false?

A \$50 item at 10% off costs the same as a \$55 item at 20% off.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 9 • FORWARD 1 • Q5/5

333

QUESTION ID

S9-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 9 • FORWARD 1 • Q5/5

SET 9 • 6.RP.A.3.b

FORWARD 2

Compare two functions represented in different ways • Grade 8 • Functions •
8.F.A.2

Plan A costs $y = 12x$ dollars for x months. Plan B is shown in the table.

Which plan has the greater rate of change?

Months (x)	2	4	6
Cost (y)	26	52	78

- A.** Plan A **B.** Plan B **C.** They are the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q1

Grade 8 • Functions • 8.F.A.2 • Set 9 • FORWARD 2 • Q1/5

335

GRADING NOTE Plan B = \$13 per month

QUESTION ID

S9-F2-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. Plan B

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

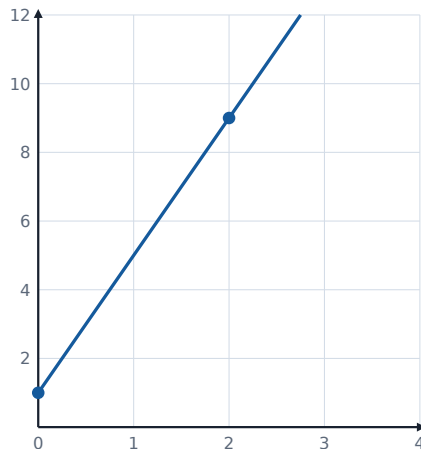
ID S9-F2-Q1

Grade 8 • Functions • 8.F.A.2 • Set 9 • FORWARD 2 • Q1/5

Function A is $y = 3x + 4$. Function B is shown in the graph.

Which function has the greater rate of change?

- A. Function A
- B. Function B
- C. They are the same.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q2

Grade 8 • Functions • 8.F.A.2 • Set 9 • FORWARD 2 • Q2/5

336

QUESTION ID

S9-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. Function B

GRADING NOTE Function B slope = 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q2

Grade 8 • Functions • 8.F.A.2 • Set 9 • FORWARD 2 • Q2/5

Function A is shown in the table. Function B is $y = -3x + 2$.

Which function has the greater y-intercept?

x	0	1	2
y	5	3	1

- A.** Function A **B.** Function B **C.** They are the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q3

Grade 8 • Functions • 8.F.A.2 • Set 9 • FORWARD 2 • Q3/5

337

GRADING NOTE Function A y-intercept = 5

QUESTION ID

S9-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. Function A

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q3

Grade 8 • Functions • 8.F.A.2 • Set 9 • FORWARD 2 • Q3/5

Phone plan A costs \$30 plus \$0.05 per text. Phone plan B is a line through (0, 20) and (100, 35).

Which plan has the greater cost per text?

Plan: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q4

Grade 8 • Functions • 8.F.A.2 • Set 9 • FORWARD 2 • Q4/5

338

QUESTION ID

S9-F2-Q4

TYPE

Short answer

CORRECT ANSWER

Plan B

GRADING NOTE Plan B = \$0.15 per text; Plan A = \$0.05 per text

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q4

Grade 8 • Functions • 8.F.A.2 • Set 9 • FORWARD 2 • Q4/5

True or false?

$y = 4x + 1$ has a greater rate of change than the function in the table.

x	0	1	2
y	0	5	10

A. True**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q5

Grade 8 • Functions • 8.F.A.2 • Set 9 • FORWARD 2 • Q5/5

339

QUESTION ID

S9-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q5

Grade 8 • Functions • 8.F.A.2 • Set 9 • FORWARD 2 • Q5/5

Find a percent of a quantity as a rate per 100

MAIN	Grade 6 main question	6.RP.A.3.c
BACKWARD 1	Hundredths as decimals	4.NF.C.6
BACKWARD 2	Tenths as hundredths	4.NF.C.5
BACKWARD 3	Fraction of a whole number	5.NF.B.4.a
BACKWARD 4	Multiplying a decimal by a whole number	5.NBT.B.7
FORWARD 1	Multi-step percent problems	7.RP.A.3
FORWARD 2	Build a linear function that uses a percent rate	8.F.B.4

SET 10 • 6.RP.A.3.c

MAIN

Grade 6 main question • Grade 6 • Ratios & Proportional Relationships •
6.RP.A.3.c

What is 35% of 60?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 10 • MAIN • Q1/5

342

QUESTION ID

S10-M-Q1

TYPE

Short answer

CORRECT ANSWER

21

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 10 • MAIN • Q1/5

A jacket costs \$60. The sales tax is 8%.

How much is the tax?

Tax: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 10 • MAIN • Q2/5

343

QUESTION ID

S10-M-Q2

TYPE

Short answer

CORRECT ANSWER

\$4.80

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 10 • MAIN • Q2/5

A class has 25 students. 40% of the students walk to school.

How many students walk to school?

Students: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 10 • MAIN • Q3/5

344

QUESTION ID

S10-M-Q3

TYPE

Short answer

CORRECT ANSWER

10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 10 • MAIN • Q3/5

A \$80 pair of shoes is 15% off.

How many dollars are taken off the price?

- A. \$1.20
- B. \$12
- C. \$15
- D. \$68

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 10 • MAIN • Q4/5

345

QUESTION ID

S10-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. \$12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 10 • MAIN • Q4/5

A school has 450 students. 12% of them play in the band.

How many students play in the band?

Students: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 10 • MAIN • Q5/5

346

QUESTION ID

S10-M-Q5

TYPE

Short answer

CORRECT ANSWER

54

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 10 • MAIN • Q5/5

SET 10 • 6.RP.A.3.c

BACKWARD 1

Hundredths as decimals • Grade 4 • Number & Operations–Fractions • 4.NF.C.6

Write $\frac{45}{100}$ as a decimal.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 10 • BACKWARD 1 • Q1/5

348

QUESTION ID

S10-B1-Q1

TYPE

Short answer

CORRECT ANSWER

0.45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 10 • BACKWARD 1 • Q1/5

True or false?

$$0.07 = \frac{7}{100}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 10 • BACKWARD 1 • Q2/5

349

QUESTION ID

S10-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 10 • BACKWARD 1 • Q2/5

Which decimal is equal to $\frac{30}{100}$?

- A. 0.03
- B. 0.30
- C. 3.0
- D. 30.0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 10 • BACKWARD 1 • Q3/5

350

QUESTION ID

S10-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 0.30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 10 • BACKWARD 1 • Q3/5

Write 0.58 as a fraction with a denominator of 100.

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 10 • BACKWARD 1 • Q4/5

351

QUESTION ID

S10-B1-Q4

TYPE

Short answer

CORRECT ANSWER

58/100

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 10 • BACKWARD 1 • Q4/5

True or false?

$$\frac{8}{100} = 0.8$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 10 • BACKWARD 1 • Q5/5

352

QUESTION ID

S10-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 10 • BACKWARD 1 • Q5/5

SET 10 • 6.RP.A.3.c

BACKWARD 2

Tenths as hundredths • Grade 4 • Number & Operations–Fractions • 4.NF.C.5

Find the missing number.

$$\frac{3}{10} = \frac{?}{100}$$

Missing number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 10 • BACKWARD 2 • Q1/5

354

QUESTION ID

S10-B2-Q1

TYPE

Short answer

CORRECT ANSWER

30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 10 • BACKWARD 2 • Q1/5

True or false?

$$\frac{6}{10} + \frac{3}{100} = \frac{63}{100}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 10 • BACKWARD 2 • Q2/5

355

QUESTION ID

S10-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 10 • BACKWARD 2 • Q2/5

Which fraction is equal to $\frac{7}{10}$?

A. $\frac{7}{100}$

B. $\frac{10}{70}$

C. $\frac{17}{100}$

D. $\frac{70}{100}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 10 • BACKWARD 2 • Q3/5

356

QUESTION ID

S10-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 70/100

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 10 • BACKWARD 2 • Q3/5

ANSWER KEY

Reduced copy of the question page, with its answer

Add.

$$\frac{2}{10} + \frac{45}{100}$$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 10 • BACKWARD 2 • Q4/5

357

QUESTION ID

S10-B2-Q4

TYPE

Short answer

CORRECT ANSWER

65/100**GRADING NOTE** Equivalent 13/20 or 0.65 is also correct

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 10 • BACKWARD 2 • Q4/5

True or false?

$$\frac{9}{10} = \frac{90}{100}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 10 • BACKWARD 2 • Q5/5

358

QUESTION ID

S10-B2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 10 • BACKWARD 2 • Q5/5

SET 10 • 6.RP.A.3.c

BACKWARD 3

Fraction of a whole number • Grade 5 • Number & Operations–Fractions •
5.NF.B.4.a

Multiply.

$$\frac{35}{100} \times 60$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B3-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 10 • BACKWARD 3 • Q1/5

360

QUESTION ID

S10-B3-Q1

TYPE

Short answer

CORRECT ANSWER

21

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B3-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 10 • BACKWARD 3 • Q1/5

ANSWER KEY

Reduced copy of the question page, with its answer

Find $\frac{1}{4}$ of 60.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B3-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 10 • BACKWARD 3 • Q2/5

361

QUESTION ID

S10-B3-Q2

TYPE

Short answer

CORRECT ANSWER

15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B3-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 10 • BACKWARD 3 • Q2/5

True or false?

$$\frac{2}{5} \times 25 = 10$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B3-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 10 • BACKWARD 3 • Q3/5

362

QUESTION ID

S10-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B3-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 10 • BACKWARD 3 • Q3/5

$$\frac{3}{4} \times 20 = ?$$

- A.** 5
- B.** 12
- C.** 15
- D.** 16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B3-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 10 • BACKWARD 3 • Q4/5

363

QUESTION ID

S10-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B3-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 10 • BACKWARD 3 • Q4/5

Find $\frac{7}{10}$ of 50.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B3-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 10 • BACKWARD 3 • Q5/5

364

QUESTION ID

S10-B3-Q5

TYPE

Short answer

CORRECT ANSWER

35

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B3-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 10 • BACKWARD 3 • Q5/5

SET 10 • 6.RP.A.3.c

BACKWARD 4

Multiplying a decimal by a whole number • Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$0.35 \times 60$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B4-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 10 • BACKWARD 4 • Q1/5

366

QUESTION ID

S10-B4-Q1

TYPE

Short answer

CORRECT ANSWER

21

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B4-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 10 • BACKWARD 4 • Q1/5

366

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$0.08 \times 60$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B4-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 10 • BACKWARD 4 • Q2/5

367

QUESTION ID

S10-B4-Q2

TYPE

Short answer

CORRECT ANSWER

4.8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B4-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 10 • BACKWARD 4 • Q2/5

367

True or false?

$$0.12 \times 450 = 54$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B4-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 10 • BACKWARD 4 • Q3/5

368

QUESTION ID

S10-B4-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B4-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 10 • BACKWARD 4 • Q3/5

$$0.4 \times 25 = ?$$

- A.** 0.1
- B.** 1
- C.** 10
- D.** 100

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B4-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 10 • BACKWARD 4 • Q4/5

369

QUESTION ID

S10-B4-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B4-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 10 • BACKWARD 4 • Q4/5

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$0.15 \times 80$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B4-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 10 • BACKWARD 4 • Q5/5

370

QUESTION ID

S10-B4-Q5

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B4-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 10 • BACKWARD 4 • Q5/5

370

SET 10 • 6.RP.A.3.c

FORWARD 1

Multi-step percent problems • Grade 7 • Ratios & Proportional Relationships •
7.RP.A.3

A meal costs \$40. You leave a 15% tip.

What is the total cost of the meal with the tip?

Total: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 10 • FORWARD 1 • Q1/5

372

QUESTION ID

S10-F1-Q1

TYPE

Short answer

CORRECT ANSWER

\$46

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 10 • FORWARD 1 • Q1/5

A pair of shoes costs \$80. They are on sale for 25% off.

What is the sale price?

Sale price: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 10 • FORWARD 1 • Q2/5

373

QUESTION ID

S10-F1-Q2

TYPE

Short answer

CORRECT ANSWER

\$60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 10 • FORWARD 1 • Q2/5

A town's population grew from 2,000 to 2,300.

What is the percent increase?

Percent increase: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 10 • FORWARD 1 • Q3/5

374

QUESTION ID

S10-F1-Q3

TYPE

Short answer

CORRECT ANSWER

15%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 10 • FORWARD 1 • Q3/5

A \$50 game is marked up 20%. Then a 5% sales tax is added to the new price.

What is the final cost?

- A.** \$60.00 **B.** \$62.00
C. \$62.50 **D.** \$63.00

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 10 • FORWARD 1 • Q4/5

375

QUESTION ID

S10-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. \$63.00

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 10 • FORWARD 1 • Q4/5

A salesperson earns a 6% commission. She sells \$3,500 in products.

How much commission does she earn?

Commission: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 10 • FORWARD 1 • Q5/5

376

QUESTION ID

S10-F1-Q5

TYPE

Short answer

CORRECT ANSWER

\$210

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 10 • FORWARD 1 • Q5/5

SET 10 • 6.RP.A.3.c

FORWARD 2

Build a linear function that uses a percent rate • Grade 8 • Functions • 8.F.B.4

A salesperson earns \$300 per week plus 5% of her sales S .

Write a function for her weekly earnings E .

$E =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q1

Grade 8 • Functions • 8.F.B.4 • Set 10 • FORWARD 2 • Q1/5

378

QUESTION ID

S10-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$E = 0.05s + 300$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q1

Grade 8 • Functions • 8.F.B.4 • Set 10 • FORWARD 2 • Q1/5

A server earns \$40 per shift plus 18% of the food sales f.

Which function gives the earnings E?

A. $E = 0.18f - 40$

B. $E = 0.18f + 40$

C. $E = 18f + 40$

D. $E = 40f + 0.18$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q2

Grade 8 • Functions • 8.F.B.4 • Set 10 • FORWARD 2 • Q2/5

379

QUESTION ID

S10-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. $E = 0.18f + 40$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q2

Grade 8 • Functions • 8.F.B.4 • Set 10 • FORWARD 2 • Q2/5

An agent earns \$500 per month plus 3% of her sales r .
What are the rate of change and the initial value of this function?

Rate of change: _____

Initial value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q3

Grade 8 • Functions • 8.F.B.4 • Set 10 • FORWARD 2 • Q3/5

380

QUESTION ID

S10-F2-Q3

TYPE

Short answer

CORRECT ANSWER

**Rate of change: 0.03
(3% of sales); Initial
value: \$500**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q3

Grade 8 • Functions • 8.F.B.4 • Set 10 • FORWARD 2 • Q3/5

A delivery driver earns \$60 per day plus 10% of the orders d delivered in dollars.

Write a function for the daily pay P.

P = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q4

Grade 8 • Functions • 8.F.B.4 • Set 10 • FORWARD 2 • Q4/5

381

QUESTION ID

S10-F2-Q4

TYPE

Short answer

CORRECT ANSWER

$P = 0.1d + 60$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q4

Grade 8 • Functions • 8.F.B.4 • Set 10 • FORWARD 2 • Q4/5

True or false?

For $E = 0.04s + 250$, earnings increase by \$0.04 for each \$1 of sales s .

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q5

Grade 8 • Functions • 8.F.B.4 • Set 10 • FORWARD 2 • Q5/5

382

QUESTION ID

S10-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q5

Grade 8 • Functions • 8.F.B.4 • Set 10 • FORWARD 2 • Q5/5

Find the whole, given a part and the percent

MAIN	Grade 6 main question	6.RP.A.3.c
BACKWARD 1	Benchmark percents as fractions	4.NF.A.1
BACKWARD 2	Finding an unknown factor	3.OA.A.4
BACKWARD 3	Dividing a whole number by a unit fraction	5.NF.B.7.b
FORWARD 1	Percent problems: find the original amount	7.RP.A.3
FORWARD 2	Linear equations with decimal coefficients	8.EE.C.7.b

SET 11 • 6.RP.A.3.c

MAIN

Grade 6 main question • Grade 6 • Ratios & Proportional Relationships •
6.RP.A.3.c

12 is 25% of what number?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 11 • MAIN • Q1/5

385

QUESTION ID

S11-M-Q1

TYPE

Short answer

CORRECT ANSWER

48

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 11 • MAIN • Q1/5

A student answered 18 questions correctly. This was 90% of the questions on the test.

How many questions were on the test?

Questions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 11 • MAIN • Q2/5

386

QUESTION ID

S11-M-Q2

TYPE

Short answer

CORRECT ANSWER

20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 11 • MAIN • Q2/5

Maria has saved \$45. This is 30% of the cost of a bike.
How much does the bike cost?

Cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 11 • MAIN • Q3/5

387

QUESTION ID

S11-M-Q3

TYPE

Short answer

CORRECT ANSWER

\$150

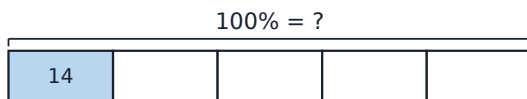
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 11 • MAIN • Q3/5

The tape diagram shows that 20% of a number is 14.

What is the number?



A. 28

B. 34

C. 70

D. 280

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 11 • MAIN • Q4/5

388

QUESTION ID

S11-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 70

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 11 • MAIN • Q4/5

In a survey, 36 students chose pizza. This was 40% of the students surveyed.

How many students were surveyed?

Students: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 11 • MAIN • Q5/5

389

QUESTION ID

S11-M-Q5

TYPE

Short answer

CORRECT ANSWER

90

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 11 • MAIN • Q5/5

SET 11 • 6.RP.A.3.c

BACKWARD 1

Benchmark percents as fractions • Grade 4 • Number & Operations–Fractions •
4.NF.A.1

Find the missing number.

$$\frac{25}{100} = \frac{1}{?}$$

Missing number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 11 • BACKWARD 1 • Q1/5

391

QUESTION ID

S11-B1-Q1

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 11 • BACKWARD 1 • Q1/5

Find the missing number.

$$\frac{50}{100} = \frac{?}{2}$$

Missing number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 11 • BACKWARD 1 • Q2/5

392

QUESTION ID

S11-B1-Q2

TYPE

Short answer

CORRECT ANSWER

1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 11 • BACKWARD 1 • Q2/5

True or false?

$$\frac{20}{100} = \frac{1}{5}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 11 • BACKWARD 1 • Q3/5

393

QUESTION ID

S11-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 11 • BACKWARD 1 • Q3/5

Which fraction is equivalent to $\frac{75}{100}$?

A. $\frac{1}{75}$

B. $\frac{7}{10}$

C. $\frac{3}{4}$

D. $\frac{4}{3}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 11 • BACKWARD 1 • Q4/5

394

QUESTION ID

S11-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 3/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 11 • BACKWARD 1 • Q4/5

Find the missing number.

$$\frac{10}{100} = \frac{1}{?}$$

Missing number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 11 • BACKWARD 1 • Q5/5

395

QUESTION ID

S11-B1-Q5

TYPE

Short answer

CORRECT ANSWER

10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 11 • BACKWARD 1 • Q5/5

SET 11 • 6.RP.A.3.c

BACKWARD 2

Finding an unknown factor • Grade 3 • Operations & Algebraic Thinking •
3.OA.A.4

What number goes in the box?

$$\square \times 4 = 36$$

$$\square = \underline{\hspace{2cm}}$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 11 • BACKWARD 2 • Q1/5

397

QUESTION ID

S11-B2-Q1

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 11 • BACKWARD 2 • Q1/5

What number goes in the box?

$$5 \times \square = 45$$

$$\square = \underline{\hspace{2cm}}$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 11 • BACKWARD 2 • Q2/5

398

QUESTION ID

S11-B2-Q2

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 11 • BACKWARD 2 • Q2/5

True or false?

If $6 \times \square = 42$, then $\square = 7$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 11 • BACKWARD 2 • Q3/5

399

QUESTION ID

S11-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 11 • BACKWARD 2 • Q3/5

$$\square \times 8 = 56$$

What number goes in the box?

- A. 6
- B. 7
- C. 8
- D. 48

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 11 • BACKWARD 2 • Q4/5

400

QUESTION ID

S11-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 11 • BACKWARD 2 • Q4/5

What number goes in the box?

$$9 \times \square = 63$$

$$\square = \underline{\hspace{2cm}}$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 11 • BACKWARD 2 • Q5/5

401

QUESTION ID

S11-B2-Q5

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 11 • BACKWARD 2 • Q5/5

SET 11 • 6.RP.A.3.c

BACKWARD 3

Dividing a whole number by a unit fraction • Grade 5 • Number & Operations–Fractions • 5.NF.B.7.b

Divide.

$$12 \div \frac{1}{4}$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B3-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.b • Set 11 • BACKWARD 3 • Q1/5

403

QUESTION ID

S11-B3-Q1

TYPE

Short answer

CORRECT ANSWER

48

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B3-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.b • Set 11 • BACKWARD 3 • Q1/5

Divide.

$$5 \div \frac{1}{3}$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B3-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.b • Set 11 • BACKWARD 3 • Q2/5

404

QUESTION ID

S11-B3-Q2

TYPE

Short answer

CORRECT ANSWER

15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B3-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.b • Set 11 • BACKWARD 3 • Q2/5

True or false?

$$6 \div \frac{1}{2} = 3$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B3-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.b • Set 11 • BACKWARD 3 • Q3/5

405

QUESTION ID

S11-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B3-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.b • Set 11 • BACKWARD 3 • Q3/5

How many fifths are in 3?

- A. $\frac{3}{5}$
- B. 5
- C. 8
- D. 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B3-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.b • Set 11 • BACKWARD 3 • Q4/5

406

QUESTION ID

S11-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B3-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.b • Set 11 • BACKWARD 3 • Q4/5

Divide.

$$8 \div \frac{1}{5}$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B3-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.b • Set 11 • BACKWARD 3 • Q5/5

407

QUESTION ID

S11-B3-Q5

TYPE

Short answer

CORRECT ANSWER

40

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B3-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.b • Set 11 • BACKWARD 3 • Q5/5

SET 11 • 6.RP.A.3.c

FORWARD 1

Percent problems: find the original amount • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3

After a 20% discount, a jacket costs \$48.

What was the original price?

Original price: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 11 • FORWARD 1 • Q1/5

409

QUESTION ID

S11-F1-Q1

TYPE

Short answer

CORRECT ANSWER

\$60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 11 • FORWARD 1 • Q1/5

A price increased by 10%. The new price is \$55.

What was the original price?

Original price: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 11 • FORWARD 1 • Q2/5

410

QUESTION ID

S11-F1-Q2

TYPE

Short answer

CORRECT ANSWER

\$50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 11 • FORWARD 1 • Q2/5

The sales tax on an item was \$3.60. The tax rate is 6%.
What was the price before tax?

Price: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 11 • FORWARD 1 • Q3/5

411

QUESTION ID

S11-F1-Q3

TYPE

Short answer

CORRECT ANSWER

\$60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 11 • FORWARD 1 • Q3/5

A town's population grew by 25% to 1,500 people.

What was the population before the increase?

- A. 1,125
- B. 1,200
- C. 1,475
- D. 1,875

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 11 • FORWARD 1 • Q4/5

412

QUESTION ID

S11-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 1,200

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 11 • FORWARD 1 • Q4/5

A store marks up the price of a bike by 40%. The selling price is \$210.

How much did the store pay for the bike?

Cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 11 • FORWARD 1 • Q5/5

413

QUESTION ID

S11-F1-Q5

TYPE

Short answer

CORRECT ANSWER

\$150

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 11 • FORWARD 1 • Q5/5

SET 11 • 6.RP.A.3.c

FORWARD 2

Linear equations with decimal coefficients • Grade 8 • Expressions & Equations
• 8.EE.C.7.b

Solve for x.

$$x - 0.2x = 48$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 11 • FORWARD 2 • Q1/5

415

QUESTION ID

S11-F2-Q1

TYPE

Short answer

CORRECT ANSWER

60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 11 • FORWARD 2 • Q1/5

Solve for x.

$$0.4x + 12 = 0.1x + 30$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 11 • FORWARD 2 • Q2/5

416

QUESTION ID

S11-F2-Q2

TYPE

Short answer

CORRECT ANSWER

60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 11 • FORWARD 2 • Q2/5

Solve for x.

$$0.25(x + 40) = 30$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 11 • FORWARD 2 • Q3/5

417

QUESTION ID

S11-F2-Q3

TYPE

Short answer

CORRECT ANSWER

80

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 11 • FORWARD 2 • Q3/5

Which value of n solves the equation?

$$n + 0.15n = 92$$

- A. 78.2
- B. 80
- C. 92.15
- D. 105.8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 11 • FORWARD 2 • Q4/5

418

QUESTION ID

S11-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 80

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 11 • FORWARD 2 • Q4/5

Solve for y.

$$2.5y - 7 = 1.5y + 3$$

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 11 • FORWARD 2 • Q5/5

419

QUESTION ID

S11-F2-Q5

TYPE

Short answer

CORRECT ANSWER

10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 11 • FORWARD 2 • Q5/5

Convert measurement units using ratio reasoning

MAIN	Grade 6 main question	6.RP.A.3.d
BACKWARD 1	Measurement equivalents	4.MD.A.1
BACKWARD 2	Multiplying and dividing by powers of 10	5.NBT.A.2
BACKWARD 3	Multiplying a decimal by a whole number	5.NBT.B.7
FORWARD 1	Rates with fractions and unit conversions	7.RP.A.1
FORWARD 2	Convert units for very large and very small quantities	8.EE.A.4

SET 12 • 6.RP.A.3.d

MAIN

Grade 6 main question • Grade 6 • Ratios & Proportional Relationships •
6.RP.A.3.d

There are 3 feet in 1 yard.

How many feet are in 14 yards?

Feet: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 12 • MAIN • Q1/5

422

QUESTION ID

S12-M-Q1

TYPE

Short answer

CORRECT ANSWER

42

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 12 • MAIN • Q1/5

There are 12 inches in 1 foot.

How many inches are in 4.5 feet?

Inches: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 12 • MAIN • Q2/5

423

QUESTION ID

S12-M-Q2

TYPE

Short answer

CORRECT ANSWER

54

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 12 • MAIN • Q2/5

There are 16 ounces in 1 pound.

How many pounds are in 56 ounces?

Pounds: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 12 • MAIN • Q3/5

424

QUESTION ID

S12-M-Q3

TYPE

Short answer

CORRECT ANSWER

3.5

GRADING NOTE 3 1/2 pounds

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 12 • MAIN • Q3/5

A recipe needs 3 quarts of milk. There are 4 cups in 1 quart.

How many cups of milk are needed?

- A.** $\frac{3}{4}$ cup **B.** $1\frac{1}{3}$ cups
C. 7 cups **D.** 12 cups

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 12 • MAIN • Q4/5

425

QUESTION ID

S12-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 12 cups

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 12 • MAIN • Q4/5

1 inch = 2.54 centimeters.

A book is 10 inches tall. How many centimeters tall is it?

Centimeters: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 12 • MAIN • Q5/5

426

QUESTION ID

S12-M-Q5

TYPE

Short answer

CORRECT ANSWER

25.4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 12 • MAIN • Q5/5

SET 12 • 6.RP.A.3.d

BACKWARD 1

Measurement equivalents • Grade 4 • Measurement & Data • 4.MD.A.1

How many ounces are in 1 pound?

Ounces: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q1

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 12 • BACKWARD 1 • Q1/5

428

QUESTION ID

S12-B1-Q1

TYPE

Short answer

CORRECT ANSWER

16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q1

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 12 • BACKWARD 1 • Q1/5

True or false?

There are 12 inches in 1 yard.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q2

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 12 • BACKWARD 1 • Q2/5

429

QUESTION ID

S12-B1-Q2

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q2

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 12 • BACKWARD 1 • Q2/5

How many cups are in 1 quart?

- A.** 2
- B.** 4
- C.** 8
- D.** 16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q3

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 12 • BACKWARD 1 • Q3/5

430

QUESTION ID

S12-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q3

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 12 • BACKWARD 1 • Q3/5

How many grams are in 1 kilogram?

Grams: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q4

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 12 • BACKWARD 1 • Q4/5

431

QUESTION ID

S12-B1-Q4

TYPE

Short answer

CORRECT ANSWER

1,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q4

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 12 • BACKWARD 1 • Q4/5

True or false?

1 meter = 100 centimeters

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q5

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 12 • BACKWARD 1 • Q5/5

432

QUESTION ID

S12-B1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q5

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 12 • BACKWARD 1 • Q5/5

SET 12 • 6.RP.A.3.d

BACKWARD 2

Multiplying and dividing by powers of 10 • Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$3.4 \times 1,000$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 12 • BACKWARD 2 • Q1/5

434

QUESTION ID

S12-B2-Q1

TYPE

Short answer

CORRECT ANSWER

3,400

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 12 • BACKWARD 2 • Q1/5

434

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$2.54 \times 10$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 12 • BACKWARD 2 • Q2/5

435

QUESTION ID

S12-B2-Q2

TYPE

Short answer

CORRECT ANSWER

25.4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 12 • BACKWARD 2 • Q2/5

435

True or false?

$$0.6 \times 1,000 = 60$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 12 • BACKWARD 2 • Q3/5

436

QUESTION ID

S12-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 12 • BACKWARD 2 • Q3/5

$$450 \div 100 = ?$$

- A.** 0.45
- B.** 4.5
- C.** 45
- D.** 45,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 12 • BACKWARD 2 • Q4/5

437

QUESTION ID

S12-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 4.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 12 • BACKWARD 2 • Q4/5

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$7.25 \times 100$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 12 • BACKWARD 2 • Q5/5

438

QUESTION ID

S12-B2-Q5

TYPE

Short answer

CORRECT ANSWER

725

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 12 • BACKWARD 2 • Q5/5

438

SET 12 • 6.RP.A.3.d

BACKWARD 3

Multiplying a decimal by a whole number • Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$4.5 \times 12$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B3-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 12 • BACKWARD 3 • Q1/5

440

QUESTION ID

S12-B3-Q1

TYPE

Short answer

CORRECT ANSWER

54

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B3-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 12 • BACKWARD 3 • Q1/5

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$3.5 \times 16$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B3-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 12 • BACKWARD 3 • Q2/5

441

QUESTION ID

S12-B3-Q2

TYPE

Short answer

CORRECT ANSWER

56

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B3-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 12 • BACKWARD 3 • Q2/5

True or false?

$$1.5 \times 4 = 6$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B3-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 12 • BACKWARD 3 • Q3/5

442

QUESTION ID

S12-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B3-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 12 • BACKWARD 3 • Q3/5

$$2.25 \times 4 = ?$$

- A.** 0.9
- B.** 8.1
- C.** 9
- D.** 90

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B3-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 12 • BACKWARD 3 • Q4/5

443

QUESTION ID

S12-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B3-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 12 • BACKWARD 3 • Q4/5

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$6 \times 0.75$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B3-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 12 • BACKWARD 3 • Q5/5

444

QUESTION ID

S12-B3-Q5

TYPE

Short answer

CORRECT ANSWER

4.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B3-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 12 • BACKWARD 3 • Q5/5

444

SET 12 • 6.RP.A.3.d

FORWARD 1

Rates with fractions and unit conversions • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1

A faucet drips $\frac{1}{4}$ cup of water every $\frac{1}{3}$ minute.

How many cups does it drip per hour?

Cups per hour: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 12 • FORWARD 1 • Q1/5

446

QUESTION ID

S12-F1-Q1

TYPE

Short answer

CORRECT ANSWER

45 cups per hour

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 12 • FORWARD 1 • Q1/5

A snail travels $\frac{3}{8}$ foot every $\frac{1}{2}$ minute.

How many feet does it travel per hour?

Feet per hour: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 12 • FORWARD 1 • Q2/5

447

QUESTION ID

S12-F1-Q2

TYPE

Short answer

CORRECT ANSWER

45 feet per hour

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 12 • FORWARD 1 • Q2/5

A plant grows $\frac{1}{2}$ inch in $\frac{2}{3}$ week.

How many inches does it grow per week?

A. $\frac{1}{6}$

B. $\frac{1}{3}$

C. $\frac{3}{4}$

D. $1\frac{1}{3}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 12 • FORWARD 1 • Q3/5

448

QUESTION ID

S12-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 3/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 12 • FORWARD 1 • Q3/5

A hose fills $\frac{3}{4}$ gallon in $\frac{1}{6}$ minute.

How many gallons does it fill per minute?

Gallons per minute: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 12 • FORWARD 1 • Q4/5

449

QUESTION ID

S12-F1-Q4

TYPE

Short answer

CORRECT ANSWER

**4 1/2 gallons per
minute**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 12 • FORWARD 1 • Q4/5

A worker paints $\frac{2}{5}$ of a room in $\frac{1}{2}$ hour.

How many rooms can the worker paint in an 8-hour day?

Rooms: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 12 • FORWARD 1 • Q5/5

450

QUESTION ID

S12-F1-Q5

TYPE

Short answer

CORRECT ANSWER

6 $\frac{2}{5}$ rooms

GRADING NOTE $(\frac{2}{5} \div \frac{1}{2}) \times 8 = 32/5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 12 • FORWARD 1 • Q5/5

SET 12 • 6.RP.A.3.d

FORWARD 2

Convert units for very large and very small quantities • Grade 8 • Expressions & Equations • 8.EE.A.4

1 km = 10^6 mm.

Write 3.2×10^6 millimeters in kilometers.

Kilometers: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 12 • FORWARD 2 • Q1/5

452

QUESTION ID

S12-F2-Q1

TYPE

Short answer

CORRECT ANSWER

3.2 km

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 12 • FORWARD 2 • Q1/5

Earth is about 1.5×10^8 km from the Sun. $1 \text{ km} = 10^3$ m.

How many meters is this? Write the answer in scientific notation.

Meters: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 12 • FORWARD 2 • Q2/5

453

QUESTION ID

S12-F2-Q2

TYPE

Short answer

CORRECT ANSWER

$1.5 \times 10^{11} \text{ m}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 12 • FORWARD 2 • Q2/5

A sheet of paper is about 1×10^{-4} meter thick.

Which unit is the most appropriate for this measurement?

- A. Miles
- B. Meters
- C. Millimeters
- D. Kilometers

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 12 • FORWARD 2 • Q3/5

454

QUESTION ID

S12-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Millimeters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 12 • FORWARD 2 • Q3/5

A virus is 1.2×10^{-7} meter long. 1 nanometer = 10^{-9} meter.

How many nanometers long is the virus?

Nanometers: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 12 • FORWARD 2 • Q4/5

455

QUESTION ID

S12-F2-Q4

TYPE

Short answer

CORRECT ANSWER

120 nm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 12 • FORWARD 2 • Q4/5

A glacier moves 3×10^{-1} meter per day. $1 \text{ m} = 10^2 \text{ cm}$.

How many centimeters does it move per day?

Centimeters: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 12 • FORWARD 2 • Q5/5

456

QUESTION ID

S12-F2-Q5

TYPE

Short answer

CORRECT ANSWER

30 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 12 • FORWARD 2 • Q5/5

Compute quotients of fractions and mixed numbers

MAIN	Grade 6 main question	6.NS.A.1
BACKWARD 1	Multiplying fractions	5.NF.B.4.a
BACKWARD 2	Whole number $\div$ unit fraction	5.NF.B.7.b
BACKWARD 3	Unit fraction $\div$ whole number	5.NF.B.7.a
BACKWARD 4	Equivalent fractions	4.NF.A.1
BACKWARD 5	Mixed numbers as fractions	4.NF.B.3.b
FORWARD 1	Multiply and divide rational numbers	7.NS.A.2.c
FORWARD 2	Equations with fraction coefficients	8.EE.C.7.b

SET 13 • 6.NS.A.1

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.A.1

Divide.

$$\frac{3}{5} \div \frac{2}{3}$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q1

Grade 6 • The Number System • 6.NS.A.1 • Set 13 • MAIN • Q1/5

459

QUESTION ID

S13-M-Q1

TYPE

Short answer

CORRECT ANSWER

9/10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q1

Grade 6 • The Number System • 6.NS.A.1 • Set 13 • MAIN • Q1/5

Divide.

$$\frac{5}{6} \div \frac{1}{3}$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q2

Grade 6 • The Number System • 6.NS.A.1 • Set 13 • MAIN • Q2/5

460

QUESTION ID

S13-M-Q2

TYPE

Short answer

CORRECT ANSWER

$5/2 = 2 \frac{1}{2}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q2

Grade 6 • The Number System • 6.NS.A.1 • Set 13 • MAIN • Q2/5

Divide.

$$1\frac{1}{2} \div \frac{3}{4}$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q3

Grade 6 • The Number System • 6.NS.A.1 • Set 13 • MAIN • Q3/5

461

QUESTION ID

S13-M-Q3

TYPE

Short answer

CORRECT ANSWER

2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q3

Grade 6 • The Number System • 6.NS.A.1 • Set 13 • MAIN • Q3/5

Which is equal to $\frac{4}{5} \div \frac{2}{3}$?

A. $\frac{2}{15}$

B. $\frac{8}{15}$

C. $\frac{5}{6}$

D. $1\frac{1}{5}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q4

Grade 6 • The Number System • 6.NS.A.1 • Set 13 • MAIN • Q4/5

462

QUESTION ID

S13-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 1 1/5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q4

Grade 6 • The Number System • 6.NS.A.1 • Set 13 • MAIN • Q4/5

Divide.

$$3\frac{1}{3} \div \frac{5}{6}$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q5

Grade 6 • The Number System • 6.NS.A.1 • Set 13 • MAIN • Q5/5

463

QUESTION ID

S13-M-Q5

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q5

Grade 6 • The Number System • 6.NS.A.1 • Set 13 • MAIN • Q5/5

SET 13 • 6.NS.A.1

BACKWARD 1

Multiplying fractions • Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a

Multiply.

$$\frac{3}{5} \times \frac{2}{7}$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 13 • BACKWARD 1 • Q1/5

465

QUESTION ID

S13-B1-Q1

TYPE

Short answer

CORRECT ANSWER

6/35

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 13 • BACKWARD 1 • Q1/5

True or false?

$$\frac{3}{4} \times \frac{2}{5} = \frac{6}{20}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 13 • BACKWARD 1 • Q2/5

466

QUESTION ID

S13-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 13 • BACKWARD 1 • Q2/5

$$\frac{1}{2} \times \frac{5}{6} = ?$$

A. $\frac{1}{3}$

B. $\frac{5}{12}$

C. $\frac{5}{8}$

D. $\frac{6}{8}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 13 • BACKWARD 1 • Q3/5

467

QUESTION ID

S13-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 5/12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 13 • BACKWARD 1 • Q3/5

Multiply.

$$\frac{4}{5} \times \frac{3}{2}$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 13 • BACKWARD 1 • Q4/5

468

QUESTION ID

S13-B1-Q4

TYPE

Short answer

CORRECT ANSWER

$6/5 = 1 \frac{1}{5}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 13 • BACKWARD 1 • Q4/5

Multiply.

$$\frac{5}{6} \times 3$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 13 • BACKWARD 1 • Q5/5

469

QUESTION ID

S13-B1-Q5

TYPE

Short answer

CORRECT ANSWER

$5/2 = 2 \frac{1}{2}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 13 • BACKWARD 1 • Q5/5

SET 13 • 6.NS.A.1

BACKWARD 2

Whole number $\div$ unit fraction • Grade 5 • Number & Operations–Fractions •
5.NF.B.7.b

Divide.

$$3 \div \frac{1}{4}$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.b • Set 13 • BACKWARD 2 • Q1/5

471

QUESTION ID

S13-B2-Q1

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.b • Set 13 • BACKWARD 2 • Q1/5

True or false?

$$4 \div \frac{1}{2} = 8$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.b • Set 13 • BACKWARD 2 • Q2/5

472

QUESTION ID

S13-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.b • Set 13 • BACKWARD 2 • Q2/5

How many thirds are in 2?

A. $\frac{1}{6}$

B. $\frac{2}{3}$

C. 5

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.b • Set 13 • BACKWARD 2 • Q3/5

473

QUESTION ID

S13-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.b • Set 13 • BACKWARD 2 • Q3/5

Divide.

$$6 \div \frac{1}{5}$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.b • Set 13 • BACKWARD 2 • Q4/5

474

QUESTION ID

S13-B2-Q4

TYPE

Short answer

CORRECT ANSWER

30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.b • Set 13 • BACKWARD 2 • Q4/5

Divide.

$$2 \div \frac{1}{8}$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.b • Set 13 • BACKWARD 2 • Q5/5

475

QUESTION ID

S13-B2-Q5

TYPE

Short answer

CORRECT ANSWER

16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.b • Set 13 • BACKWARD 2 • Q5/5

SET 13 • 6.NS.A.1

BACKWARD 3

Unit fraction $\div$ whole number • Grade 5 • Number & Operations–Fractions •
5.NF.B.7.a

Divide.

$$\frac{1}{4} \div 3$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B3-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.a • Set 13 • BACKWARD 3 • Q1/5

477

QUESTION ID

S13-B3-Q1

TYPE

Short answer

CORRECT ANSWER

1/12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B3-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.a • Set 13 • BACKWARD 3 • Q1/5

True or false?

$$\frac{1}{2} \div 3 = \frac{1}{6}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B3-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.a • Set 13 • BACKWARD 3 • Q2/5

478

QUESTION ID

S13-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B3-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.a • Set 13 • BACKWARD 3 • Q2/5

$$\frac{1}{5} \div 2 = ?$$

A. $\frac{1}{10}$

B. $\frac{2}{5}$

C. $\frac{5}{2}$

D. 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B3-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.a • Set 13 • BACKWARD 3 • Q3/5

479

QUESTION ID

S13-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 1/10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B3-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.a • Set 13 • BACKWARD 3 • Q3/5

Divide.

$$\frac{1}{6} \div 3$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B3-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.a • Set 13 • BACKWARD 3 • Q4/5

480

QUESTION ID

S13-B3-Q4

TYPE

Short answer

CORRECT ANSWER

1/18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B3-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.a • Set 13 • BACKWARD 3 • Q4/5

True or false?

$$\frac{1}{6} \div 2 = \frac{2}{6}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B3-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.a • Set 13 • BACKWARD 3 • Q5/5

481

QUESTION ID

S13-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B3-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.a • Set 13 • BACKWARD 3 • Q5/5

SET 13 • 6.NS.A.1

BACKWARD 4

Equivalent fractions • Grade 4 • Number & Operations–Fractions • 4.NF.A.1

True or false?

$\frac{8}{12}$ is equivalent to $\frac{2}{3}$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B4-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 13 • BACKWARD 4 • Q1/5

483

QUESTION ID

S13-B4-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B4-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 13 • BACKWARD 4 • Q1/5

Find the missing number.

$$\frac{10}{15} = \frac{?}{3}$$

Missing number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B4-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 13 • BACKWARD 4 • Q2/5

484

QUESTION ID

S13-B4-Q2

TYPE

Short answer

CORRECT ANSWER

2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B4-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 13 • BACKWARD 4 • Q2/5

Which fraction is equivalent to $\frac{9}{12}$?

A. $\frac{2}{3}$

B. $\frac{3}{4}$

C. $\frac{12}{9}$

D. $\frac{9}{4}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B4-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 13 • BACKWARD 4 • Q3/5

485

QUESTION ID

S13-B4-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 3/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B4-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 13 • BACKWARD 4 • Q3/5

Find the missing number.

$$\frac{6}{8} = \frac{?}{4}$$

Missing number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B4-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 13 • BACKWARD 4 • Q4/5

486

QUESTION ID

S13-B4-Q4

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B4-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 13 • BACKWARD 4 • Q4/5

True or false?

$\frac{4}{6}$ is equivalent to $\frac{6}{8}$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B4-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 13 • BACKWARD 4 • Q5/5

487

QUESTION ID

S13-B4-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B4-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 13 • BACKWARD 4 • Q5/5

SET 13 • 6.NS.A.1

BACKWARD 5

Mixed numbers as fractions • Grade 4 • Number & Operations–Fractions •
4.NF.B.3.b

Write $1\frac{1}{2}$ as a fraction.

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B5-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.b • Set 13 • BACKWARD 5 • Q1/5

489

QUESTION ID

S13-B5-Q1

TYPE

Short answer

CORRECT ANSWER

3/2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B5-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.b • Set 13 • BACKWARD 5 • Q1/5

Write $3\frac{1}{3}$ as a fraction.

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B5-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.b • Set 13 • BACKWARD 5 • Q2/5

490

QUESTION ID

S13-B5-Q2

TYPE

Short answer

CORRECT ANSWER

10/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B5-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.b • Set 13 • BACKWARD 5 • Q2/5

True or false?

$$2\frac{1}{4} = \frac{9}{4}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B5-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.b • Set 13 • BACKWARD 5 • Q3/5

491

QUESTION ID

S13-B5-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B5-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.b • Set 13 • BACKWARD 5 • Q3/5

Which fraction is equal to $1\frac{2}{3}$?

A. $\frac{3}{3}$

B. $\frac{5}{3}$

C. $\frac{7}{3}$

D. $\frac{12}{3}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B5-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.b • Set 13 • BACKWARD 5 • Q4/5

492

QUESTION ID

S13-B5-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 5/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B5-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.b • Set 13 • BACKWARD 5 • Q4/5

Write $\frac{11}{4}$ as a mixed number.

Mixed number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B5-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.b • Set 13 • BACKWARD 5 • Q5/5

493

QUESTION ID

S13-B5-Q5

TYPE

Short answer

CORRECT ANSWER

2 3/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B5-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.b • Set 13 • BACKWARD 5 • Q5/5

SET 13 • 6.NS.A.1

FORWARD 1

Multiply and divide rational numbers • Grade 7 • The Number System •
7.NS.A.2.c

Find the value.

$$\left(-\frac{2}{3}\right) \div \frac{4}{5}$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q1

Grade 7 • The Number System • 7.NS.A.2.c • Set 13 • FORWARD 1 • Q1/5

495

QUESTION ID

S13-F1-Q1

TYPE

Short answer

CORRECT ANSWER

−5/6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q1

Grade 7 • The Number System • 7.NS.A.2.c • Set 13 • FORWARD 1 • Q1/5

Find the value.

$$-\frac{3}{4} \times \left(-\frac{8}{9}\right)$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q2

Grade 7 • The Number System • 7.NS.A.2.c • Set 13 • FORWARD 1 • Q2/5

496

QUESTION ID

S13-F1-Q2

TYPE

Short answer

CORRECT ANSWER

2/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q2

Grade 7 • The Number System • 7.NS.A.2.c • Set 13 • FORWARD 1 • Q2/5

$$(-1\frac{1}{2}) \div (-\frac{3}{8}) = ?$$

A. -4

B. $-\frac{9}{16}$

C. $\frac{9}{16}$

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q3

Grade 7 • The Number System • 7.NS.A.2.c • Set 13 • FORWARD 1 • Q3/5

497

QUESTION ID

S13-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q3

Grade 7 • The Number System • 7.NS.A.2.c • Set 13 • FORWARD 1 • Q3/5

Find the value.

$$\frac{5}{6} \div \left(-\frac{5}{12}\right)$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q4

Grade 7 • The Number System • 7.NS.A.2.c • Set 13 • FORWARD 1 • Q4/5

498

QUESTION ID

S13-F1-Q4

TYPE

Short answer

CORRECT ANSWER

-2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q4

Grade 7 • The Number System • 7.NS.A.2.c • Set 13 • FORWARD 1 • Q4/5

True or false?

$$\left(-\frac{1}{2}\right) \times \frac{4}{5} = -\frac{2}{5}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q5

Grade 7 • The Number System • 7.NS.A.2.c • Set 13 • FORWARD 1 • Q5/5

499

QUESTION ID

S13-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q5

Grade 7 • The Number System • 7.NS.A.2.c • Set 13 • FORWARD 1 • Q5/5

SET 13 • 6.NS.A.1

FORWARD 2

Equations with fraction coefficients • Grade 8 • Expressions & Equations •
8.EE.C.7.b

Solve for x.

$$\frac{3}{4}(x - 2) = \frac{1}{2}x + 5$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 13 • FORWARD 2 • Q1/5

501

QUESTION ID

S13-F2-Q1

TYPE

Short answer

CORRECT ANSWER

26

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 13 • FORWARD 2 • Q1/5

ANSWER KEY

Reduced copy of the question page, with its answer

Solve for x.

$$\frac{2}{3}x + 4 = \frac{1}{6}x + 7$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 13 • FORWARD 2 • Q2/5

502

QUESTION ID

S13-F2-Q2

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 13 • FORWARD 2 • Q2/5

502

Solve for x.

$$\frac{x}{2} + \frac{x}{3} = 10$$

- A. 5
- B. 6
- C. 12
- D. 60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 13 • FORWARD 2 • Q3/5

503

QUESTION ID

S13-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 13 • FORWARD 2 • Q3/5

Solve for x.

$$\frac{1}{2}(4x + 6) = 3x - 5$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 13 • FORWARD 2 • Q4/5

504

QUESTION ID

S13-F2-Q4

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 13 • FORWARD 2 • Q4/5

Solve for y.

$$\frac{5}{8}y - 1 = \frac{3}{8}y + 2$$

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 13 • FORWARD 2 • Q5/5

505

QUESTION ID

S13-F2-Q5

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 13 • FORWARD 2 • Q5/5

Solve word problems involving division of fractions

MAIN	Grade 6 main question	6.NS.A.1
BACKWARD 1	Division as "how many groups?"	3.OA.A.2
BACKWARD 2	Unit fractions build a fraction	3.NF.A.1
BACKWARD 3	Area with fractional side lengths	5.NF.B.4.b
BACKWARD 4	Word problems: dividing with unit fractions	5.NF.B.7.c
FORWARD 1	Real-world problems with rational numbers	7.NS.A.3
FORWARD 2	Solve linear equations from contexts	8.EE.C.7.b

SET 14 • 6.NS.A.1

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.A.1

How many $\frac{2}{3}$ -cup scoops are in $\frac{5}{6}$ cup of flour?

Scoops: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q1

Grade 6 • The Number System • 6.NS.A.1 • Set 14 • MAIN • Q1/5

508

QUESTION ID

S14-M-Q1

TYPE

Short answer

CORRECT ANSWER

$5/4 = 1 \frac{1}{4}$ scoops

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q1

Grade 6 • The Number System • 6.NS.A.1 • Set 14 • MAIN • Q1/5

A path is $\frac{3}{4}$ mile long. It is divided into sections that are each $\frac{1}{8}$ mile long.

How many sections are there?

Sections: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q2

Grade 6 • The Number System • 6.NS.A.1 • Set 14 • MAIN • Q2/5

509

QUESTION ID

S14-M-Q2

TYPE

Short answer

CORRECT ANSWER

6 sections

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q2

Grade 6 • The Number System • 6.NS.A.1 • Set 14 • MAIN • Q2/5

A rectangular patio has an area of $\frac{3}{5}$ square yard. Its length is $\frac{9}{10}$ yard.

What is its width?

Width: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q3

Grade 6 • The Number System • 6.NS.A.1 • Set 14 • MAIN • Q3/5

510

QUESTION ID

S14-M-Q3

TYPE

Short answer

CORRECT ANSWER

$\frac{2}{3}$ yard

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q3

Grade 6 • The Number System • 6.NS.A.1 • Set 14 • MAIN • Q3/5

A pitcher holds $\frac{9}{10}$ liter of juice. Each glass holds $\frac{3}{10}$ liter.

How many glasses can be filled?

A. $\frac{27}{100}$

B. $\frac{1}{3}$

C. 3

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q4

Grade 6 • The Number System • 6.NS.A.1 • Set 14 • MAIN • Q4/5

511

QUESTION ID

S14-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q4

Grade 6 • The Number System • 6.NS.A.1 • Set 14 • MAIN • Q4/5

Ana has $\frac{5}{6}$ pound of trail mix. She fills bags that each hold $\frac{5}{12}$ pound.

How many bags can she fill?

Bags: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q5

Grade 6 • The Number System • 6.NS.A.1 • Set 14 • MAIN • Q5/5

512

QUESTION ID

S14-M-Q5

TYPE

Short answer

CORRECT ANSWER

2 bags

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q5

Grade 6 • The Number System • 6.NS.A.1 • Set 14 • MAIN • Q5/5

SET 14 • 6.NS.A.1

BACKWARD 1

Division as "how many groups?" • Grade 3 • Operations & Algebraic Thinking •
3.OA.A.2

How many groups of 4 are in 20?

Groups: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 14 • BACKWARD 1 • Q1/5

514

QUESTION ID

S14-B1-Q1

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 14 • BACKWARD 1 • Q1/5

True or false?

$12 \div 3$ can mean "How many groups of 3 are in 12?"

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 14 • BACKWARD 1 • Q2/5

515

QUESTION ID

S14-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 14 • BACKWARD 1 • Q2/5

Tom has 18 cookies. He puts 6 cookies in each bag.

Which equation finds the number of bags?

A. $18 \times 6 = \square$

B. $18 - 6 = \square$

C. $18 \div 6 = \square$

D. $6 \div 18 = \square$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 14 • BACKWARD 1 • Q3/5

516

QUESTION ID

S14-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $18 \div 6 = \square$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 14 • BACKWARD 1 • Q3/5

24 students are split into teams of 8.

How many teams are there?

Teams: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 14 • BACKWARD 1 • Q4/5

517

QUESTION ID

S14-B1-Q4

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 14 • BACKWARD 1 • Q4/5

How many 5s are in 35?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 14 • BACKWARD 1 • Q5/5

518

QUESTION ID

S14-B1-Q5

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 14 • BACKWARD 1 • Q5/5

SET 14 • 6.NS.A.1

BACKWARD 2

Unit fractions build a fraction • Grade 3 • Number & Operations–Fractions •
3.NF.A.1

How many $\frac{1}{4}$ s make $\frac{3}{4}$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q1

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 14 • BACKWARD 2 • Q1/5

520

QUESTION ID

S14-B2-Q1

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q1

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 14 • BACKWARD 2 • Q1/5

How many $\frac{1}{8}$ s make 1 whole?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q2

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 14 • BACKWARD 2 • Q2/5

521

QUESTION ID

S14-B2-Q2

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q2

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 14 • BACKWARD 2 • Q2/5

True or false?

$\frac{5}{6}$ is 5 parts of size $\frac{1}{6}$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q3

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 14 • BACKWARD 2 • Q3/5

522

QUESTION ID

S14-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q3

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 14 • BACKWARD 2 • Q3/5

$\frac{3}{8}$ is made of how many parts of size $\frac{1}{8}$?

- A. $\frac{1}{3}$
- B. 3
- C. 5
- D. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q4

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 14 • BACKWARD 2 • Q4/5

523

QUESTION ID

S14-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q4

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 14 • BACKWARD 2 • Q4/5

How many $\frac{1}{3}$ s make $\frac{2}{3}$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q5

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 14 • BACKWARD 2 • Q5/5

524

QUESTION ID

S14-B2-Q5

TYPE

Short answer

CORRECT ANSWER

2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q5

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 14 • BACKWARD 2 • Q5/5

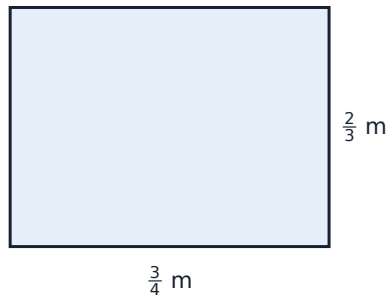
SET 14 • 6.NS.A.1

BACKWARD 3

Area with fractional side lengths • Grade 5 • Number & Operations–Fractions •
5.NF.B.4.b

What is the area of the rectangle?

Area: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B3-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.b • Set 14 • BACKWARD 3 • Q1/5

526

QUESTION ID

S14-B3-Q1

TYPE

Short answer

CORRECT ANSWER

$\frac{1}{2}$ square meter

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B3-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.b • Set 14 • BACKWARD 3 • Q1/5

True or false?

A rectangle that is $\frac{1}{2}$ foot by $\frac{1}{3}$ foot has an area of $\frac{1}{6}$ square foot.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B3-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.b • Set 14 • BACKWARD 3 • Q2/5

527

QUESTION ID

S14-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B3-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.b • Set 14 • BACKWARD 3 • Q2/5

A rectangle is $\frac{2}{5}$ unit by $\frac{1}{2}$ unit.

What is its area?

- A.** $\frac{1}{5}$ square unit **B.** $\frac{2}{7}$ square unit
C. $\frac{3}{7}$ square unit **D.** $\frac{9}{10}$ square unit

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B3-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.b • Set 14 • BACKWARD 3 • Q3/5

528

QUESTION ID

S14-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 1/5 square unit

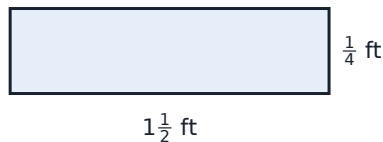
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B3-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.b • Set 14 • BACKWARD 3 • Q3/5

What is the area of the rectangle?

Area: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B3-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.b • Set 14 • BACKWARD 3 • Q4/5

529

QUESTION ID

S14-B3-Q4

TYPE

Short answer

CORRECT ANSWER

$\frac{3}{8}$ square foot

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B3-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.b • Set 14 • BACKWARD 3 • Q4/5

A garden is $\frac{5}{6}$ yard long and $\frac{3}{5}$ yard wide.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B3-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.b • Set 14 • BACKWARD 3 • Q5/5

530

QUESTION ID

S14-B3-Q5

TYPE

Short answer

CORRECT ANSWER

$\frac{1}{2}$ square yard

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B3-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.b • Set 14 • BACKWARD 3 • Q5/5

SET 14 • 6.NS.A.1

BACKWARD 4

Word problems: dividing with unit fractions • Grade 5 • Number & Operations–Fractions • 5.NF.B.7.c

How many $\frac{1}{4}$ -pound burgers can be made from 3 pounds of beef?

Burgers: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B4-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.c • Set 14 • BACKWARD 4 • Q1/5

532

QUESTION ID

S14-B4-Q1

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B4-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.c • Set 14 • BACKWARD 4 • Q1/5

A 4-pound bag of rice is divided into $\frac{1}{2}$ -pound portions.

How many portions are there?

Portions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B4-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.c • Set 14 • BACKWARD 4 • Q2/5

533

QUESTION ID

S14-B4-Q2

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B4-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.c • Set 14 • BACKWARD 4 • Q2/5

Which expression finds how many $\frac{1}{5}$ -mile laps are in 3 miles?

A. $\frac{1}{5} \div 3$

B. $3 - \frac{1}{5}$

C. $3 \div \frac{1}{5}$

D. $3 \times \frac{1}{5}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B4-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.c • Set 14 • BACKWARD 4 • Q3/5

534

QUESTION ID

S14-B4-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $3 \div 1/5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B4-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.c • Set 14 • BACKWARD 4 • Q3/5

4 friends share $\frac{1}{3}$ pound of cheese equally.

How much cheese does each friend get?

Each friend: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B4-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.c • Set 14 • BACKWARD 4 • Q4/5

535

QUESTION ID

S14-B4-Q4

TYPE

Short answer

CORRECT ANSWER

$\frac{1}{12}$ pound

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B4-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.c • Set 14 • BACKWARD 4 • Q4/5

True or false?

There are 10 half-hours in 5 hours.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B4-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.c • Set 14 • BACKWARD 4 • Q5/5

536

QUESTION ID

S14-B4-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B4-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.c • Set 14 • BACKWARD 4 • Q5/5

SET 14 • 6.NS.A.1

FORWARD 1

Real-world problems with rational numbers • Grade 7 • The Number System •
7.NS.A.3

A diver descends $\frac{3}{4}$ foot each second.

How many seconds does it take to descend $5\frac{1}{4}$ feet?

Seconds: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q1

Grade 7 • The Number System • 7.NS.A.3 • Set 14 • FORWARD 1 • Q1/5

538

QUESTION ID

S14-F1-Q1

TYPE

Short answer

CORRECT ANSWER

7 seconds

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q1

Grade 7 • The Number System • 7.NS.A.3 • Set 14 • FORWARD 1 • Q1/5

The temperature was 4°F . It dropped $2\frac{1}{2}^{\circ}\text{F}$ each hour for 3 hours.

What was the final temperature?

Temperature: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q2

Grade 7 • The Number System • 7.NS.A.3 • Set 14 • FORWARD 1 • Q2/5

539

QUESTION ID

S14-F1-Q2

TYPE

Short answer

CORRECT ANSWER

-3.5°F

GRADING NOTE $-3\frac{1}{2}$ degrees

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q2

Grade 7 • The Number System • 7.NS.A.3 • Set 14 • FORWARD 1 • Q2/5

A stock's value changed by $-\$1\frac{1}{4}$ each day for 4 days.

What was the total change?

- A. $-\$5$
- B. $-\$4\frac{1}{4}$
- C. $-\$1$
- D. $\$5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q3

Grade 7 • The Number System • 7.NS.A.3 • Set 14 • FORWARD 1 • Q3/5

540

QUESTION ID

S14-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $-\$5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q3

Grade 7 • The Number System • 7.NS.A.3 • Set 14 • FORWARD 1 • Q3/5

A recipe needs $\frac{2}{3}$ cup of oil per batch. Jo has $3\frac{1}{3}$ cups of oil.

How many batches can Jo make?

Batches: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q4

Grade 7 • The Number System • 7.NS.A.3 • Set 14 • FORWARD 1 • Q4/5

541

QUESTION ID

S14-F1-Q4

TYPE

Short answer

CORRECT ANSWER

5 batches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q4

Grade 7 • The Number System • 7.NS.A.3 • Set 14 • FORWARD 1 • Q4/5

A hiker descends 1,250 feet in $\frac{5}{6}$ hour at a constant rate.

What is the change in elevation per hour?

Change per hour: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q5

Grade 7 • The Number System • 7.NS.A.3 • Set 14 • FORWARD 1 • Q5/5

542

QUESTION ID

S14-F1-Q5

TYPE

Short answer

CORRECT ANSWER

–1,500 feet per hour

GRADING NOTE A descent of 1,500 feet per hour

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q5

Grade 7 • The Number System • 7.NS.A.3 • Set 14 • FORWARD 1 • Q5/5

SET 14 • 6.NS.A.1

FORWARD 2

Solve linear equations from contexts • Grade 8 • Expressions & Equations •
8.EE.C.7.b

Solve for x.

$$\frac{1}{2}(x + 6) = \frac{1}{3}(x + 12)$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 14 • FORWARD 2 • Q1/5

544

QUESTION ID

S14-F2-Q1

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 14 • FORWARD 2 • Q1/5

Lee's age plus $\frac{1}{3}$ of his age is 24.

Write and solve an equation to find Lee's age.

Age: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 14 • FORWARD 2 • Q2/5

545

QUESTION ID

S14-F2-Q2

TYPE

Short answer

CORRECT ANSWER

18 years old

GRADING NOTE Equation: $a + (1/3)a = 24$, or equivalent

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 14 • FORWARD 2 • Q2/5

Tank A has 40 gallons and drains $\frac{3}{4}$ gallon per minute.

Tank B has 10 gallons and fills at $1\frac{1}{4}$ gallons per minute.

After how many minutes will the tanks hold the same amount?

Minutes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 14 • FORWARD 2 • Q3/5

546

QUESTION ID

S14-F2-Q3

TYPE

Short answer

CORRECT ANSWER

15 minutes

GRADING NOTE $40 - (3/4)t = 10 + (5/4)t$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 14 • FORWARD 2 • Q3/5

Solve for x.

$$\frac{2}{5}x - 3 = \frac{1}{5}x + 1$$

A. -10

B. 4

C. 10

D. 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 14 • FORWARD 2 • Q4/5

547

QUESTION ID

S14-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 14 • FORWARD 2 • Q4/5

Solve for n.

$$\frac{3}{4}n + \frac{1}{2} = \frac{1}{4}n + 4$$

n = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 14 • FORWARD 2 • Q5/5

548

QUESTION ID

S14-F2-Q5

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 14 • FORWARD 2 • Q5/5

Divide multi-digit numbers using the standard algorithm

MAIN	Grade 6 main question	6.NS.B.2
BACKWARD 1	Multiplying multi-digit numbers	5.NBT.B.5
BACKWARD 2	Dividing by a one-digit number	4.NBT.B.6
BACKWARD 3	Estimating quotients	5.NBT.B.6
BACKWARD 4	Multi-digit subtraction	4.NBT.B.4
FORWARD 1	Long division: fractions to decimals	7.NS.A.2.d
FORWARD 2	Repeating decimals and rational numbers	8.NS.A.1

SET 15 • 6.NS.B.2

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.B.2

Use the standard algorithm to divide.

$$4,508 \div 23$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q1

Grade 6 • The Number System • 6.NS.B.2 • Set 15 • MAIN • Q1/5

551

QUESTION ID

S15-M-Q1

TYPE

Short answer

CORRECT ANSWER

196

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q1

Grade 6 • The Number System • 6.NS.B.2 • Set 15 • MAIN • Q1/5

Use the standard algorithm to divide.

$$7,236 \div 36$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q2

Grade 6 • The Number System • 6.NS.B.2 • Set 15 • MAIN • Q2/5

552

QUESTION ID

S15-M-Q2

TYPE

Short answer

CORRECT ANSWER

201

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q2

Grade 6 • The Number System • 6.NS.B.2 • Set 15 • MAIN • Q2/5

Use the standard algorithm to divide.

$$9,975 \div 75$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q3

Grade 6 • The Number System • 6.NS.B.2 • Set 15 • MAIN • Q3/5

553

QUESTION ID

S15-M-Q3

TYPE

Short answer

CORRECT ANSWER

133

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q3

Grade 6 • The Number System • 6.NS.B.2 • Set 15 • MAIN • Q3/5

Use the standard algorithm to divide.

$$1,680 \div 48$$

- A.** 30
- B.** 35
- C.** 36
- D.** 350

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q4

Grade 6 • The Number System • 6.NS.B.2 • Set 15 • MAIN • Q4/5

554

QUESTION ID

S15-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 35

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q4

Grade 6 • The Number System • 6.NS.B.2 • Set 15 • MAIN • Q4/5

Use the standard algorithm to divide.

$$25,494 \div 42$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q5

Grade 6 • The Number System • 6.NS.B.2 • Set 15 • MAIN • Q5/5

555

QUESTION ID

S15-M-Q5

TYPE

Short answer

CORRECT ANSWER

607

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q5

Grade 6 • The Number System • 6.NS.B.2 • Set 15 • MAIN • Q5/5

SET 15 • 6.NS.B.2

BACKWARD 1

Multiplying multi-digit numbers • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.5

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$46 \times 23$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 15 • BACKWARD 1 • Q1/5

557

QUESTION ID

S15-B1-Q1

TYPE

Short answer

CORRECT ANSWER

1,058

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 15 • BACKWARD 1 • Q1/5

557

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$196 \times 23$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 15 • BACKWARD 1 • Q2/5

558

QUESTION ID

S15-B1-Q2

TYPE

Short answer

CORRECT ANSWER

4,508

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 15 • BACKWARD 1 • Q2/5

558

True or false?

$$35 \times 48 = 1,680$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 15 • BACKWARD 1 • Q3/5

559

QUESTION ID

S15-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 15 • BACKWARD 1 • Q3/5

$$133 \times 75 = ?$$

- A.** 1,596
- B.** 9,775
- C.** 9,875
- D.** 9,975

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 15 • BACKWARD 1 • Q4/5

560

QUESTION ID

S15-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 9,975

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 15 • BACKWARD 1 • Q4/5

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$607 \times 42$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 15 • BACKWARD 1 • Q5/5

561

QUESTION ID

S15-B1-Q5

TYPE

Short answer

CORRECT ANSWER

25,494

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 15 • BACKWARD 1 • Q5/5

561

SET 15 • 6.NS.B.2

BACKWARD 2

Dividing by a one-digit number • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.6

Divide.

$$852 \div 4$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 15 • BACKWARD 2 • Q1/5

563

QUESTION ID

S15-B2-Q1

TYPE

Short answer

CORRECT ANSWER

213

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 15 • BACKWARD 2 • Q1/5

ANSWER KEY

Reduced copy of the question page, with its answer

Divide.

$$1,256 \div 8$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 15 • BACKWARD 2 • Q2/5

564

QUESTION ID

S15-B2-Q2

TYPE

Short answer

CORRECT ANSWER

157

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 15 • BACKWARD 2 • Q2/5

True or false?

$$936 \div 3 = 312$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 15 • BACKWARD 2 • Q3/5

565

QUESTION ID

S15-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 15 • BACKWARD 2 • Q3/5

$$714 \div 7 = ?$$

- A.** 12
- B.** 102
- C.** 120
- D.** 1,002

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 15 • BACKWARD 2 • Q4/5

566

QUESTION ID

S15-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 102

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 15 • BACKWARD 2 • Q4/5

ANSWER KEY

Reduced copy of the question page, with its answer

Divide.

$$2,345 \div 5$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 15 • BACKWARD 2 • Q5/5

567

QUESTION ID

S15-B2-Q5

TYPE

Short answer

CORRECT ANSWER

469

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 15 • BACKWARD 2 • Q5/5

567

SET 15 • 6.NS.B.2

BACKWARD 3

Estimating quotients • Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6

Which is the best estimate for $4,508 \div 23$?

- A. 2
- B. 20
- C. 200
- D. 2,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B3-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 15 • BACKWARD 3 • Q1/5

569

QUESTION ID

S15-B3-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. 200

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B3-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 15 • BACKWARD 3 • Q1/5

Which is the best estimate for $7,236 \div 36$?

- A. 2
- B. 20
- C. 200
- D. 2,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B3-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 15 • BACKWARD 3 • Q2/5

570

QUESTION ID

S15-B3-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. 200

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B3-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 15 • BACKWARD 3 • Q2/5

True or false?

$9,975 \div 75$ is between 100 and 200.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B3-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 15 • BACKWARD 3 • Q3/5

571

QUESTION ID

S15-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B3-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 15 • BACKWARD 3 • Q3/5

Estimate $6,120 \div 29$ by rounding 29 to 30.

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B3-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 15 • BACKWARD 3 • Q4/5

572

QUESTION ID

S15-B3-Q4

TYPE

Short answer

CORRECT ANSWER

204

GRADING NOTE $6,120 \div 30 = 204$; "about 200" is also acceptable

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B3-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 15 • BACKWARD 3 • Q4/5

How many digits are in the quotient of $1,680 \div 48$?

- A. 1
- B. 2
- C. 3
- D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B3-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 15 • BACKWARD 3 • Q5/5

573

QUESTION ID

S15-B3-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B3-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 15 • BACKWARD 3 • Q5/5

SET 15 • 6.NS.B.2

BACKWARD 4

Multi-digit subtraction • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.4

ANSWER KEY

Reduced copy of the question page, with its answer

Subtract.

$$5,012 - 4,698$$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B4-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 15 • BACKWARD 4 • Q1/5

575

QUESTION ID

S15-B4-Q1

TYPE

Short answer

CORRECT ANSWER

314

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B4-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 15 • BACKWARD 4 • Q1/5

575

ANSWER KEY

Reduced copy of the question page, with its answer

Subtract.

$$3,000 - 1,847$$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B4-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 15 • BACKWARD 4 • Q2/5

576

QUESTION ID

S15-B4-Q2

TYPE

Short answer

CORRECT ANSWER

1,153

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B4-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 15 • BACKWARD 4 • Q2/5

576

True or false?

$$7,236 - 7,200 = 36$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B4-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 15 • BACKWARD 4 • Q3/5

577

QUESTION ID

S15-B4-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B4-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 15 • BACKWARD 4 • Q3/5

$$9,975 - 9,000 = ?$$

- A. 75
- B. 975
- C. 1,975
- D. 9,075

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B4-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 15 • BACKWARD 4 • Q4/5

578

QUESTION ID

S15-B4-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 975

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B4-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 15 • BACKWARD 4 • Q4/5

ANSWER KEY

Reduced copy of the question page, with its answer

Subtract.

$$6,203 - 2,875$$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B4-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 15 • BACKWARD 4 • Q5/5

579

QUESTION ID

S15-B4-Q5

TYPE

Short answer

CORRECT ANSWER

3,328

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B4-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 15 • BACKWARD 4 • Q5/5

SET 15 • 6.NS.B.2

FORWARD 1

Long division: fractions to decimals • Grade 7 • The Number System •
7.NS.A.2.d

Use long division to write $\frac{5}{11}$ as a decimal.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q1

Grade 7 • The Number System • 7.NS.A.2.d • Set 15 • FORWARD 1 • Q1/5

581

QUESTION ID

S15-F1-Q1

TYPE

Short answer

CORRECT ANSWER

**0.454545... (0.45
repeating)**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q1

Grade 7 • The Number System • 7.NS.A.2.d • Set 15 • FORWARD 1 • Q1/5

Use long division to write $\frac{3}{8}$ as a decimal.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q2

Grade 7 • The Number System • 7.NS.A.2.d • Set 15 • FORWARD 1 • Q2/5

582

QUESTION ID

S15-F1-Q2

TYPE

Short answer

CORRECT ANSWER

0.375

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q2

Grade 7 • The Number System • 7.NS.A.2.d • Set 15 • FORWARD 1 • Q2/5

Which fraction has a repeating decimal?

A. $\frac{1}{5}$

B. $\frac{2}{3}$

C. $\frac{3}{4}$

D. $\frac{7}{8}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q3

Grade 7 • The Number System • 7.NS.A.2.d • Set 15 • FORWARD 1 • Q3/5

583

QUESTION ID

S15-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 2/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q3

Grade 7 • The Number System • 7.NS.A.2.d • Set 15 • FORWARD 1 • Q3/5

True or false?

The decimal form of $\frac{7}{20}$ terminates.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q4

Grade 7 • The Number System • 7.NS.A.2.d • Set 15 • FORWARD 1 • Q4/5

584

QUESTION ID

S15-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q4

Grade 7 • The Number System • 7.NS.A.2.d • Set 15 • FORWARD 1 • Q4/5

Use long division to write $\frac{5}{6}$ as a decimal.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q5

Grade 7 • The Number System • 7.NS.A.2.d • Set 15 • FORWARD 1 • Q5/5

585

QUESTION ID

S15-F1-Q5

TYPE

Short answer

CORRECT ANSWER

**0.8333... (0.83, with
the 3 repeating)**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q5

Grade 7 • The Number System • 7.NS.A.2.d • Set 15 • FORWARD 1 • Q5/5

SET 15 • 6.NS.B.2

FORWARD 2

Repeating decimals and rational numbers • Grade 8 • The Number System •
8.NS.A.1

Write 0.272727... as a fraction in simplest form.

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q1

Grade 8 • The Number System • 8.NS.A.1 • Set 15 • FORWARD 2 • Q1/5

587

QUESTION ID

S15-F2-Q1

TYPE

Short answer

CORRECT ANSWER

3/11

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q1

Grade 8 • The Number System • 8.NS.A.1 • Set 15 • FORWARD 2 • Q1/5

Write 0.444... as a fraction.

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q2

Grade 8 • The Number System • 8.NS.A.1 • Set 15 • FORWARD 2 • Q2/5

588

QUESTION ID

S15-F2-Q2

TYPE

Short answer

CORRECT ANSWER

4/9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q2

Grade 8 • The Number System • 8.NS.A.1 • Set 15 • FORWARD 2 • Q2/5

Which number is irrational?

A. 0.333...

B. $\frac{5}{8}$

C. $\sqrt{16}$

D. $\sqrt{7}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q3

Grade 8 • The Number System • 8.NS.A.1 • Set 15 • FORWARD 2 • Q3/5

589

QUESTION ID

S15-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $\sqrt{7}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q3

Grade 8 • The Number System • 8.NS.A.1 • Set 15 • FORWARD 2 • Q3/5

True or false?

Every repeating decimal is a rational number.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q4

Grade 8 • The Number System • 8.NS.A.1 • Set 15 • FORWARD 2 • Q4/5

590

QUESTION ID

S15-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q4

Grade 8 • The Number System • 8.NS.A.1 • Set 15 • FORWARD 2 • Q4/5

Write 1.1666... as a fraction.

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q5

Grade 8 • The Number System • 8.NS.A.1 • Set 15 • FORWARD 2 • Q5/5

591

QUESTION ID

S15-F2-Q5

TYPE

Short answer

CORRECT ANSWER

7/6

GRADING NOTE 1 1/6 is also correct

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q5

Grade 8 • The Number System • 8.NS.A.1 • Set 15 • FORWARD 2 • Q5/5

Add multi-digit decimals

MAIN	Grade 6 main question
BACKWARD 1	Place value of decimal digits
BACKWARD 2	Equivalent decimals
BACKWARD 3	Lining up decimal points to add
BACKWARD 4	Multi-digit addition
FORWARD 1	Add rational numbers
FORWARD 2	Linear equations with decimal coefficients

6.NS.B.3

5.NBT.A.3.a

5.NBT.A.3.b

5.NBT.B.7

4.NBT.B.4

7.NS.A.1.d

8.EE.C.7.b

SET 16 • 6.NS.B.3

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.B.3

Find the sum.

$$315.4 + 28.75 + 6.309$$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 16 • MAIN • Q1/5

594

QUESTION ID

S16-M-Q1

TYPE

Short answer

CORRECT ANSWER

350.459

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 16 • MAIN • Q1/5

Find the sum.

$$47.08 + 9.6$$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 16 • MAIN • Q2/5

595

QUESTION ID

S16-M-Q2

TYPE

Short answer

CORRECT ANSWER

56.68

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 16 • MAIN • Q2/5

Find the sum.

$$128.5 + 3.47 + 60.009$$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 16 • MAIN • Q3/5

596

QUESTION ID

S16-M-Q3

TYPE

Short answer

CORRECT ANSWER

191.979

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 16 • MAIN • Q3/5

Find the sum.

$$6.75 + 12.9 + 0.48$$

- A.** 7.23
- B.** 19.23
- C.** 20.03
- D.** 20.13

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 16 • MAIN • Q4/5

597

QUESTION ID

S16-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 20.13

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 16 • MAIN • Q4/5

Find the sum.

$$0.875 + 14.2 + 3.06$$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 16 • MAIN • Q5/5

598

QUESTION ID

S16-M-Q5

TYPE

Short answer

CORRECT ANSWER

18.135

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 16 • MAIN • Q5/5

SET 16 • 6.NS.B.3

BACKWARD 1

Place value of decimal digits • Grade 5 • Number & Operations in Base Ten •
5.NBT.A.3.a

In 7.358, what place value does the digit 5 have?

- A.** Tenths
- B.** Hundredths
- C.** Ones
- D.** Thousandths

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 16 • BACKWARD 1 • Q1/5

600

QUESTION ID

S16-B1-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. Hundredths

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 16 • BACKWARD 1 • Q1/5

In 42.906, which digit is in the tenths place?

Digit: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 16 • BACKWARD 1 • Q2/5

601

QUESTION ID

S16-B1-Q2

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 16 • BACKWARD 1 • Q2/5

True or false?

In 3.47, the digit 7 is in the hundredths place.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 16 • BACKWARD 1 • Q3/5

602

QUESTION ID

S16-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 16 • BACKWARD 1 • Q3/5

What is the value of the 8 in 0.584?

- A. 0.008
- B. 0.08
- C. 0.8
- D. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 16 • BACKWARD 1 • Q4/5

603

QUESTION ID

S16-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 0.08

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 16 • BACKWARD 1 • Q4/5

Write $5 + 0.3 + 0.07$ as a decimal.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 16 • BACKWARD 1 • Q5/5

604

QUESTION ID

S16-B1-Q5

TYPE

Short answer

CORRECT ANSWER

5.37

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 16 • BACKWARD 1 • Q5/5

SET 16 • 6.NS.B.3

BACKWARD 2

Equivalent decimals • Grade 5 • Number & Operations in Base Ten •
5.NBT.A.3.b

True or false?

2.4 and 2.40 have the same value.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 16 • BACKWARD 2 • Q1/5

606

QUESTION ID

S16-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 16 • BACKWARD 2 • Q1/5

True or false?

5.6 and 5.06 have the same value.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 16 • BACKWARD 2 • Q2/5

607

QUESTION ID

S16-B2-Q2

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 16 • BACKWARD 2 • Q2/5

Which decimal is equal to 7.5?

- A.** 0.75
- B.** 7.005
- C.** 7.05
- D.** 7.50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 16 • BACKWARD 2 • Q3/5

608

QUESTION ID

S16-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 7.50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 16 • BACKWARD 2 • Q3/5

Write 9.6 with a zero in the hundredths place.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 16 • BACKWARD 2 • Q4/5

609

QUESTION ID

S16-B2-Q4

TYPE

Short answer

CORRECT ANSWER

9.60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 16 • BACKWARD 2 • Q4/5

True or false?

$$0.300 = 0.3$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 16 • BACKWARD 2 • Q5/5

610

QUESTION ID

S16-B2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 16 • BACKWARD 2 • Q5/5

SET 16 • 6.NS.B.3

BACKWARD 3

Lining up decimal points to add • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.7

Which setup lines up the decimal points for $36.2 + 4.58$?

A.

$$\begin{array}{r} 36.2 \\ +4.58 \\ \hline \end{array}$$

B.

$$\begin{array}{r} 36.20 \\ + 4.58 \\ \hline \end{array}$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B3-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 16 • BACKWARD 3 • Q1/5

612

QUESTION ID

S16-B3-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. Choice B (picture)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B3-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 16 • BACKWARD 3 • Q1/5

Which setup lines up the decimal points for $5.07 + 12.3$?

A.

$$\begin{array}{r} 5.07 \\ +12.30 \\ \hline \end{array}$$

B.

$$\begin{array}{r} 5.07 \\ +12.3 \\ \hline \end{array}$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B3-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 16 • BACKWARD 3 • Q2/5

613

QUESTION ID

S16-B3-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. Choice A (picture)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B3-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 16 • BACKWARD 3 • Q2/5

True or false?

The decimal points are lined up correctly.

$$\begin{array}{r} 8.4 \\ +2.75 \\ \hline \end{array}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B3-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 16 • BACKWARD 3 • Q3/5

614

QUESTION ID

S16-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B3-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 16 • BACKWARD 3 • Q3/5

Which setup lines up the decimal points for $128.5 + 3.47$?

A.

$$\begin{array}{r} 128.50 \\ + 3.47 \\ \hline \end{array}$$

B.

$$\begin{array}{r} 128.5 \\ + 3.47 \\ \hline \end{array}$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B3-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 16 • BACKWARD 3 • Q4/5

615

QUESTION ID

S16-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. Choice A (picture)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B3-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 16 • BACKWARD 3 • Q4/5

True or false?

The decimal points are lined up correctly.

- A.** True
- B.** False

$$\begin{array}{r} 9.30 \\ + 0.45 \\ \hline \end{array}$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B3-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 16 • BACKWARD 3 • Q5/5

616

QUESTION ID

S16-B3-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B3-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 16 • BACKWARD 3 • Q5/5

SET 16 • 6.NS.B.3

BACKWARD 4

Multi-digit addition • Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4

ANSWER KEY

Reduced copy of the question page, with its answer

Add.

$$3,154 + 2,875$$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B4-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 16 • BACKWARD 4 • Q1/5

618

QUESTION ID

S16-B4-Q1

TYPE

Short answer

CORRECT ANSWER

6,029

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B4-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 16 • BACKWARD 4 • Q1/5

618

ANSWER KEY

Reduced copy of the question page, with its answer

Add.

$$4,605 + 2,798$$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B4-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 16 • BACKWARD 4 • Q2/5

619

QUESTION ID

S16-B4-Q2

TYPE

Short answer

CORRECT ANSWER

7,403

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B4-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 16 • BACKWARD 4 • Q2/5

619

True or false?

$$1,975 + 3,489 = 5,464$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B4-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 16 • BACKWARD 4 • Q3/5

620

QUESTION ID

S16-B4-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B4-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 16 • BACKWARD 4 • Q3/5

$$6,758 + 1,265 = ?$$

- A.** 7,013
- B.** 7,923
- C.** 8,013
- D.** 8,023

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B4-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 16 • BACKWARD 4 • Q4/5

621

QUESTION ID

S16-B4-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 8,023

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B4-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 16 • BACKWARD 4 • Q4/5

ANSWER KEY

Reduced copy of the question page, with its answer

Add.

$$12,480 + 9,635$$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B4-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 16 • BACKWARD 4 • Q5/5

622

QUESTION ID

S16-B4-Q5

TYPE

Short answer

CORRECT ANSWER

22,115

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B4-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 16 • BACKWARD 4 • Q5/5

622

SET 16 • 6.NS.B.3

FORWARD 1

Add rational numbers • Grade 7 • The Number System • 7.NS.A.1.d

Find the value.

$$-18.6 + 4.35 =$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q1

Grade 7 • The Number System • 7.NS.A.1.d • Set 16 • FORWARD 1 • Q1/5

624

QUESTION ID

S16-F1-Q1

TYPE

Short answer

CORRECT ANSWER

-14.25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q1

Grade 7 • The Number System • 7.NS.A.1.d • Set 16 • FORWARD 1 • Q1/5

Find the value.

$$-24.5 + 3.27 =$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q2

Grade 7 • The Number System • 7.NS.A.1.d • Set 16 • FORWARD 1 • Q2/5

625

QUESTION ID

S16-F1-Q2

TYPE

Short answer

CORRECT ANSWER

-21.23

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q2

Grade 7 • The Number System • 7.NS.A.1.d • Set 16 • FORWARD 1 • Q2/5

Find the value.

$$-3.4 + (-9.15) =$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q3

Grade 7 • The Number System • 7.NS.A.1.d • Set 16 • FORWARD 1 • Q3/5

626

QUESTION ID

S16-F1-Q3

TYPE

Short answer

CORRECT ANSWER

-12.55

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q3

Grade 7 • The Number System • 7.NS.A.1.d • Set 16 • FORWARD 1 • Q3/5

Find the value.

$$-6.25 + (-2.8) = ?$$

- A. -9.05
- B. -3.45
- C. 3.45
- D. 9.05

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q4

Grade 7 • The Number System • 7.NS.A.1.d • Set 16 • FORWARD 1 • Q4/5

627

QUESTION ID

S16-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. -9.05

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q4

Grade 7 • The Number System • 7.NS.A.1.d • Set 16 • FORWARD 1 • Q4/5

Find the value.

$$-15.75 + 20.3 =$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q5

Grade 7 • The Number System • 7.NS.A.1.d • Set 16 • FORWARD 1 • Q5/5

628

QUESTION ID

S16-F1-Q5

TYPE

Short answer

CORRECT ANSWER

4.55

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q5

Grade 7 • The Number System • 7.NS.A.1.d • Set 16 • FORWARD 1 • Q5/5

SET 16 • 6.NS.B.3

FORWARD 2

Linear equations with decimal coefficients • Grade 8 • Expressions & Equations
• 8.EE.C.7.b

Solve for x.

$$-2.4x + 7.8 = 1.2x - 1.2$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 16 • FORWARD 2 • Q1/5

630

QUESTION ID

S16-F2-Q1

TYPE

Short answer

CORRECT ANSWER

2.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 16 • FORWARD 2 • Q1/5

Solve for x.

$$-1.5x + 6.4 = 0.5x - 1.6$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 16 • FORWARD 2 • Q2/5

631

QUESTION ID

S16-F2-Q2

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 16 • FORWARD 2 • Q2/5

Solve for x.

$$0.8x + 7.2 = -1.2x + 1.2$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 16 • FORWARD 2 • Q3/5

632

QUESTION ID

S16-F2-Q3

TYPE

Short answer

CORRECT ANSWER

-3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 16 • FORWARD 2 • Q3/5

Solve for x.

$$-4.5x - 3.25 = -2.5x + 0.75$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 16 • FORWARD 2 • Q4/5

633

QUESTION ID

S16-F2-Q4

TYPE

Short answer

CORRECT ANSWER

-2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 16 • FORWARD 2 • Q4/5

Solve for x.

$$2.4(x + 1.5) = 1.4x + 6.6$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 16 • FORWARD 2 • Q5/5

634

QUESTION ID

S16-F2-Q5

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 16 • FORWARD 2 • Q5/5

Subtract multi-digit decimals

MAIN

Grade 6 main question

6.NS.B.3

BACKWARD 1

Place value of decimal digits

5.NBT.A.3.a

BACKWARD 2

Writing zeros to make equal decimal places

5.NBT.A.3.b

BACKWARD 3

Lining up decimal points to subtract

5.NBT.B.7

BACKWARD 4

Subtracting across zeros

4.NBT.B.4

FORWARD 1

Subtract rational numbers by adding the additive inverse

7.NS.A.1.c

FORWARD 2

Solve equations by collecting like terms

8.EE.C.7.b

SET 17 • 6.NS.B.3

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.B.3

Find the difference.

$$52.3 - 18.47$$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 17 • MAIN • Q1/5

637

QUESTION ID

S17-M-Q1

TYPE

Short answer

CORRECT ANSWER

33.83

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 17 • MAIN • Q1/5

ANSWER KEY

Reduced copy of the question page, with its answer

Find the difference.

$$100 - 36.58$$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 17 • MAIN • Q2/5

638

QUESTION ID

S17-M-Q2

TYPE

Short answer

CORRECT ANSWER

63.42

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 17 • MAIN • Q2/5

638

Find the difference.

$$7.4 - 2.068$$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 17 • MAIN • Q3/5

639

QUESTION ID

S17-M-Q3

TYPE

Short answer

CORRECT ANSWER

5.332

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 17 • MAIN • Q3/5

Find the difference.

$$15.02 - 6.9$$

- A.** 8.12
- B.** 8.93
- C.** 9.12
- D.** 14.33

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 17 • MAIN • Q4/5

640

QUESTION ID

S17-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 8.12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 17 • MAIN • Q4/5

Find the difference.

$$243.6 - 87.95$$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 17 • MAIN • Q5/5

641

QUESTION ID

S17-M-Q5

TYPE

Short answer

CORRECT ANSWER

155.65

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 17 • MAIN • Q5/5

SET 17 • 6.NS.B.3

BACKWARD 1

Place value of decimal digits • Grade 5 • Number & Operations in Base Ten •
5.NBT.A.3.a

In 2.068, what place value does the digit 6 have?

- A. Tenths
- B. Thousandths
- C. Hundredths
- D. Ones

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 17 • BACKWARD 1 • Q1/5

643

QUESTION ID

S17-B1-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. Hundredths

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 17 • BACKWARD 1 • Q1/5

In 87.95, which digit is in the hundredths place?

Digit: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 17 • BACKWARD 1 • Q2/5

644

QUESTION ID

S17-B1-Q2

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 17 • BACKWARD 1 • Q2/5

True or false?

In 36.58, the digit 5 is in the tenths place.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 17 • BACKWARD 1 • Q3/5

645

QUESTION ID

S17-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 17 • BACKWARD 1 • Q3/5

What is the value of the 4 in 7.04?

- A.** 0.004
- B.** 0.04
- C.** 0.4
- D.** 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 17 • BACKWARD 1 • Q4/5

646

QUESTION ID

S17-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 0.04

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 17 • BACKWARD 1 • Q4/5

Write $20 + 0.6 + 0.009$ as a decimal.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 17 • BACKWARD 1 • Q5/5

647

QUESTION ID

S17-B1-Q5

TYPE

Short answer

CORRECT ANSWER

20.609

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 17 • BACKWARD 1 • Q5/5

SET 17 • 6.NS.B.3

BACKWARD 2

Writing zeros to make equal decimal places • Grade 5 • Number & Operations
in Base Ten • 5.NBT.A.3.b

True or false?

$$7.4 = 7.400$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 17 • BACKWARD 2 • Q1/5

649

QUESTION ID

S17-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 17 • BACKWARD 2 • Q1/5

Write 100 with two decimal places.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 17 • BACKWARD 2 • Q2/5

650

QUESTION ID

S17-B2-Q2

TYPE

Short answer

CORRECT ANSWER

100.00

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 17 • BACKWARD 2 • Q2/5

Which decimal is equal to 15.1?

- A.** 1.51
- B.** 15.001
- C.** 15.01
- D.** 15.10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 17 • BACKWARD 2 • Q3/5

651

QUESTION ID

S17-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 15.10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 17 • BACKWARD 2 • Q3/5

Write 52.3 with a digit in the hundredths place, without changing its value.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 17 • BACKWARD 2 • Q4/5

652

QUESTION ID

S17-B2-Q4

TYPE

Short answer

CORRECT ANSWER

52.30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 17 • BACKWARD 2 • Q4/5

True or false?

$$6.9 = 6.09$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 17 • BACKWARD 2 • Q5/5

653

QUESTION ID

S17-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 17 • BACKWARD 2 • Q5/5

SET 17 • 6.NS.B.3

BACKWARD 3

Lining up decimal points to subtract • Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7

Which setup lines up the decimal points for $15.6 - 3.42$?

A.

$$\begin{array}{r} 15.60 \\ - 3.42 \\ \hline \end{array}$$

B.

$$\begin{array}{r} 15.6 \\ - 3.42 \\ \hline \end{array}$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B3-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 17 • BACKWARD 3 • Q1/5

655

QUESTION ID

S17-B3-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. Choice A (picture)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B3-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 17 • BACKWARD 3 • Q1/5

Which setup lines up the decimal points for
 $52.3 - 18.47$?

A.

$$\begin{array}{r} 52.3 \\ -18.47 \\ \hline \end{array}$$

B.

$$\begin{array}{r} 52.30 \\ -18.47 \\ \hline \end{array}$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B3-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 17 • BACKWARD 3 • Q2/5

656

QUESTION ID

S17-B3-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. Choice B (picture)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B3-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 17 • BACKWARD 3 • Q2/5

True or false?

The decimal points are lined up correctly.

- A.** True
- B.** False

$$\begin{array}{r} 7.4 \\ -2.068 \\ \hline \end{array}$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B3-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 17 • BACKWARD 3 • Q3/5

657

QUESTION ID

S17-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B3-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 17 • BACKWARD 3 • Q3/5

True or false?

The decimal points are lined up correctly.

$$\begin{array}{r} 100.00 \\ - 36.58 \\ \hline \end{array}$$

- A.** True
B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B3-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 17 • BACKWARD 3 • Q4/5

658

QUESTION ID

S17-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B3-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 17 • BACKWARD 3 • Q4/5

Which setup lines up the decimal points for
 $243.6 - 87.95$?

A.

$$\begin{array}{r} 243.60 \\ - 87.95 \\ \hline \end{array}$$

B.

$$\begin{array}{r} 243.6 \\ - 87.95 \\ \hline \end{array}$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B3-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 17 • BACKWARD 3 • Q5/5

659

QUESTION ID

S17-B3-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. Choice A (picture)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B3-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 17 • BACKWARD 3 • Q5/5

SET 17 • 6.NS.B.3

BACKWARD 4

Subtracting across zeros • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.4

ANSWER KEY

Reduced copy of the question page, with its answer

Subtract.

$$5,000 - 2,847$$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B4-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 17 • BACKWARD 4 • Q1/5

661

QUESTION ID

S17-B4-Q1

TYPE

Short answer

CORRECT ANSWER

2,153

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B4-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 17 • BACKWARD 4 • Q1/5

661

Subtract.

$$3,006 - 1,478$$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B4-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 17 • BACKWARD 4 • Q2/5

662

QUESTION ID

S17-B4-Q2

TYPE

Short answer

CORRECT ANSWER

1,528

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B4-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 17 • BACKWARD 4 • Q2/5

True or false?

$$6,000 - 2,458 = 3,542$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B4-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 17 • BACKWARD 4 • Q3/5

663

QUESTION ID

S17-B4-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B4-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 17 • BACKWARD 4 • Q3/5

$$8,014 - 3,276 = ?$$

- A.** 4,738
- B.** 4,748
- C.** 4,838
- D.** 5,262

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B4-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 17 • BACKWARD 4 • Q4/5

664

QUESTION ID

S17-B4-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 4,738

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B4-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 17 • BACKWARD 4 • Q4/5

ANSWER KEY

Reduced copy of the question page, with its answer

Subtract.

$$10,000 - 3,658$$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B4-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 17 • BACKWARD 4 • Q5/5

665

QUESTION ID

S17-B4-Q5

TYPE

Short answer

CORRECT ANSWER

6,342

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B4-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 17 • BACKWARD 4 • Q5/5

665

SET 17 • 6.NS.B.3

FORWARD 1

Subtract rational numbers by adding the additive inverse • Grade 7 • The Number System • 7.NS.A.1.c

Find the value.

$$-4.2 - 3.75$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q1

Grade 7 • The Number System • 7.NS.A.1.c • Set 17 • FORWARD 1 • Q1/5

667

QUESTION ID

S17-F1-Q1

TYPE

Short answer

CORRECT ANSWER

-7.95

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q1

Grade 7 • The Number System • 7.NS.A.1.c • Set 17 • FORWARD 1 • Q1/5

Find the value.

$$2.5 - (-6.8)$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q2

Grade 7 • The Number System • 7.NS.A.1.c • Set 17 • FORWARD 1 • Q2/5

668

QUESTION ID

S17-F1-Q2

TYPE

Short answer

CORRECT ANSWER

9.3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q2

Grade 7 • The Number System • 7.NS.A.1.c • Set 17 • FORWARD 1 • Q2/5

Which expression is equal to $8.1 - 12.4$?

- A. $12.4 - 8.1$
- B. $-8.1 + 12.4$
- C. $8.1 + (-12.4)$
- D. $8.1 + 12.4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q3

Grade 7 • The Number System • 7.NS.A.1.c • Set 17 • FORWARD 1 • Q3/5

669

QUESTION ID

S17-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $8.1 + (-12.4)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q3

Grade 7 • The Number System • 7.NS.A.1.c • Set 17 • FORWARD 1 • Q3/5

ANSWER KEY

Reduced copy of the question page, with its answer

Find the value.

$$-0.65 - (-1.2)$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q4

Grade 7 • The Number System • 7.NS.A.1.c • Set 17 • FORWARD 1 • Q4/5

670

QUESTION ID

S17-F1-Q4

TYPE

Short answer

CORRECT ANSWER

0.55

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q4

Grade 7 • The Number System • 7.NS.A.1.c • Set 17 • FORWARD 1 • Q4/5

670

True or false?

$$-7.5 - 2.5 = -7.5 + (-2.5)$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q5

Grade 7 • The Number System • 7.NS.A.1.c • Set 17 • FORWARD 1 • Q5/5

671

QUESTION ID

S17-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q5

Grade 7 • The Number System • 7.NS.A.1.c • Set 17 • FORWARD 1 • Q5/5

SET 17 • 6.NS.B.3

FORWARD 2

Solve equations by collecting like terms • Grade 8 • Expressions & Equations •
8.EE.C.7.b

Solve for x.

$$5.2x - 3.75 = 2.2x + 8.25$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • FORWARD 2 • Q1/5

673

QUESTION ID

S17-F2-Q1

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • FORWARD 2 • Q1/5

Solve for x.

$$0.9x + 12.5 = 0.4x + 10$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • FORWARD 2 • Q2/5

674

QUESTION ID

S17-F2-Q2

TYPE

Short answer

CORRECT ANSWER

-5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • FORWARD 2 • Q2/5

Solve for x.

$$6.3 - 1.2x = 0.8x - 1.7$$

- A.** -4
- B.** 0.4
- C.** 2.5
- D.** 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • FORWARD 2 • Q3/5

675

QUESTION ID

S17-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • FORWARD 2 • Q3/5

Solve for x.

$$4.75 - x = 2x - 1.25$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • FORWARD 2 • Q4/5

676

QUESTION ID

S17-F2-Q4

TYPE

Short answer

CORRECT ANSWER

2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • FORWARD 2 • Q4/5

Solve for x.

$$3.4x - 9.6 = 1.4x - 2.6$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • FORWARD 2 • Q5/5

677

QUESTION ID

S17-F2-Q5

TYPE

Short answer

CORRECT ANSWER

3.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • FORWARD 2 • Q5/5

Find and correct decimal alignment errors in addition and subtraction

MAIN	Grade 6 main question	6.NS.B.3
BACKWARD 1	Lining up decimal points	5.NBT.B.7
BACKWARD 2	Place value of decimal digits	5.NBT.A.3.a
BACKWARD 3	Estimating to check reasonableness	5.NBT.A.4
FORWARD 1	Find errors when adding and subtracting rational numbers	7.NS.A.1.d
FORWARD 2	Find errors in solving linear equations	8.EE.C.7.b

SET 18 • 6.NS.B.3

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.B.3

Jon put this example on the board.

What mistake was made?

What should the answer be?

Mistake: _____

Correct answer: _____

$$\begin{array}{r} 315.2 \\ 24.8 \\ +7.936 \\ \hline 11.336 \end{array}$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 18 • MAIN • Q1/5

680

QUESTION ID

S18-M-Q1

TYPE

Short answer

CORRECT ANSWER

Mistake: the numbers were lined up by their last digits instead of by their decimal points. Correct answer: 347.936

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 18 • MAIN • Q1/5

Ava put this example on the board.

What mistake was made?
What should the answer be?

$$\begin{array}{r} 56.4 \\ +3.25 \\ \hline 8.89 \end{array}$$

Mistake: _____

Correct answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 18 • MAIN • Q2/5

681

QUESTION ID

S18-M-Q2

TYPE

Short answer

CORRECT ANSWER

Mistake: the numbers were lined up by their last digits instead of by their decimal points. Correct answer: 59.65

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 18 • MAIN • Q2/5

Leo put this example on the board.

What mistake was made?
What should the answer be?

$$\begin{array}{r} 48.6 \\ -2.15 \\ \hline 2.71 \end{array}$$

Mistake: _____

Correct answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 18 • MAIN • Q3/5

682

QUESTION ID

S18-M-Q3

TYPE

Short answer

CORRECT ANSWER

Mistake: the numbers were lined up by their last digits instead of by their decimal points. Correct answer: 46.45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 18 • MAIN • Q3/5

Mia put this example on the board.

What mistake was made?
What should the answer be?

$$\begin{array}{r} 7.08 \\ +12.6 \\ \hline 8.34 \end{array}$$

Mistake: _____

Correct answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 18 • MAIN • Q4/5

683

QUESTION ID

S18-M-Q4

TYPE

Short answer

CORRECT ANSWER

Mistake: the numbers were lined up by their last digits instead of by their decimal points. Correct answer: 19.68

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 18 • MAIN • Q4/5

Ben put this example on the board.

What mistake was made?
What should the answer be?

$$\begin{array}{r} 125.6 \\ -3.48 \\ \hline 9.08 \end{array}$$

Mistake: _____

Correct answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 18 • MAIN • Q5/5

684

QUESTION ID

S18-M-Q5

TYPE

Short answer

CORRECT ANSWER

Mistake: the numbers were lined up by their last digits instead of by their decimal points. Correct answer: 122.12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 18 • MAIN • Q5/5

SET 18 • 6.NS.B.3

BACKWARD 1

Lining up decimal points • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.7

Which setup lines up the decimal points for $56.4 + 3.25$?

A.

$$\begin{array}{r} 56.4 \\ +3.25 \\ \hline \end{array}$$

B.

$$\begin{array}{r} 56.40 \\ + 3.25 \\ \hline \end{array}$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 18 • BACKWARD 1 • Q1/5

686

QUESTION ID

S18-B1-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. Choice B (picture)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 18 • BACKWARD 1 • Q1/5

Which setup lines up the decimal points for $48.6 - 2.15$?

A.

$$\begin{array}{r} 48.60 \\ - 2.15 \\ \hline \end{array}$$

B.

$$\begin{array}{r} 48.6 \\ - 2.15 \\ \hline \end{array}$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 18 • BACKWARD 1 • Q2/5

687

QUESTION ID

S18-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. Choice A (picture)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 18 • BACKWARD 1 • Q2/5

True or false?

The decimal points are lined up correctly.

$$\begin{array}{r} 7.08 \\ +12.6 \\ \hline \end{array}$$

- A.** True
B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 18 • BACKWARD 1 • Q3/5

688

QUESTION ID

S18-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 18 • BACKWARD 1 • Q3/5

True or false?

The decimal points are lined up correctly.

- A.** True
- B.** False

$$\begin{array}{r} 125.60 \\ - 3.48 \\ \hline \end{array}$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 18 • BACKWARD 1 • Q4/5

689

QUESTION ID

S18-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 18 • BACKWARD 1 • Q4/5

Which setup lines up the decimal points for
 $24.8 + 7.936$?

A.

$$\begin{array}{r} 24.8 \\ +7.936 \\ \hline \end{array}$$

B.

$$\begin{array}{r} 24.800 \\ + 7.936 \\ \hline \end{array}$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 18 • BACKWARD 1 • Q5/5

690

QUESTION ID

S18-B1-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. Choice B (picture)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 18 • BACKWARD 1 • Q5/5

SET 18 • 6.NS.B.3

BACKWARD 2

Place value of decimal digits • Grade 5 • Number & Operations in Base Ten •
5.NBT.A.3.a

In 7.936, what place value does the digit 3 have?

- A.** Hundredths
- B.** Tenths
- C.** Thousandths
- D.** Ones

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 18 • BACKWARD 2 • Q1/5

692

QUESTION ID

S18-B2-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. Hundredths

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 18 • BACKWARD 2 • Q1/5

In 315.2, which digit is in the ones place?

Digit: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 18 • BACKWARD 2 • Q2/5

693

QUESTION ID

S18-B2-Q2

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 18 • BACKWARD 2 • Q2/5

True or false?

In 3.25, the digit 2 is in the tenths place.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 18 • BACKWARD 2 • Q3/5

694

QUESTION ID

S18-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 18 • BACKWARD 2 • Q3/5

What is the value of the 6 in 12.56?

- A.** 0.06
- B.** 0.6
- C.** 6
- D.** 60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 18 • BACKWARD 2 • Q4/5

695

QUESTION ID

S18-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 0.06

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 18 • BACKWARD 2 • Q4/5

In 48.6, which digit is in the tenths place?

Digit: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 18 • BACKWARD 2 • Q5/5

696

QUESTION ID

S18-B2-Q5

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 18 • BACKWARD 2 • Q5/5

SET 18 • 6.NS.B.3

BACKWARD 3

Estimating to check reasonableness • Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4

Round each number to the nearest whole number. Then add to estimate.

$$315.2 + 24.8 + 7.936$$

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B3-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 18 • BACKWARD 3 • Q1/5

698

QUESTION ID

S18-B3-Q1

TYPE

Short answer

CORRECT ANSWER

348

GRADING NOTE $315 + 25 + 8$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B3-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 18 • BACKWARD 3 • Q1/5

Round 48.6 to the nearest whole number.

Rounded: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B3-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 18 • BACKWARD 3 • Q2/5

699

QUESTION ID

S18-B3-Q2

TYPE

Short answer

CORRECT ANSWER

49

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B3-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 18 • BACKWARD 3 • Q2/5

True or false?

7.08 + 12.6 is about 20.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B3-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 18 • BACKWARD 3 • Q3/5

700

QUESTION ID

S18-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B3-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 18 • BACKWARD 3 • Q3/5

Which is the best estimate for $125.6 - 3.48$?

- A. 9
- B. 90
- C. 122
- D. 130

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B3-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 18 • BACKWARD 3 • Q4/5

701

QUESTION ID

S18-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 122

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B3-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 18 • BACKWARD 3 • Q4/5

Round 56.4 and 3.25 to the nearest whole numbers.
Then add.

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B3-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 18 • BACKWARD 3 • Q5/5

702

QUESTION ID

S18-B3-Q5

TYPE

Short answer

CORRECT ANSWER

59

GRADING NOTE 56 + 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B3-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 18 • BACKWARD 3 • Q5/5

SET 18 • 6.NS.B.3

FORWARD 1

Find errors when adding and subtracting rational numbers • Grade 7 • The Number System • 7.NS.A.1.d

Ray wrote: $-3.5 + 1.25 = -4.75$

What mistake was made? What is the correct value?

Mistake: _____

Correct answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q1

Grade 7 • The Number System • 7.NS.A.1.d • Set 18 • FORWARD 1 • Q1/5

704

QUESTION ID

S18-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Mistake: the numbers have different signs, so their absolute values should be subtracted, not added. Correct value: -2.25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q1

Grade 7 • The Number System • 7.NS.A.1.d • Set 18 • FORWARD 1 • Q1/5

Tia wrote: $-6.2 - 2.4 = -3.8$

What mistake was made? What is the correct value?

Mistake: _____

Correct answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q2

Grade 7 • The Number System • 7.NS.A.1.d • Set 18 • FORWARD 1 • Q2/5

705

QUESTION ID

S18-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Mistake: subtracting 2.4 means adding -2.4 ; the result should be more negative, not closer to 0. Correct value: -8.6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q2

Grade 7 • The Number System • 7.NS.A.1.d • Set 18 • FORWARD 1 • Q2/5

Sam wrote: $4.5 - (-1.5) = 3$

What mistake was made? What is the correct value?

Mistake: _____

Correct answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q3

Grade 7 • The Number System • 7.NS.A.1.d • Set 18 • FORWARD 1 • Q3/5

706

QUESTION ID

S18-F1-Q3

TYPE

Short answer

CORRECT ANSWER

**Mistake: subtracting
–1.5 means adding
1.5, not subtracting
1.5. Correct value: 6**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q3

Grade 7 • The Number System • 7.NS.A.1.d • Set 18 • FORWARD 1 • Q3/5

Kai wrote: $-7.25 + (-2.5) = -4.75$

What mistake was made? What is the correct value?

Mistake: _____

Correct answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q4

Grade 7 • The Number System • 7.NS.A.1.d • Set 18 • FORWARD 1 • Q4/5

707

QUESTION ID

S18-F1-Q4

TYPE

Short answer

CORRECT ANSWER

Mistake: both numbers are negative, so their absolute values should be added, not subtracted. Correct value: -9.75

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q4

Grade 7 • The Number System • 7.NS.A.1.d • Set 18 • FORWARD 1 • Q4/5

Lia wrote: $-0.8 + 2.3 = -1.5$

What mistake was made? What is the correct value?

Mistake: _____

Correct answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q5

Grade 7 • The Number System • 7.NS.A.1.d • Set 18 • FORWARD 1 • Q5/5

708

QUESTION ID

S18-F1-Q5

TYPE

Short answer

CORRECT ANSWER

Mistake: the sign is wrong; 2.3 has the greater absolute value, so the sum is positive. Correct value: 1.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q5

Grade 7 • The Number System • 7.NS.A.1.d • Set 18 • FORWARD 1 • Q5/5

SET 18 • 6.NS.B.3

FORWARD 2

Find errors in solving linear equations • Grade 8 • Expressions & Equations •
8.EE.C.7.b

Kim solved $2.4x + 1.2 = 0.4x - 3.8$.

Step 1: $2x + 1.2 = -3.8$

Step 2: $2x = -2.6$

Step 3: $x = -1.3$

Which step has the mistake? What is the correct solution?

Step with the mistake: _____

Correct solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • FORWARD 2 • Q1/5

710

QUESTION ID

S18-F2-Q1

TYPE

Short answer

CORRECT ANSWER

**Step 2 (subtracting
1.2 gives $2x = -5$).
Correct solution:
 $x = -2.5$**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • FORWARD 2 • Q1/5

Jo solved $3.5x - 2 = 1.5x + 6$.

Step 1: $2x - 2 = 6$

Step 2: $2x = 4$

Step 3: $x = 2$

Which step has the mistake? What is the correct solution?

Step with the mistake: _____

Correct solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • FORWARD 2 • Q2/5

711

QUESTION ID

S18-F2-Q2

TYPE

Short answer

CORRECT ANSWER

**Step 2 (adding 2
gives $2x = 8$).
Correct solution:
 $x = 4$**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • FORWARD 2 • Q2/5

Ali solved $-1.2x + 5 = 0.8x - 3$.

Step 1: $-2x + 5 = -3$

Step 2: $-2x = -8$

Step 3: $x = -4$

Which step has the mistake? What is the correct solution?

Step with the mistake: _____

Correct solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • FORWARD 2 • Q3/5

712

QUESTION ID

S18-F2-Q3

TYPE

Short answer

CORRECT ANSWER

**Step 3 (dividing -8
by -2 gives 4).**

**Correct solution:
 $x = 4$**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • FORWARD 2 • Q3/5

Ned solved $0.5(x + 6) = 2x - 3$.

Step 1: $0.5x + 6 = 2x - 3$

Step 2: $9 = 1.5x$

Step 3: $x = 6$

Which step has the mistake? What is the correct solution?

Step with the mistake: _____

Correct solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • FORWARD 2 • Q4/5

713

QUESTION ID

S18-F2-Q4

TYPE

Short answer

CORRECT ANSWER

Step 1 (0.5 must multiply both terms: $0.5x + 3$). Correct solution: $x = 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • FORWARD 2 • Q4/5

Eve solved $4.2x - 1.5 = 2.2x + 2.5$.

Step 1: $2x - 1.5 = 2.5$

Step 2: $2x = 1$

Step 3: $x = 0.5$

Which step has the mistake? What is the correct solution?

Step with the mistake: _____

Correct solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • FORWARD 2 • Q5/5

714

QUESTION ID

S18-F2-Q5

TYPE

Short answer

CORRECT ANSWER

**Step 2 (adding 1.5
gives $2x = 4$).
Correct solution:
 $x = 2$**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • FORWARD 2 • Q5/5

Multiply multi-digit decimals

- MAIN** Grade 6 main question
- BACKWARD 1** Multiplying whole numbers
- BACKWARD 2** Powers of 10 and the decimal point
- BACKWARD 3** Multiplying tenths and hundredths
- BACKWARD 4** Rounding decimals to estimate
- FORWARD 1** Multiply rational numbers
- FORWARD 2** Multiply numbers in scientific notation

6.NS.B.3

5.NBT.B.5

5.NBT.A.2

5.NBT.B.7

5.NBT.A.4

7.NS.A.2.c

8.EE.A.4

SET 19 • 6.NS.B.3

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.B.3

Multiply.

$$3.6 \times 0.45$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 19 • MAIN • Q1/5

717

QUESTION ID

S19-M-Q1

TYPE

Short answer

CORRECT ANSWER

1.62

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 19 • MAIN • Q1/5

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$12.5 \times 3.8$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 19 • MAIN • Q2/5

718

QUESTION ID

S19-M-Q2

TYPE

Short answer

CORRECT ANSWER

47.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 19 • MAIN • Q2/5

718

Multiply.

$$4.25 \times 2.4$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 19 • MAIN • Q3/5

719

QUESTION ID

S19-M-Q3

TYPE

Short answer

CORRECT ANSWER

10.2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 19 • MAIN • Q3/5

Multiply.

$$0.07 \times 0.6$$

- A.** 0.0042
- B.** 0.042
- C.** 0.42
- D.** 4.2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 19 • MAIN • Q4/5

720

QUESTION ID

S19-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 0.042

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 19 • MAIN • Q4/5

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$18.4 \times 0.35$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 19 • MAIN • Q5/5

721

QUESTION ID

S19-M-Q5

TYPE

Short answer

CORRECT ANSWER

6.44

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 19 • MAIN • Q5/5

721

SET 19 • 6.NS.B.3

BACKWARD 1

Multiplying whole numbers • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.5

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$36 \times 45$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 19 • BACKWARD 1 • Q1/5

723

QUESTION ID

S19-B1-Q1

TYPE

Short answer

CORRECT ANSWER

1,620

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 19 • BACKWARD 1 • Q1/5

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$125 \times 38$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 19 • BACKWARD 1 • Q2/5

724

QUESTION ID

S19-B1-Q2

TYPE

Short answer

CORRECT ANSWER

4,750

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 19 • BACKWARD 1 • Q2/5

724

True or false?

$$425 \times 24 = 10,200$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 19 • BACKWARD 1 • Q3/5

725

QUESTION ID

S19-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 19 • BACKWARD 1 • Q3/5

$$184 \times 35 = ?$$

- A.** 1,472
- B.** 5,440
- C.** 6,340
- D.** 6,440

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 19 • BACKWARD 1 • Q4/5

726

QUESTION ID

S19-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 6,440

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 19 • BACKWARD 1 • Q4/5

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$208 \times 15$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 19 • BACKWARD 1 • Q5/5

727

QUESTION ID

S19-B1-Q5

TYPE

Short answer

CORRECT ANSWER

3,120

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 19 • BACKWARD 1 • Q5/5

727

SET 19 • 6.NS.B.3

BACKWARD 2

Powers of 10 and the decimal point • Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$0.45 \times 10$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 19 • BACKWARD 2 • Q1/5

729

QUESTION ID

S19-B2-Q1

TYPE

Short answer

CORRECT ANSWER

4.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 19 • BACKWARD 2 • Q1/5

729

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$3.6 \times 100$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 19 • BACKWARD 2 • Q2/5

730

QUESTION ID

S19-B2-Q2

TYPE

Short answer

CORRECT ANSWER

360

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 19 • BACKWARD 2 • Q2/5

730

True or false?

$$24 \div 10 = 2.4$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 19 • BACKWARD 2 • Q3/5

731

QUESTION ID

S19-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 19 • BACKWARD 2 • Q3/5

$$1,620 \div 1,000 = ?$$

- A.** 0.162
- B.** 1.62
- C.** 16.2
- D.** 162

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 19 • BACKWARD 2 • Q4/5

732

QUESTION ID

S19-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 1.62

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 19 • BACKWARD 2 • Q4/5

ANSWER KEY

Reduced copy of the question page, with its answer

Divide.

$$475 \div 10$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 19 • BACKWARD 2 • Q5/5

733

QUESTION ID

S19-B2-Q5

TYPE

Short answer

CORRECT ANSWER

47.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 19 • BACKWARD 2 • Q5/5

733

SET 19 • 6.NS.B.3

BACKWARD 3

Multiplying tenths and hundredths • Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$0.3 \times 0.2$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B3-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 19 • BACKWARD 3 • Q1/5

735

QUESTION ID

S19-B3-Q1

TYPE

Short answer

CORRECT ANSWER

0.06

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B3-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 19 • BACKWARD 3 • Q1/5

735

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$0.5 \times 0.8$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B3-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 19 • BACKWARD 3 • Q2/5

736

QUESTION ID

S19-B3-Q2

TYPE

Short answer

CORRECT ANSWER

0.4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B3-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 19 • BACKWARD 3 • Q2/5

736

True or false?

$$0.4 \times 0.4 = 1.6$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B3-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 19 • BACKWARD 3 • Q3/5

737

QUESTION ID

S19-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B3-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 19 • BACKWARD 3 • Q3/5

$$0.2 \times 0.03 = ?$$

- A.** 0.006
- B.** 0.06
- C.** 0.6
- D.** 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B3-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 19 • BACKWARD 3 • Q4/5

738

QUESTION ID

S19-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 0.006

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B3-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 19 • BACKWARD 3 • Q4/5

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$1.2 \times 0.3$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B3-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 19 • BACKWARD 3 • Q5/5

739

QUESTION ID

S19-B3-Q5

TYPE

Short answer

CORRECT ANSWER

0.36

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B3-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 19 • BACKWARD 3 • Q5/5

SET 19 • 6.NS.B.3

BACKWARD 4

Rounding decimals to estimate • Grade 5 • Number & Operations in Base Ten •
5.NBT.A.4

Round 3.62 to the nearest whole number.

Rounded: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B4-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 19 • BACKWARD 4 • Q1/5

741

QUESTION ID

S19-B4-Q1

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B4-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 19 • BACKWARD 4 • Q1/5

Round 12.48 to the nearest tenth.

Rounded: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B4-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 19 • BACKWARD 4 • Q2/5

742

QUESTION ID

S19-B4-Q2

TYPE

Short answer

CORRECT ANSWER

12.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B4-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 19 • BACKWARD 4 • Q2/5

True or false?

4.25 rounded to the nearest whole number is 4.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B4-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 19 • BACKWARD 4 • Q3/5

743

QUESTION ID

S19-B4-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B4-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 19 • BACKWARD 4 • Q3/5

Round 0.456 to the nearest hundredth.

- A. 0.4
- B. 0.45
- C. 0.46
- D. 0.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B4-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 19 • BACKWARD 4 • Q4/5

744

QUESTION ID

S19-B4-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 0.46

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B4-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 19 • BACKWARD 4 • Q4/5

Round 8.75 to the nearest whole number.

Rounded: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B4-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 19 • BACKWARD 4 • Q5/5

745

QUESTION ID

S19-B4-Q5

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B4-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 19 • BACKWARD 4 • Q5/5

SET 19 • 6.NS.B.3

FORWARD 1

Multiply rational numbers • Grade 7 • The Number System • 7.NS.A.2.c

Find the value.

$$(-3.6)(0.45)$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q1

Grade 7 • The Number System • 7.NS.A.2.c • Set 19 • FORWARD 1 • Q1/5

747

QUESTION ID

S19-F1-Q1

TYPE

Short answer

CORRECT ANSWER

-1.62

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q1

Grade 7 • The Number System • 7.NS.A.2.c • Set 19 • FORWARD 1 • Q1/5

Find the value.

$$(-2.5)(-4.2)$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q2

Grade 7 • The Number System • 7.NS.A.2.c • Set 19 • FORWARD 1 • Q2/5

748

QUESTION ID

S19-F1-Q2

TYPE

Short answer

CORRECT ANSWER

10.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q2

Grade 7 • The Number System • 7.NS.A.2.c • Set 19 • FORWARD 1 • Q2/5

$$-0.8 \times 1.5 = ?$$

- A.** -1.2
- B.** -0.12
- C.** 0.12
- D.** 1.2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q3

Grade 7 • The Number System • 7.NS.A.2.c • Set 19 • FORWARD 1 • Q3/5

749

QUESTION ID

S19-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. -1.2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q3

Grade 7 • The Number System • 7.NS.A.2.c • Set 19 • FORWARD 1 • Q3/5

Find the value.

$$(-1.2)(-0.5)(-3)$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q4

Grade 7 • The Number System • 7.NS.A.2.c • Set 19 • FORWARD 1 • Q4/5

750

QUESTION ID

S19-F1-Q4

TYPE

Short answer

CORRECT ANSWER

-1.8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q4

Grade 7 • The Number System • 7.NS.A.2.c • Set 19 • FORWARD 1 • Q4/5

True or false?

$$(-0.6)(0.7) = 0.42$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q5

Grade 7 • The Number System • 7.NS.A.2.c • Set 19 • FORWARD 1 • Q5/5

751

QUESTION ID

S19-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q5

Grade 7 • The Number System • 7.NS.A.2.c • Set 19 • FORWARD 1 • Q5/5

SET 19 • 6.NS.B.3

FORWARD 2

Multiply numbers in scientific notation • Grade 8 • Expressions & Equations •
8.EE.A.4

Multiply. Write the answer in scientific notation.

$$(3.6 \times 10^4)(4.5 \times 10^{-2})$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 19 • FORWARD 2 • Q1/5

753

QUESTION ID

S19-F2-Q1

TYPE

Short answer

CORRECT ANSWER

1.62×10^3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 19 • FORWARD 2 • Q1/5

Multiply. Write the answer in scientific notation.

$$(2.5 \times 10^3)(4 \times 10^5)$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 19 • FORWARD 2 • Q2/5

754

QUESTION ID

S19-F2-Q2

TYPE

Short answer

CORRECT ANSWER

1×10^9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 19 • FORWARD 2 • Q2/5

$$(1.2 \times 10^6)(3 \times 10^{-4}) = ?$$

- A. 4.2×10^2
- B. 3.6×10^{10}
- C. 3.6×10^2
- D. 3.6×10^{-2}

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 19 • FORWARD 2 • Q3/5

755

QUESTION ID

S19-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 3.6×10^2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 19 • FORWARD 2 • Q3/5

A light-year is about 9.5×10^{12} km.

About how many kilometers are in 4.2 light-years? Write the answer in scientific notation.

Kilometers: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 19 • FORWARD 2 • Q4/5

756

QUESTION ID

S19-F2-Q4

TYPE

Short answer

CORRECT ANSWER

3.99×10^{13} km

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 19 • FORWARD 2 • Q4/5

True or false?

$$(5 \times 10^3)(6 \times 10^2) = 3 \times 10^6$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 19 • FORWARD 2 • Q5/5

757

QUESTION ID

S19-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 19 • FORWARD 2 • Q5/5

Divide multi-digit decimals

MAIN

Grade 6 main question

6.NS.B.3

BACKWARD 1

Dividing by a two-digit number

5.NBT.B.6

BACKWARD 2

Using powers of 10 to shift the decimal point

5.NBT.A.2

BACKWARD 3

Dividing tenths and hundredths

5.NBT.B.7

FORWARD 1

Divide rational numbers

7.NS.A.2.b

FORWARD 2

Divide numbers in scientific notation

8.EE.A.4

SET 20 • 6.NS.B.3

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.B.3

ANSWER KEY

Reduced copy of the question page, with its answer

Divide.

$$15.75 \div 0.35$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 20 • MAIN • Q1/5

760

QUESTION ID

S20-M-Q1

TYPE

Short answer

CORRECT ANSWER

45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 20 • MAIN • Q1/5

760

ANSWER KEY

Reduced copy of the question page, with its answer

Divide.

$$8.64 \div 2.4$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 20 • MAIN • Q2/5

761

QUESTION ID

S20-M-Q2

TYPE

Short answer

CORRECT ANSWER

3.6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 20 • MAIN • Q2/5

761

ANSWER KEY

Reduced copy of the question page, with its answer

Divide.

$$12.6 \div 0.45$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 20 • MAIN • Q3/5

762

QUESTION ID

S20-M-Q3

TYPE

Short answer

CORRECT ANSWER

28

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 20 • MAIN • Q3/5

762

Divide.

$$0.72 \div 0.08$$

- A.** 0.09
- B.** 0.9
- C.** 9
- D.** 90

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 20 • MAIN • Q4/5

763

QUESTION ID

S20-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 20 • MAIN • Q4/5

ANSWER KEY

Reduced copy of the question page, with its answer

Divide.

$$283.5 \div 13.5$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 20 • MAIN • Q5/5

764

QUESTION ID

S20-M-Q5

TYPE

Short answer

CORRECT ANSWER

21

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 20 • MAIN • Q5/5

764

SET 20 • 6.NS.B.3

BACKWARD 1

Dividing by a two-digit number • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.6

Divide.

$$1,575 \div 35$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 20 • BACKWARD 1 • Q1/5

766

QUESTION ID

S20-B1-Q1

TYPE

Short answer

CORRECT ANSWER

45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 20 • BACKWARD 1 • Q1/5

ANSWER KEY

Reduced copy of the question page, with its answer

Divide.

$$864 \div 24$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 20 • BACKWARD 1 • Q2/5

767

QUESTION ID

S20-B1-Q2

TYPE

Short answer

CORRECT ANSWER

36

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 20 • BACKWARD 1 • Q2/5

767

True or false?

$$1,260 \div 45 = 28$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 20 • BACKWARD 1 • Q3/5

768

QUESTION ID

S20-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 20 • BACKWARD 1 • Q3/5

$$672 \div 32 = ?$$

- A.** 12
- B.** 21
- C.** 22
- D.** 210

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 20 • BACKWARD 1 • Q4/5

769

QUESTION ID

S20-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 21

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 20 • BACKWARD 1 • Q4/5

ANSWER KEY

Reduced copy of the question page, with its answer

Divide.

$$2,835 \div 15$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 20 • BACKWARD 1 • Q5/5

770

QUESTION ID

S20-B1-Q5

TYPE

Short answer

CORRECT ANSWER

189

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 20 • BACKWARD 1 • Q5/5

770

SET 20 • 6.NS.B.3

BACKWARD 2

Using powers of 10 to shift the decimal point • Grade 5 • Number & Operations
in Base Ten • 5.NBT.A.2

Which has the same quotient as $15.75 \div 0.35$?

- A. $1,575 \div 35$
- B. $157.5 \div 35$
- C. $1,575 \div 3.5$
- D. $15.75 \div 35$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 20 • BACKWARD 2 • Q1/5

772

QUESTION ID

S20-B2-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. 1,575 ÷ 35

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 20 • BACKWARD 2 • Q1/5

True or false?

$8.64 \div 2.4$ has the same quotient as $86.4 \div 24$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 20 • BACKWARD 2 • Q2/5

773

QUESTION ID

S20-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 20 • BACKWARD 2 • Q2/5

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$0.45 \times 100$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 20 • BACKWARD 2 • Q3/5

774

QUESTION ID

S20-B2-Q3

TYPE

Short answer

CORRECT ANSWER

45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 20 • BACKWARD 2 • Q3/5

774

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$12.6 \times 100$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 20 • BACKWARD 2 • Q4/5

775

QUESTION ID

S20-B2-Q4

TYPE

Short answer

CORRECT ANSWER

1,260

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 20 • BACKWARD 2 • Q4/5

775

$$0.08 \times \underline{\quad} = 8$$

- A.** 0.1
- B.** 10
- C.** 100
- D.** 1,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 20 • BACKWARD 2 • Q5/5

776

QUESTION ID

S20-B2-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. 100

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 20 • BACKWARD 2 • Q5/5

SET 20 • 6.NS.B.3

BACKWARD 3

Dividing tenths and hundredths • Grade 5 • Number & Operations in Base Ten
• 5.NBT.B.7

ANSWER KEY

Reduced copy of the question page, with its answer

Divide.

$$2.4 \div 0.6$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B3-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 20 • BACKWARD 3 • Q1/5

778

QUESTION ID

S20-B3-Q1

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B3-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 20 • BACKWARD 3 • Q1/5

778

ANSWER KEY

Reduced copy of the question page, with its answer

Divide.

$$3.5 \div 0.5$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B3-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 20 • BACKWARD 3 • Q2/5

779

QUESTION ID

S20-B3-Q2

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B3-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 20 • BACKWARD 3 • Q2/5

779

True or false?

$$1.2 \div 0.3 = 0.4$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B3-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 20 • BACKWARD 3 • Q3/5

780

QUESTION ID

S20-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B3-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 20 • BACKWARD 3 • Q3/5

$$0.9 \div 0.3 = ?$$

- A.** 0.03
- B.** 0.3
- C.** 3
- D.** 30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B3-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 20 • BACKWARD 3 • Q4/5

781

QUESTION ID

S20-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B3-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 20 • BACKWARD 3 • Q4/5

ANSWER KEY

Reduced copy of the question page, with its answer

Divide.

$$4.8 \div 4$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B3-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 20 • BACKWARD 3 • Q5/5

782

QUESTION ID

S20-B3-Q5

TYPE

Short answer

CORRECT ANSWER

1.2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B3-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 20 • BACKWARD 3 • Q5/5

782

SET 20 • 6.NS.B.3

FORWARD 1

Divide rational numbers • Grade 7 • The Number System • 7.NS.A.2.b

Find the value.

$$(-15.75) \div 0.35$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q1

Grade 7 • The Number System • 7.NS.A.2.b • Set 20 • FORWARD 1 • Q1/5

784

QUESTION ID

S20-F1-Q1

TYPE

Short answer

CORRECT ANSWER

-45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q1

Grade 7 • The Number System • 7.NS.A.2.b • Set 20 • FORWARD 1 • Q1/5

Find the value.

$$-8.64 \div (-2.4)$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q2

Grade 7 • The Number System • 7.NS.A.2.b • Set 20 • FORWARD 1 • Q2/5

785

QUESTION ID

S20-F1-Q2

TYPE

Short answer

CORRECT ANSWER

3.6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q2

Grade 7 • The Number System • 7.NS.A.2.b • Set 20 • FORWARD 1 • Q2/5

True or false?

$$-(12 \div 4) = (-12) \div 4$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q3

Grade 7 • The Number System • 7.NS.A.2.b • Set 20 • FORWARD 1 • Q3/5

786

QUESTION ID

S20-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q3

Grade 7 • The Number System • 7.NS.A.2.b • Set 20 • FORWARD 1 • Q3/5

$$7.2 \div (-0.9) = ?$$

- A.** -8
- B.** -0.8
- C.** 0.8
- D.** 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q4

Grade 7 • The Number System • 7.NS.A.2.b • Set 20 • FORWARD 1 • Q4/5

787

QUESTION ID

S20-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. -8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q4

Grade 7 • The Number System • 7.NS.A.2.b • Set 20 • FORWARD 1 • Q4/5

Find the value.

$$-0.72 \div 0.08$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q5

Grade 7 • The Number System • 7.NS.A.2.b • Set 20 • FORWARD 1 • Q5/5

788

QUESTION ID

S20-F1-Q5

TYPE

Short answer

CORRECT ANSWER

-9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q5

Grade 7 • The Number System • 7.NS.A.2.b • Set 20 • FORWARD 1 • Q5/5

SET 20 • 6.NS.B.3

FORWARD 2

Divide numbers in scientific notation • Grade 8 • Expressions & Equations •
8.EE.A.4

Divide. Write the answer in scientific notation.

$$(6.3 \times 10^8) \div (2.1 \times 10^3)$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 20 • FORWARD 2 • Q1/5

790

QUESTION ID

S20-F2-Q1

TYPE

Short answer

CORRECT ANSWER

3×10^5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 20 • FORWARD 2 • Q1/5

Divide. Write the answer in scientific notation.

$$(4.8 \times 10^{-2}) \div (1.6 \times 10^{-5})$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 20 • FORWARD 2 • Q2/5

791

QUESTION ID

S20-F2-Q2

TYPE

Short answer

CORRECT ANSWER

3×10^3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 20 • FORWARD 2 • Q2/5

$$(9 \times 10^6) \div (3 \times 10^9) = ?$$

- A. 6×10^{-3}
- B. 3×10^{15}
- C. 3×10^3
- D. 3×10^{-3}

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 20 • FORWARD 2 • Q3/5

792

QUESTION ID

S20-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 3×10^{-3}

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 20 • FORWARD 2 • Q3/5

Divide. Write the answer in scientific notation.

$$(7.5 \times 10^4) \div (2.5 \times 10^{-1})$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 20 • FORWARD 2 • Q4/5

793

QUESTION ID

S20-F2-Q4

TYPE

Short answer

CORRECT ANSWER

3×10^5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 20 • FORWARD 2 • Q4/5

True or false?

$$(8.4 \times 10^5) \div (4.2 \times 10^2) = 2 \times 10^3$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 20 • FORWARD 2 • Q5/5

794

QUESTION ID

S20-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 20 • FORWARD 2 • Q5/5

Find the greatest common factor of two whole numbers

MAIN	Grade 6 main question	6.NS.B.4
BACKWARD 1	Factors	4.OA.B.4
BACKWARD 2	Prime and composite numbers	4.OA.B.4
BACKWARD 3	Division facts	3.OA.C.7
FORWARD 1	Factor linear expressions using a common factor	7.EE.A.1
FORWARD 2	Solve equations using the distributive property	8.EE.C.7.b

SET 21 • 6.NS.B.4

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.B.4

Find the greatest common factor of 24 and 36.

GCF: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q1

Grade 6 • The Number System • 6.NS.B.4 • Set 21 • MAIN • Q1/5

797

QUESTION ID

S21-M-Q1

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q1

Grade 6 • The Number System • 6.NS.B.4 • Set 21 • MAIN • Q1/5

What is the greatest common factor of 18 and 45?

- A.** 3
- B.** 9
- C.** 15
- D.** 90

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q2

Grade 6 • The Number System • 6.NS.B.4 • Set 21 • MAIN • Q2/5

798

QUESTION ID

S21-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q2

Grade 6 • The Number System • 6.NS.B.4 • Set 21 • MAIN • Q2/5

Find the greatest common factor of 28 and 42.

GCF: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q3

Grade 6 • The Number System • 6.NS.B.4 • Set 21 • MAIN • Q3/5

799

QUESTION ID

S21-M-Q3

TYPE

Short answer

CORRECT ANSWER

14

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q3

Grade 6 • The Number System • 6.NS.B.4 • Set 21 • MAIN • Q3/5

Find the greatest common factor of 30 and 75.

GCF: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q4

Grade 6 • The Number System • 6.NS.B.4 • Set 21 • MAIN • Q4/5

800

QUESTION ID

S21-M-Q4

TYPE

Short answer

CORRECT ANSWER

15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q4

Grade 6 • The Number System • 6.NS.B.4 • Set 21 • MAIN • Q4/5